
QFT Real Scalar Field Notes

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(Dated: Saturday 19th September, 2026)

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Part I

Free Field

1. CLASSICAL SCALAR FIELD

The Lagrangian density is

$$2\mathcal{L} = \eta^{\mu\nu}(\partial_\mu\phi)(\partial_\nu\phi) - m^2\phi^2, \quad (1)$$

where the metric tensor in Cartesian coordinates is

$$\eta_{\mu\nu} = \eta^{\mu\nu} = \text{diag}(1, -1, -1, -1), \quad (2)$$

and $\phi = \phi(t, \mathbf{r}) = \phi(\vec{x})$ is the scalar field and m is the field's mass. Classical fields don't have mass, but when the scalar field is quantized m is the rest mass its particle excitations have, so the name is appropriate. In vector calculus notation the Lagrangian density is

$$2\mathcal{L} = \dot{\phi}^2 - (\nabla\phi)^2 - m^2\phi^2, \quad (3)$$

where $\dot{\phi} = \partial_t\phi$. By the way, in GR the coordinates x^μ are not vectors, $\vec{x}$. But in Euclidean space and Minkowski spacetime, they are.

In classical physics ϕ is analogous to the particle position. The conjugate momentum density is

$$\pi(\vec{x}) = \frac{\partial\mathcal{L}}{\partial\dot{\phi}(\vec{x})}, \quad (4)$$

$$\begin{aligned} \pi(\vec{x}) &= \frac{\partial\mathcal{L}}{\partial\dot{\phi}} = \frac{1}{2} \frac{\partial 2\mathcal{L}}{\partial\dot{\phi}} = \frac{1}{2} \frac{\partial}{\partial\dot{\phi}} [\dot{\phi}^2 - (\nabla\phi)^2 - m^2\phi^2] \\ &= \frac{1}{2} \frac{\partial\dot{\phi}^2}{\partial\dot{\phi}} = \dot{\phi}, \end{aligned} \quad (5)$$

$$\pi(\vec{x}) = \dot{\phi}. \quad (6)$$

The Euler-Lagrange equation is

$$\partial_\mu \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} = \frac{\partial\mathcal{L}}{\partial\phi}, \quad (7)$$

where

$$\frac{\partial\mathcal{L}}{\partial\phi} = \frac{\partial}{\partial\phi} \frac{\dot{\phi}^2 - (\nabla\phi)^2 - m^2\phi^2}{2} = -m^2\phi. \quad (8)$$

The other derivative is trickier. Sean Carroll gives the rules for evaluating it in Sec. 1.10 of his GR textbook. First I write the Lagrangian as

$$2\mathcal{L} = \eta^{\rho\sigma}(\partial_\rho\phi)(\partial_\sigma\phi) - m^2\phi^2. \quad (9)$$

Then

$$\begin{aligned} \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} &= \frac{1}{2} \frac{\partial}{\partial(\partial_\mu\phi)} [\eta^{\rho\sigma}(\partial_\rho\phi)(\partial_\sigma\phi) - m^2\phi^2] = \\ &= \frac{1}{2} \frac{\partial}{\partial(\partial_\mu\phi)} \eta^{\rho\sigma}(\partial_\rho\phi)(\partial_\sigma\phi) = \frac{\eta^{\rho\sigma}}{2} \frac{\partial}{\partial(\partial_\mu\phi)} (\partial_\rho\phi)(\partial_\sigma\phi) = \\ &= \frac{\eta^{\rho\sigma}}{2} (\partial_\sigma\phi) \frac{\partial(\partial_\rho\phi)}{\partial(\partial_\mu\phi)} + \frac{\eta^{\rho\sigma}}{2} (\partial_\rho\phi) \frac{\partial(\partial_\sigma\phi)}{\partial(\partial_\mu\phi)}. \end{aligned} \quad (10)$$

I use the rule

$$\frac{dV_\alpha}{dV_\beta} = \delta_\alpha^\beta, \quad (11)$$

to calculate

$$\begin{aligned} \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} &= \frac{\eta^{\rho\sigma} \delta_\rho^\mu \partial_\sigma\phi + \eta^{\rho\sigma} \delta_\sigma^\mu \partial_\rho\phi}{2} = \\ &= \frac{\eta^{\mu\sigma} \partial_\sigma\phi + \eta^{\rho\mu} \partial_\rho\phi}{2} = \eta^{\mu\nu} \partial_\nu\phi. \end{aligned} \quad (12)$$

Then

$$\partial_\mu \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} = \partial_\mu \eta^{\mu\nu} \partial_\nu\phi = \eta^{\mu\nu} \partial_\mu \partial_\nu\phi. \quad (13)$$

I put everything together to get the equation of motion

$$\eta^{\mu\nu} \partial_\mu \partial_\nu\phi = -m^2\phi, \quad (14)$$

$$\partial_t^2\phi - \partial_x^2\phi - \partial_y^2\phi - \partial_z^2\phi = -m^2\phi, \quad (15)$$

$$(\square + m^2)\phi(\vec{x}) = 0, \quad \square = \partial_t^2 - \nabla^2 \quad (16)$$

This is the Klein-Gordon equation.

The Hamiltonian density is

$$\mathcal{H} = \pi(\vec{x})\dot{\phi} - \mathcal{L}, \quad (17)$$

$$2\mathcal{H} = \dot{\phi}^2 + (\nabla\phi)^2 + m^2\phi^2. \quad (18)$$

The Lagrangian ($\mathcal{L}$) and Hamiltonian ($\mathcal{H}$) are related to their densities by

$$\mathcal{L} = \int d^3x \mathcal{L}, \quad \mathcal{H} = \int d^3x \mathcal{H}. \quad (19)$$

I can't bring myself to write them or their densities with anything but script or calligraphic letters. Doing otherwise feels like a sin. So I defy standard convention and do what feels good.

The action is the integral of the Lagrangian

$$S = \int d^4x \mathcal{L}. \quad (20)$$

The action is more real than the Lagrangian, because it is what determines the dynamics. For example the classical

equation of motion comes from the principle of least action. All Lagrangians that yield the same action are equivalent. If we assume, as we always do here, that the fields are 0 at spacetime infinity, we can integrate by parts (move derivatives around $\mathcal{L}$ at the cost of a minus sign) and not worry about boundary terms. This yields another expression for the action

$$2\mathcal{L} = \dot{\phi}^2 - (\nabla\phi)^2 - m^2\phi^2 = -\phi\partial_t^2\phi + \phi\nabla^2\phi - m^2\phi^2 = -\phi(\partial_t^2 - \nabla^2)\phi - m^2\phi^2, \quad (21)$$

$$2\mathcal{L} = -\phi(\square + m^2)\phi. \quad (22)$$

1.1. Noether's Theorem & Conserved Quantities

Noether's theorem says that if a continuous transformation of the field (not the coordinates)

$$\phi(\vec{x}) \rightarrow \phi(\vec{x}) + \alpha \Delta\phi(\vec{x}), \quad (23)$$

where α is infinitesimal, exists and does not change the equation of motion then it is a symmetry, and the Lagrangian transforms like

$$\mathcal{L}(\vec{x}) \rightarrow \mathcal{L}(\vec{x}) + \alpha \partial_\mu J^\mu(\vec{x}). \quad (24)$$

Then

$$\partial_\mu j^\mu = 0, \quad j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \Delta\phi - J^\mu, \quad (25)$$

where j^μ is a conserved current with a conserved "charge"

$$Q = \int d^3x j^0(\vec{x}), \quad \dot{Q} = 0. \quad (26)$$

Each continuous transformation yields one conserved charge and associated current. I stated this theorem in a very particular way. I assumed only one scalar field and I did not make the assumptions about boundary conditions explicit. I will expand it later, if I need to.

Consider the coordinate transformation

$$x^{\bar{\mu}} = x^\mu - a^\mu. \quad (27)$$

where all of the components of the 4-vector $\vec{a}$ are infinitesimal. This is a translation in spacetime. Its effect on a scalar field is

$$\phi(\vec{x}) = \phi(\vec{x} + \vec{a}) = \phi(\vec{x}) + a^\mu \partial_\mu \phi(\vec{x}) \quad (28)$$

where I used a Taylor series. The Lagrangian is also a scalar so it transforms just like ϕ

$$2\mathcal{L}(\vec{x}) = \dot{\phi}(\vec{x})^2 - [\nabla\phi(\vec{x})]^2 - m^2\phi(\vec{x})^2, \\ \mathcal{L}(\vec{x}) = \mathcal{L}(\vec{x}) + a^\mu \partial_\mu \mathcal{L}(\vec{x}) = \mathcal{L}(\vec{x}) + a^\nu \partial_\mu \delta_\nu^\mu \mathcal{L}(\vec{x}). \quad (29)$$

Even though $\mathcal{L}$ is the Lagrangian density I often call it the Lagrangian. This is standard practice. Comparing Eqs. 23 & 24 with 28 & 29 yields $J^\mu_\nu = \delta_\nu^\mu \mathcal{L}$, $\alpha = a^\mu$, and $\Delta\phi = \partial_\mu \phi(\vec{x})$. This corresponds to 4 conserved currents which can be combined into the QFT momentum-energy tensor (not the same as the GR momentum-energy tensor)

$$T^\mu_\nu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \Delta\phi - J^\mu_\nu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\nu \phi - \delta_\nu^\mu \mathcal{L} = \boxed{T^\mu_\nu = \eta^{\mu\beta}(\partial_\beta \phi)(\partial_\nu \phi) - \delta_\nu^\mu \mathcal{L}}. \quad (30)$$

Note my careful index placement while constructing T^μ_ν . The conserved current associated with time translation is

$$T^\mu_0 = \eta^{\mu\beta}(\partial_\beta \phi)(\partial_0 \phi) - \delta_0^\mu \mathcal{L} = \eta^{\mu\beta}(\partial_\beta \phi)\dot{\phi} - \delta_0^\mu \mathcal{L}. \quad (31)$$

It's conserved charge density is

$$T^0_0 = \eta^{0\beta}(\partial_\beta \phi)\dot{\phi} - \delta_0^0 \mathcal{L} = \eta^{00}(\partial_0 \phi)\dot{\phi} - \mathcal{L} = \dot{\phi}^2 - \mathcal{L} = \frac{\dot{\phi}^2 + (\nabla\phi)^2 + m^2\phi^2}{2} = \mathcal{H}. \quad (32)$$

So the conserved charge density is the Hamiltonian density, i.e. the energy density, and the conserved charge is the energy of ϕ . Which is exactly what I expected from time translation symmetry. Repeating for the spatial components yields the currents

$$T^\mu_i = \eta^{\mu\beta}(\partial_\beta \phi)(\partial_i \phi) - \delta_i^\mu \mathcal{L} \quad (33)$$

and charge densities

$$T^0_i = \eta^{0\beta}(\partial_\beta \phi)(\partial_i \phi) - \delta_i^0 \mathcal{L} = \eta^{00}(\partial_0 \phi)(\partial_i \phi) = K_i = \dot{\phi} \partial_i \phi. \quad (34)$$

I prefer the vector over the one-form

$$K^i = \eta^{i\mu} K_\mu = \eta^{ii} K_i = -K_i = -\dot{\phi} \partial_i \phi \implies \boxed{K^i = -\dot{\phi} \partial_i \phi}, \quad (35)$$

where $\mathbf{K} = (K^x, K^y, K^z)$ is the momentum density of ϕ and the conserved charge is momentum, which is exactly what I expected for spatial translation.

2. SCHRÖDINGER PICTURE FIELD OPERATORS

When I quantize this free scalar theory I get the field operator and its conjugate momentum density operator

$$\phi(\mathbf{r}) = \int \frac{d^3p}{(2\pi)^3} \frac{a(\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{r}} + a^\dagger(\mathbf{p})e^{-i\mathbf{p}\cdot\mathbf{r}}}{\sqrt{2\omega(p)}}, \\ \pi(\mathbf{r}) = -i \int \frac{d^3p}{(2\pi)^3} \sqrt{\frac{\omega(p)}{2}} [a(\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{r}} - a^\dagger(\mathbf{p})e^{-i\mathbf{p}\cdot\mathbf{r}}]. \quad (36)$$

These equations are very important and maybe later I will explain where these equations come from. For now I will only describe them. I am working in the Schrödinger picture where the wavefunction evolves in time but the operators are constant. So even though the classical field and its conjugate momenta are dynamical these operators are not functions of time, they are only functions of space and the spacial components of momentum. That is why I am using bold face, which is my standard notation for 3-vectors. The operator $a(\mathbf{p})$ annihilates a particle with momentum $\mathbf{p}$, and $a^\dagger(\mathbf{p})$ creates one. The angular frequency function is

$$\omega(p) = \sqrt{p^2 + m^2}, \quad p = |\mathbf{p}|. \quad (37)$$

These operators obey the commutation relations

$$\begin{aligned} [\phi(\mathbf{r}_1), \phi(\mathbf{r}_2)] &= [\pi(\mathbf{r}_1), \pi(\mathbf{r}_2)] = 0, \\ [\phi(\mathbf{r}_1), \pi(\mathbf{r}_2)] &= i\delta^{(3)}(\mathbf{r}_1 - \mathbf{r}_2). \end{aligned} \quad (38)$$

The creation and annihilation operators obey the commutation relation

$$[a(\mathbf{p}_1), a^\dagger(\mathbf{p}_2)] = (2\pi)^3 \delta^{(3)}(\mathbf{p}_1 - \mathbf{p}_2). \quad (39)$$

To practice using these QFT operators I will use Eq. 39 to compute $[\phi(\mathbf{r}_1), \pi(\mathbf{r}_2)]$. First I derive a new expression for $\phi(\mathbf{r})$ by using u-substitution and relabeling integration variables.

$$\begin{aligned} \phi(\mathbf{r}) &= \int \frac{d^3p}{(2\pi)^3} \frac{a(\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{r}} + a^\dagger(\mathbf{p})e^{-i\mathbf{p}\cdot\mathbf{r}}}{\sqrt{2\omega(p)}} = \\ &\quad \int \frac{d^3p}{(2\pi)^3} \frac{a(\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{r}}}{\sqrt{2\omega(p)}} + I, \end{aligned} \quad (40)$$

where

$$I = \int \frac{d^3p}{(2\pi)^3} \frac{a^\dagger(\mathbf{p})e^{-i\mathbf{p}\cdot\mathbf{r}}}{\sqrt{2\omega(p)}}. \quad (41)$$

$$\begin{aligned} I &= \int_{-\infty}^{\infty} dp_x \int_{-\infty}^{\infty} dp_y \int_{-\infty}^{\infty} dp_z \\ &\quad \frac{a^\dagger(\mathbf{p})e^{-i\mathbf{p}\cdot\mathbf{r}}}{(2\pi)^3 \sqrt{2\omega(\sqrt{p_x^2 + p_y^2 + p_z^2})}} \end{aligned} \quad (42)$$

$$\begin{aligned} I &= \int_{-\infty}^{\infty} dk_x \int_{-\infty}^{\infty} dk_y \int_{-\infty}^{\infty} dk_z \\ &\quad \frac{a^\dagger(\mathbf{k})e^{-i\mathbf{k}\cdot\mathbf{r}}}{(2\pi)^3 \sqrt{2\omega(\sqrt{k_x^2 + k_y^2 + k_z^2})}} \end{aligned} \quad (43)$$

Now I substitute $\mathbf{p} = -\mathbf{k}$

$$\begin{aligned} I &= - \int_{-\infty}^{\infty} dp_x \int_{-\infty}^{\infty} dp_y \int_{-\infty}^{\infty} dp_z \\ &\quad \frac{a^\dagger(-\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{r}}}{(2\pi)^3 \sqrt{2\omega(\sqrt{p_x^2 + p_y^2 + p_z^2})}} \end{aligned} \quad (44)$$

$$I = \int_{-\infty}^{\infty} dp_x \int_{-\infty}^{\infty} dp_y \int_{-\infty}^{\infty} dp_z \frac{a^\dagger(-\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{r}}}{(2\pi)^3 \sqrt{2\omega(p)}} \quad (45)$$

$$I = \int d^3p \frac{a^\dagger(-\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{r}}}{(2\pi)^3 \sqrt{2\omega(p)}} \quad (46)$$

I plug this back into $\phi(\mathbf{r})$.

$$\phi(\mathbf{r}) = \int d^3p \frac{a(\mathbf{p}) + a^\dagger(-\mathbf{p})}{(2\pi)^3 \sqrt{2\omega(p)}} e^{i\mathbf{p}\cdot\mathbf{r}} \quad (47)$$

A similar derivation yields

$$\pi(\mathbf{r}) = -i \int \frac{d^3p}{(2\pi)^3} \sqrt{\frac{\omega(p)}{2}} [a(\mathbf{p}) - a^\dagger(-\mathbf{p})] e^{i\mathbf{p}\cdot\mathbf{r}}. \quad (48)$$

Using these expressions I compute $[\phi(\mathbf{r}), \pi(\mathbf{q})]$.

$$[\phi(\mathbf{r}), \pi(\mathbf{q})] = \phi(\mathbf{r})\pi(\mathbf{q}) - \pi(\mathbf{q})\phi(\mathbf{r}) \quad (49)$$

$$\begin{aligned} [\phi(\mathbf{r}), \pi(\mathbf{q})] &= -i \left[\int d^3p \frac{a(\mathbf{p}) + a^\dagger(-\mathbf{p})}{(2\pi)^3 \sqrt{2\omega(p)}} e^{i\mathbf{p}\cdot\mathbf{r}} \right] \\ &\quad \left[\int \frac{d^3p}{(2\pi)^3} \sqrt{\frac{\omega(p)}{2}} [a(\mathbf{p}) - a^\dagger(-\mathbf{p})] e^{i\mathbf{p}\cdot\mathbf{q}} \right] \\ &\quad + i \left[\int \frac{d^3p}{(2\pi)^3} \sqrt{\frac{\omega(p)}{2}} [a(\mathbf{p}) - a^\dagger(-\mathbf{p})] e^{i\mathbf{p}\cdot\mathbf{q}} \right] \\ &\quad \left[\int d^3p \frac{a(\mathbf{p}) + a^\dagger(-\mathbf{p})}{(2\pi)^3 \sqrt{2\omega(p)}} e^{i\mathbf{p}\cdot\mathbf{r}} \right] \end{aligned} \quad (50)$$

$$\begin{aligned} [\phi(\mathbf{r}), \pi(\mathbf{q})] &= -i \int d^3p \frac{a(\mathbf{p}) + a^\dagger(-\mathbf{p})}{(2\pi)^3 \sqrt{2\omega(p)}} e^{i\mathbf{p}\cdot\mathbf{r}} \\ &\quad \int \frac{d^3k}{(2\pi)^3} \sqrt{\frac{\omega(k)}{2}} [a(\mathbf{k}) - a^\dagger(-\mathbf{k})] e^{i\mathbf{k}\cdot\mathbf{q}} \\ &\quad + i \int \frac{d^3k}{(2\pi)^3} \sqrt{\frac{\omega(k)}{2}} [a(\mathbf{k}) - a^\dagger(-\mathbf{k})] e^{i\mathbf{k}\cdot\mathbf{q}} \\ &\quad \int d^3p \frac{a(\mathbf{p}) + a^\dagger(-\mathbf{p})}{(2\pi)^3 \sqrt{2\omega(p)}} e^{i\mathbf{p}\cdot\mathbf{r}} \end{aligned} \quad (51)$$

$$\begin{aligned} 2(2\pi)^6 i [\phi(\mathbf{r}), \pi(\mathbf{q})] &= \int d^3p [a(\mathbf{p}) + a^\dagger(-\mathbf{p})] e^{i\mathbf{p}\cdot\mathbf{r}} \\ &\quad \int d^3k \sqrt{\frac{\omega(k)}{\omega(p)}} [a(\mathbf{k}) - a^\dagger(-\mathbf{k})] e^{i\mathbf{k}\cdot\mathbf{q}} \\ &\quad - \int d^3k \sqrt{\frac{\omega(k)}{\omega(p)}} [a(\mathbf{k}) - a^\dagger(-\mathbf{k})] e^{i\mathbf{k}\cdot\mathbf{q}} \\ &\quad \int d^3p [a(\mathbf{p}) + a^\dagger(-\mathbf{p})] e^{i\mathbf{p}\cdot\mathbf{r}} \end{aligned} \quad (52)$$

The momenta $\mathbf{p}$ and $\mathbf{k}$ are not operators, they are just numbers, so they commute with everything.

$$\begin{aligned}
 2(2\pi)^6 i [\phi(\mathbf{r}), \pi(\mathbf{q})] &= \int d^3p d^3k \sqrt{\frac{\omega(k)}{\omega(p)}} e^{i(\mathbf{k}\cdot\mathbf{q}+\mathbf{p}\cdot\mathbf{r})} \\
 &\quad [a(\mathbf{p}) + a^\dagger(-\mathbf{p})][a(\mathbf{k}) - a^\dagger(-\mathbf{k})] \\
 &\quad - \int d^3k d^3p \sqrt{\frac{\omega(k)}{\omega(p)}} e^{i(\mathbf{k}\cdot\mathbf{q}+\mathbf{p}\cdot\mathbf{r})} \\
 &\quad [a(\mathbf{k}) - a^\dagger(-\mathbf{k})][a(\mathbf{p}) + a^\dagger(-\mathbf{p})] \quad (53)
 \end{aligned}$$

$$2(2\pi)^6 i [\phi(\mathbf{r}), \pi(\mathbf{q})] = \int d^3p d^3k \sqrt{\frac{\omega(k)}{\omega(p)}} e^{i(\mathbf{k}\cdot\mathbf{q}+\mathbf{p}\cdot\mathbf{r})} \Psi \quad (54)$$

$$\begin{aligned}
 \Psi &= [a(\mathbf{p}) + a^\dagger(-\mathbf{p})][a(\mathbf{k}) - a^\dagger(-\mathbf{k})] - \\
 &\quad [a(\mathbf{k}) - a^\dagger(-\mathbf{k})][a(\mathbf{p}) + a^\dagger(-\mathbf{p})] \quad (55)
 \end{aligned}$$

$$\begin{aligned}
 \Psi &= a^\dagger(-\mathbf{p})a(\mathbf{k}) - a(\mathbf{p})a^\dagger(-\mathbf{k}) - a(\mathbf{k})a^\dagger(-\mathbf{p}) + a^\dagger(-\mathbf{k})a(\mathbf{p}) \\
 &\quad + \cancel{a(\mathbf{p})a(\mathbf{k}) - a(\mathbf{k})a(\mathbf{p}) - a^\dagger(-\mathbf{p})a^\dagger(-\mathbf{k}) + a^\dagger(-\mathbf{k})a^\dagger(-\mathbf{p})} \quad (56)
 \end{aligned}$$

$$[a(\mathbf{p}), a(\mathbf{k})] = [a^\dagger(\mathbf{p}), a^\dagger(\mathbf{k})] = 0 \quad (57)$$

$$\Psi = a^\dagger(-\mathbf{k})a(\mathbf{p}) - a(\mathbf{p})a^\dagger(-\mathbf{k}) + a^\dagger(-\mathbf{p})a(\mathbf{k}) - a(\mathbf{k})a^\dagger(-\mathbf{p}) \quad (58)$$

$$\Psi = [a^\dagger(-\mathbf{k}), a(\mathbf{p})] + [a^\dagger(-\mathbf{p}), a(\mathbf{k})] \quad (59)$$

$$\Psi = -(2\pi)^3 [\delta^{(3)}(-\mathbf{k} - \mathbf{p}) + \delta^{(3)}(-\mathbf{p} - \mathbf{k})] \quad (60)$$

$$\delta(x) = \delta(-x) \quad (61)$$

$$\Psi = -2(2\pi)^3 \delta^{(3)}(-\mathbf{p} - \mathbf{k}) \quad (62)$$

$$\begin{aligned}
 2(2\pi)^6 i [\phi(\mathbf{r}), \pi(\mathbf{q})] &= \\
 -2(2\pi)^3 \int d^3p d^3k \sqrt{\frac{\omega(k)}{\omega(p)}} e^{i(\mathbf{k}\cdot\mathbf{q}+\mathbf{p}\cdot\mathbf{r})} \delta^{(3)}(-\mathbf{p} - \mathbf{k}) \quad (63)
 \end{aligned}$$

$$\begin{aligned}
 (2\pi)^3 [\phi(\mathbf{r}), \pi(\mathbf{q})] &= \\
 i \int d^3p d^3k \sqrt{\frac{\omega(k)}{\omega(p)}} e^{i(\mathbf{k}\cdot\mathbf{q}+\mathbf{p}\cdot\mathbf{r})} \delta^{(3)}(-\mathbf{p} - \mathbf{k}) \quad (64)
 \end{aligned}$$

$$-\mathbf{p} - \mathbf{k} = 0 \implies \mathbf{p} + \mathbf{k} = 0 \implies \mathbf{k} = -\mathbf{p} \quad (65)$$

$$(2\pi)^3 [\phi(\mathbf{r}), \pi(\mathbf{q})] = i \int d^3p \sqrt{\frac{\omega(p)}{\omega(p)}} e^{i\mathbf{p}\cdot(\mathbf{r}-\mathbf{q})} \quad (66)$$

$$(2\pi)^3 [\phi(\mathbf{r}), \pi(\mathbf{q})] = i \int d^3p e^{i\mathbf{p}\cdot\mathbf{u}} \quad (67)$$

$$\mathbf{u} = \mathbf{r} - \mathbf{q} \quad (68)$$

$$\mathbf{p} \cdot \mathbf{u} = p^i u_i = p^x u_x + p^y u_y + p^z u_z \quad (69)$$

$$(2\pi)^3 [\phi(\mathbf{r}), \pi(\mathbf{q})] = i \int d^3p e^{i(p^x u_x + p^y u_y + p^z u_z)} \quad (70)$$

$$\begin{aligned}
 (2\pi)^3 [\phi(\mathbf{r}), \pi(\mathbf{q})] &= \\
 i \int_{-\infty}^{\infty} dp^x e^{ip^x u_x} \int_{-\infty}^{\infty} dp^y e^{ip^y u_y} \int_{-\infty}^{\infty} dp^z e^{ip^z u_z} \quad (71)
 \end{aligned}$$

In Eitan Geva's spectroscopy class I proved that (so I won't prove it here)

$$2\pi \delta(x) = \int_{-\infty}^{\infty} dp e^{ipx}. \quad (72)$$

$$\cancel{(2\pi)^3} [\phi(\mathbf{r}), \pi(\mathbf{q})] = i \cancel{(2\pi)^3} \delta(u_x) \delta(u_y) \delta(u_z) \quad (73)$$

$$[\phi(\mathbf{r}), \pi(\mathbf{q})] = i \delta(u_x) \delta(u_y) \delta(u_z) \quad (74)$$

$$[\phi(\mathbf{r}), \pi(\mathbf{q})] = i \delta(u_x) \delta(u_y) \delta(u_z) \quad (75)$$

$$\delta^{(3)}(\mathbf{r}) = \delta(x) \delta(y) \delta(z), \mathbf{r} = (x, y, z) \quad (76)$$

$$[\phi(\mathbf{r}), \pi(\mathbf{q})] = i \delta^{(3)}(\mathbf{u}) \quad (77)$$

$$[\phi(\mathbf{r}), \pi(\mathbf{q})] = i \delta^{(3)}(\mathbf{r} - \mathbf{q}) \quad (78)$$

QED.

2.1. Hamiltonian and Total Momentum Operators

I derive the Hamiltonian operator for the QFT of a real valued scalar field. First I will need this result when I go into the Heisenberg picture. Also it's a good exercise that will help me practice manipulating field operators. The Hamiltonian density (Eq. 18) is

$$2\mathcal{H} = \dot{\phi}^2 + (\nabla\phi)^2 + m^2\phi^2. \quad (79)$$

At this point I must write it as a functional of $\phi(\vec{x})$ and $\pi(\vec{x})$ (i.e. it should not contain $\dot{\phi}$)

$$2\mathcal{H}[\phi, \pi](\vec{x}) = \pi(\vec{x})^2 + [\nabla\phi(\vec{x})]^2 + m^2\phi(\vec{x})^2. \quad (80)$$

This is still a classical quantity with time dependence, $\vec{x} = (t, \mathbf{r})$. Now I turn it into an operator by replacing the classical dynamical quantities $\phi(\vec{x})$ and $\pi(\vec{x})$ with the time independent Schrödinger operators $\phi(\mathbf{r})$ and $\pi(\mathbf{r})$

$$2\mathcal{H} = \pi(\mathbf{r})^2 + [\nabla\phi(\mathbf{r})]^2 + m^2\phi(\mathbf{r})^2. \quad (81)$$

Now $\mathcal{H}$ is a time independent QFT operator. The Hamiltonian is

$$2\mathcal{H} = 2 \int d^3x \mathcal{H} = \int d^3x [\pi(\mathbf{r})^2 + [\nabla\phi(\mathbf{r})]^2 + m^2\phi(\mathbf{r})^2]. \quad (82)$$

I calculate the explicit form of $\mathcal{H}$.

$$\phi(\mathbf{r}) = \int d^3p \frac{a(\mathbf{p}) + a^\dagger(-\mathbf{p})}{(2\pi)^3 \sqrt{2\omega(p)}} e^{i\mathbf{p}\cdot\mathbf{r}} \quad (83)$$

$$\phi(\mathbf{r})^2 = \int d^3p d^3k \frac{[a(\mathbf{p}) + a^\dagger(-\mathbf{p})][a(\mathbf{k}) + a^\dagger(-\mathbf{k})]}{2(2\pi)^6 \sqrt{\omega(k)\omega(p)}} e^{i(\mathbf{p}+\mathbf{k})\cdot\mathbf{r}} \quad (84)$$

I introduce the less cumbersome notation $a_{\mathbf{p}} = a(\mathbf{p})$ and $\omega(p) = \omega_p$.

$$\phi(\mathbf{r})^2 = \int d^3p d^3k \frac{(a_{\mathbf{p}} + a_{-\mathbf{p}}^\dagger)(a_{\mathbf{k}} + a_{-\mathbf{k}}^\dagger)}{2(2\pi)^6 \sqrt{\omega_k\omega_p}} e^{i(\mathbf{p}+\mathbf{k})\cdot\mathbf{r}} \quad (85)$$

$$\nabla\phi(\mathbf{r}) = \int d^3p \frac{a_{\mathbf{p}} + a_{-\mathbf{p}}^\dagger}{(2\pi)^3 \sqrt{2\omega_p}} \nabla e^{i\mathbf{p}\cdot\mathbf{r}} \quad (86)$$

$$\nabla\phi(\mathbf{r}) = i \int d^3p \frac{a_{\mathbf{p}} + a_{-\mathbf{p}}^\dagger}{(2\pi)^3 \sqrt{2\omega_p}} \mathbf{p} e^{i\mathbf{p}\cdot\mathbf{r}} \quad (87)$$

$$[\nabla\phi(\mathbf{r})]^2 = \nabla\phi(\mathbf{r}) \cdot \nabla\phi(\mathbf{r}) \quad (88)$$

$$[\nabla\phi(\mathbf{r})]^2 = - \int d^3p d^3k \frac{(a_{\mathbf{p}} + a_{-\mathbf{p}}^\dagger)(a_{\mathbf{k}} + a_{-\mathbf{k}}^\dagger)}{2(2\pi)^6 \sqrt{\omega_k\omega_p}} e^{i(\mathbf{p}+\mathbf{k})\cdot\mathbf{r}} \mathbf{p} \cdot \mathbf{k} \quad (89)$$

$$\pi(\mathbf{r}) = -i \int \frac{d^3p}{(2\pi)^3} \sqrt{\frac{\omega_p}{2}} (a_{\mathbf{p}} - a_{-\mathbf{p}}^\dagger) e^{i\mathbf{p}\cdot\mathbf{r}} \quad (90)$$

$$\pi(\mathbf{r})^2 = - \int \frac{d^3p d^3k}{2(2\pi)^6} \sqrt{\omega_p\omega_k} (a_{\mathbf{p}} - a_{-\mathbf{p}}^\dagger)(a_{\mathbf{k}} - a_{-\mathbf{k}}^\dagger) e^{i(\mathbf{p}+\mathbf{k})\cdot\mathbf{r}} \quad (91)$$

$$\mathcal{H} = \int \frac{d^3x d^3p d^3k}{4(2\pi)^6} e^{i(\mathbf{p}+\mathbf{k})\cdot\mathbf{r}} \left[-\sqrt{\omega_p\omega_k} (a_{\mathbf{p}} - a_{-\mathbf{p}}^\dagger)(a_{\mathbf{k}} - a_{-\mathbf{k}}^\dagger) + \frac{m^2 - \mathbf{p} \cdot \mathbf{k}}{\sqrt{\omega_k\omega_p}} (a_{\mathbf{p}} + a_{-\mathbf{p}}^\dagger)(a_{\mathbf{k}} + a_{-\mathbf{k}}^\dagger) \right] \quad (92)$$

$$\mathcal{H} = \int \frac{d^3p d^3k}{4(2\pi)^6} \left[-\sqrt{\omega_p\omega_k} (a_{\mathbf{p}} - a_{-\mathbf{p}}^\dagger)(a_{\mathbf{k}} - a_{-\mathbf{k}}^\dagger) + \frac{m^2 - \mathbf{p} \cdot \mathbf{k}}{\sqrt{\omega_k\omega_p}} (a_{\mathbf{p}} + a_{-\mathbf{p}}^\dagger)(a_{\mathbf{k}} + a_{-\mathbf{k}}^\dagger) \right] \int d^3x e^{i(\mathbf{p}+\mathbf{k})\cdot\mathbf{r}} \quad (93)$$

$$\int d^3x e^{i(\mathbf{p}+\mathbf{k})\cdot\mathbf{r}} = (2\pi)^3 \delta^{(3)}(\mathbf{p} + \mathbf{k}) \quad (94)$$

$$\mathcal{H} = \int \frac{d^3p d^3k}{4(2\pi)^3} \left[-\sqrt{\omega_p\omega_k} (a_{\mathbf{p}} - a_{-\mathbf{p}}^\dagger)(a_{\mathbf{k}} - a_{-\mathbf{k}}^\dagger) + \frac{m^2 - \mathbf{p} \cdot \mathbf{k}}{\sqrt{\omega_k\omega_p}} (a_{\mathbf{p}} + a_{-\mathbf{p}}^\dagger)(a_{\mathbf{k}} + a_{-\mathbf{k}}^\dagger) \right] \delta^{(3)}(\mathbf{p} + \mathbf{k}) \quad (95)$$

$$\mathcal{H} = \int \frac{d^3p}{4(2\pi)^3} \left[-\omega_p (a_{\mathbf{p}} - a_{-\mathbf{p}}^\dagger)(a_{-\mathbf{p}} - a_{\mathbf{p}}^\dagger) + \frac{m^2 + \mathbf{p} \cdot \mathbf{p}}{\omega_p} (a_{\mathbf{p}} + a_{-\mathbf{p}}^\dagger)(a_{-\mathbf{p}} + a_{\mathbf{p}}^\dagger) \right] \quad (96)$$

$$\mathcal{H} = \int \frac{d^3p}{4(2\pi)^3} \left[-\omega_p (a_{\mathbf{p}} - a_{-\mathbf{p}}^\dagger)(a_{-\mathbf{p}} - a_{\mathbf{p}}^\dagger) + \frac{m^2 + p^2}{\omega_p} \right], \quad \omega_p^2 = p^2 + m^2 \quad (97)$$

$$\mathcal{H} = \int \frac{d^3p \omega_p}{4(2\pi)^3} \Phi \quad (98)$$

$$\Phi = (a_{\mathbf{p}} + a_{-\mathbf{p}}^\dagger)(a_{-\mathbf{p}} + a_{\mathbf{p}}^\dagger) - (a_{\mathbf{p}} - a_{-\mathbf{p}}^\dagger)(a_{-\mathbf{p}} - a_{\mathbf{p}}^\dagger) \quad (99)$$

$$\Phi = \cancel{a_{\mathbf{p}} a_{-\mathbf{p}}} - \cancel{a_{-\mathbf{p}} a_{\mathbf{p}}} + a_{\mathbf{p}} a_{\mathbf{p}}^\dagger + a_{-\mathbf{p}}^\dagger a_{-\mathbf{p}} + a_{-\mathbf{p}}^\dagger a_{-\mathbf{p}} + \cancel{a_{\mathbf{p}}^\dagger a_{-\mathbf{p}}} + \cancel{a_{-\mathbf{p}}^\dagger a_{\mathbf{p}}} + a_{\mathbf{p}} a_{\mathbf{p}}^\dagger \quad (100)$$

$$\Phi = 2a_{\mathbf{p}} a_{\mathbf{p}}^\dagger + 2a_{-\mathbf{p}}^\dagger a_{-\mathbf{p}} \quad (101)$$

$$\mathcal{H} = \int \frac{d^3p \omega_p}{2(2\pi)^3} (a_{\mathbf{p}} a_{\mathbf{p}}^\dagger + a_{-\mathbf{p}}^\dagger a_{-\mathbf{p}}) \quad (102)$$

$$2(2\pi)^3 \mathcal{H} = \int d^3p \omega_p (a_{\mathbf{p}} a_{\mathbf{p}}^\dagger + a_{-\mathbf{p}}^\dagger a_{-\mathbf{p}}) \quad (103)$$

$$2(2\pi)^3 \mathcal{H} = \int d^3p \omega_p a_{\mathbf{p}} a_{\mathbf{p}}^\dagger + I \quad (104)$$

$$I = \int d^3p \omega_p a_{-\mathbf{p}}^\dagger a_{-\mathbf{p}} \quad (105)$$

I use the substitution $\mathbf{p} = -\mathbf{k}$.

$$I = \int_{-\infty}^{\infty} dp_x \int_{-\infty}^{\infty} dp_y \int_{-\infty}^{\infty} dp_z \omega_p a_{-\mathbf{p}}^\dagger a_{-\mathbf{p}} \quad (106)$$

$$I = \int_{-\infty}^{\infty} dk_x \int_{-\infty}^{\infty} dk_y \int_{-\infty}^{\infty} dk_z \omega_k a_{\mathbf{k}}^\dagger a_{\mathbf{k}} \quad (107)$$

$$I = \int d^3k \omega_k a_{\mathbf{k}}^\dagger a_{\mathbf{k}} = \int d^3p \omega_p a_{\mathbf{p}}^\dagger a_{\mathbf{p}} \quad (108)$$

$$2(2\pi)^3 \mathcal{H} = \int d^3p \omega_p (a_{\mathbf{p}} a_{\mathbf{p}}^\dagger + a_{\mathbf{p}}^\dagger a_{\mathbf{p}}) \quad (109)$$

$$[a_{\mathbf{p}}, a_{\mathbf{p}}^\dagger] = a_{\mathbf{p}} a_{\mathbf{p}}^\dagger - a_{\mathbf{p}}^\dagger a_{\mathbf{p}} \implies [a_{\mathbf{p}}, a_{\mathbf{p}}^\dagger] + a_{\mathbf{p}}^\dagger a_{\mathbf{p}} = a_{\mathbf{p}} a_{\mathbf{p}}^\dagger \quad (110)$$

$$2(2\pi)^3 \mathcal{H} = \int d^3p \omega_p (2a_{\mathbf{p}}^\dagger a_{\mathbf{p}} + [a_{\mathbf{p}}, a_{\mathbf{p}}^\dagger]) \quad (111)$$

$$[a_{\mathbf{p}}, a_{\mathbf{p}}^\dagger] = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}) = (2\pi)^3 \delta^{(3)}(0) \quad (112)$$

$$2(2\pi)^3 \mathcal{H} = \int d^3p [\omega_p 2a_{\mathbf{p}}^\dagger a_{\mathbf{p}} + (2\pi)^3 \delta^{(3)}(0)] \quad (113)$$

Since $\delta^{(3)}(0) = \infty$ the Hamiltonian diverges. This is not surprising: the Hamiltonian density $\mathcal{H}$ is the energy density per unit volume of space. Since space is infinite and $\mathcal{H}$ is finite the total energy $\mathcal{H}$ is infinite. The standard way for dealing with this singularity is: ignore it. The final result is

$$\boxed{\mathcal{H} = \frac{1}{(2\pi)^3} \int d^3p \omega_p a_{\mathbf{p}}^\dagger a_{\mathbf{p}}}. \quad (114)$$

I integrated the field's Hamiltonian (energy) density to get its total Hamiltonian (energy). Likewise I can integrate its momentum density (Eq. 35) to get its total momentum

$$\mathbf{P} = \int d^3x \mathbf{K} = \frac{1}{(2\pi)^3} \int d^3p \mathbf{p} a_{\mathbf{p}}^\dagger a_{\mathbf{p}}. \quad (115)$$

I won't actually do the calculation to save time, but I can always work out the algebra if I need to later.

Two important commutators are

$$[\mathcal{H}, a_{\mathbf{p}}^\dagger] = \omega_p a_{\mathbf{p}}^\dagger, \quad [a_{\mathbf{p}}, \mathcal{H}] = \omega_p a_{\mathbf{p}} \quad (116)$$

I will only prove one of these.

$$\begin{aligned} [a_{\mathbf{p}}, \mathcal{H}] &= a_{\mathbf{p}} \mathcal{H} - \mathcal{H} a_{\mathbf{p}} = \\ a_{\mathbf{p}} \frac{1}{(2\pi)^3} \int d^3p \omega_p a_{\mathbf{p}}^\dagger a_{\mathbf{p}} - \left[\frac{1}{(2\pi)^3} \int d^3p \omega_p a_{\mathbf{p}}^\dagger a_{\mathbf{p}} \right] a_{\mathbf{p}} &= \\ \frac{1}{(2\pi)^3} \int d^3k \omega_k (a_{\mathbf{p}} a_{\mathbf{k}}^\dagger a_{\mathbf{k}} - a_{\mathbf{k}}^\dagger a_{\mathbf{k}} a_{\mathbf{p}}) &= \\ \frac{1}{(2\pi)^3} \int d^3k \omega_k (a_{\mathbf{p}} a_{\mathbf{k}}^\dagger - a_{\mathbf{k}}^\dagger a_{\mathbf{p}}) a_{\mathbf{k}} &= \\ \frac{1}{(2\pi)^3} \int d^3k \omega_k [a_{\mathbf{p}}, a_{\mathbf{k}}^\dagger] a_{\mathbf{k}} = \int d^3k \omega_k \delta^{(3)}(\mathbf{p} - \mathbf{k}) a_{\mathbf{k}} &= \\ \omega_p a_{\mathbf{p}} \end{aligned} \quad (117)$$

QED.

3. HEISENBERG PICTURE

In QFT the wavefunction is less accessible than in non-relativistic quantum mechanics. So we usually work in the Heisenberg picture, where the fields evolve dynamically in time, just like in classical field theory (CFT).

3.1. Nonrelativistic Quantum Mechanics

For a time-independent Hamiltonian, $\mathcal{H}$, the time evolution operator is $\exp[-i\mathcal{H}(t - t_0)]$ where t_0 is the initial time and t is the final time. For simplicity I take $t_0 = 0$ so that wavefunctions evolve like

$$|\Psi(t)\rangle = e^{-i\mathcal{H}t} |\Psi(0)\rangle. \quad (118)$$

Let A be an operator in the Schrödinger picture. I assume A has no explicit time dependence. It's expectation value as a function of time is

$$\begin{aligned} \langle A \rangle(t) &= \langle \Psi(t) | A | \Psi(t) \rangle = \langle \Psi(0) | e^{i\mathcal{H}t} A e^{-i\mathcal{H}t} | \Psi(0) \rangle \\ &= \langle \Psi(0) | \mathcal{A}(t) | \Psi(0) \rangle, \end{aligned} \quad (119)$$

where

$$\mathcal{A}(t) = e^{i\mathcal{H}t} A e^{-i\mathcal{H}t} \quad (120)$$

is the dynamical operator corresponding to A in the Heisenberg picture. Its equation of motion is

$$i\dot{\mathcal{A}}(t) = [\mathcal{A}(t), \mathcal{H}]. \quad (121)$$

3.2. QFT

The Schrödinger field operator and its conjugate momentum operator are (Eq. 36)

$$\begin{aligned} \phi(\mathbf{r}) &= \int \frac{d^3p}{(2\pi)^3} \frac{a(\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{r}} + a^\dagger(\mathbf{p})e^{-i\mathbf{p}\cdot\mathbf{r}}}{\sqrt{2\omega(p)}}, \\ \pi(\mathbf{r}) &= -i \int \frac{d^3p}{(2\pi)^3} \sqrt{\frac{\omega(p)}{2}} [a(\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{r}} - a^\dagger(\mathbf{p})e^{-i\mathbf{p}\cdot\mathbf{r}}]. \end{aligned} \quad (122)$$

In CFT the field ϕ is analogous to particle position in classical mechanics. And the conjugate momentum of the field $\pi = \dot{\phi}$ is analogous to the particle conjugate momentum. In the Schrödinger picture of quantum mechanics position and momentum are operators with no time dependence. So it's not surprising that in Schrödinger QFT $\phi(\mathbf{r})$ and $\pi(\mathbf{r})$ have no time dependence. I say that because it's a little disconcerting that in CFT $\pi = \dot{\phi}$ but in Schrödinger QFT ϕ has no time dependence ($\dot{\phi} = 0$).

The time dependent Heisenberg field operator is

$$\phi(\vec{x}) = \phi(t, \mathbf{r}) = e^{i\mathcal{H}t} \phi(\mathbf{r}) e^{-i\mathcal{H}t}, \quad \phi(\mathbf{r}) = \phi(0, \mathbf{r}), \quad (123)$$

where I overload ϕ . Likewise

$$\pi(\vec{x}) = e^{i\mathcal{H}t} \pi(\mathbf{r}) e^{-i\mathcal{H}t}. \quad (124)$$

Let $\mathcal{A}$ be a Schrödinger operator with no explicit time dependence, $\partial_t \mathcal{A} = 0$. It's Heisenberg version's, $\mathcal{A}(t)$, equation of motion is

$$i\partial_t \mathcal{A}(t) = [\mathcal{A}(t), \mathcal{H}]. \quad (125)$$

I assume the Hamiltonian is time independent but I think this equation holds even if it is time dependent.

The Heisenberg creation & annihilation operators are

$$a_{\mathbf{p}}(t) = e^{i\mathcal{H}t} a_{\mathbf{p}} e^{-i\mathcal{H}t}, \quad a_{\mathbf{p}}^\dagger(t) = e^{i\mathcal{H}t} a_{\mathbf{p}}^\dagger e^{-i\mathcal{H}t}. \quad (126)$$

Following Peskin & Schroeder (PS) I introduce

$$E_{\mathbf{p}} = \omega_{\mathbf{p}}. \quad (127)$$

This notation makes sense because $\omega_{\mathbf{p}} = \sqrt{p^2 + m^2}$ is the energy of a particle with momentum $\mathbf{p}$ and mass m . When $\hbar = 1$ the frequency is an energy. From Eq. 116 I derive some identities.

$$\mathcal{H} a_{\mathbf{p}} = a_{\mathbf{p}}(\mathcal{H} - E_{\mathbf{p}}), \quad \mathcal{H} a_{\mathbf{p}}^\dagger = a_{\mathbf{p}}^\dagger(\mathcal{H} + E_{\mathbf{p}}) \quad (128)$$

$$\begin{aligned} \mathcal{H}^2 a_{\mathbf{p}} &= \mathcal{H} \mathcal{H} a_{\mathbf{p}} = \mathcal{H} a_{\mathbf{p}}(\mathcal{H} - E_{\mathbf{p}}) = a_{\mathbf{p}}(\mathcal{H} - E_{\mathbf{p}})(\mathcal{H} - E_{\mathbf{p}}) \\ &= a_{\mathbf{p}}(\mathcal{H} - E_{\mathbf{p}})^2 \end{aligned} \quad (129)$$

So

$$\mathcal{H}^n a_{\mathbf{p}} = a_{\mathbf{p}}(\mathcal{H} - E_{\mathbf{p}})^n, \quad \mathcal{H}^n a_{\mathbf{p}}^\dagger = a_{\mathbf{p}}^\dagger(\mathcal{H} + E_{\mathbf{p}})^n, \quad (130)$$

where n is a positive integer. I derive more identities using these.

$$\begin{aligned} a_{\mathbf{p}}(t) &= e^{i\mathcal{H}t} a_{\mathbf{p}} e^{-i\mathcal{H}t} = \sum_{n=0}^{\infty} \frac{(i\mathcal{H}t)^n}{n!} a_{\mathbf{p}} e^{-i\mathcal{H}t} = \\ \sum_{n=0}^{\infty} \frac{(it)^n}{n!} \mathcal{H}^n a_{\mathbf{p}} e^{-i\mathcal{H}t} &= \sum_{n=0}^{\infty} \frac{(it)^n}{n!} a_{\mathbf{p}}(\mathcal{H} - E_{\mathbf{p}})^n e^{-i\mathcal{H}t} = \\ a_{\mathbf{p}} \sum_{n=0}^{\infty} \frac{[it(\mathcal{H} - E_{\mathbf{p}})]^n}{n!} e^{-i\mathcal{H}t} &= a_{\mathbf{p}} e^{i\mathcal{H}t - iE_{\mathbf{p}}t} e^{-i\mathcal{H}t} = \\ a_{\mathbf{p}} e^{i\mathcal{H}t - iE_{\mathbf{p}}t - i\mathcal{H}t} &= a_{\mathbf{p}} e^{-iE_{\mathbf{p}}t} \end{aligned} \quad (131)$$

$$a_{\mathbf{p}}(t) = a_{\mathbf{p}} e^{-iE_{\mathbf{p}}t} \implies a_{\mathbf{p}}^\dagger(t) = a_{\mathbf{p}}^\dagger e^{iE_{\mathbf{p}}t} \quad (132)$$

Using these identities I rewrite the time dependent field operator.

$$\begin{aligned} \phi(\vec{x}) &= e^{i\mathcal{H}t} \phi(\mathbf{r}) e^{-i\mathcal{H}t} = \\ \int \frac{d^3p}{(2\pi)^3} \frac{e^{i\mathcal{H}t} a_{\mathbf{p}} e^{-i\mathcal{H}t} e^{i\mathbf{p}\cdot\mathbf{r}} + e^{i\mathcal{H}t} a_{\mathbf{p}}^\dagger e^{-i\mathcal{H}t} e^{-i\mathbf{p}\cdot\mathbf{r}}}{\sqrt{2E_{\mathbf{p}}}} &= \\ \int \frac{d^3p}{(2\pi)^3} \frac{a_{\mathbf{p}}(t) e^{i\mathbf{p}\cdot\mathbf{r}} + a_{\mathbf{p}}^\dagger(t) e^{-i\mathbf{p}\cdot\mathbf{r}}}{\sqrt{2E_{\mathbf{p}}}} &= \\ \int \frac{d^3p}{(2\pi)^3} \frac{a_{\mathbf{p}} e^{-iE_{\mathbf{p}}t} e^{i\mathbf{p}\cdot\mathbf{r}} + a_{\mathbf{p}}^\dagger e^{iE_{\mathbf{p}}t} e^{-i\mathbf{p}\cdot\mathbf{r}}}{\sqrt{2E_{\mathbf{p}}}} \end{aligned} \quad (133)$$

Consider the momentum 4-vector in the integrand. It is the momentum of a momentum eigenstate particle. In the rest frame of this particle $p^\mu = (m, 0, 0, 0)$. Then $\vec{p} \cdot \vec{p} = p^\mu p_\mu = m^2$. The inner product is invariant across all frames. In the frame of the integrand $p^\mu = (p^0, p^1, p^2, p^3) = (p^0, \mathbf{p})$ and

$$\begin{aligned} p^\mu p_\mu &= m^2 = (p^0)^2 - \mathbf{p}^2 = (p^0)^2 - p^2 \implies \\ (p^0)^2 &= m^2 + p^2 = E_p^2 \implies p^0 = E_p. \end{aligned} \quad (134)$$

In the last step I assumed the energy p^0 must be positive in all frames. Then the 4-vector, $\vec{p}$, referenced in the integrand is $p^\mu = (E_p, \mathbf{p})$ so

$$\phi(\vec{x}) = \int \frac{d^3p}{(2\pi)^3} \frac{a_{\mathbf{p}} e^{-i(E_p t - \mathbf{p} \cdot \mathbf{r})} + a_{\mathbf{p}}^\dagger e^{i(E_p t - \mathbf{p} \cdot \mathbf{r})}}{\sqrt{2E_p}}, \quad (135)$$

$$\vec{p} \cdot \vec{x} = p^\mu x_\mu = E_p t - \mathbf{p} \cdot \mathbf{r}, \quad (136)$$

$$\boxed{\phi(\vec{x}) = \int \frac{d^3p}{(2\pi)^3} \frac{a_{\mathbf{p}} e^{-i\vec{p} \cdot \vec{x}} + a_{\mathbf{p}}^\dagger e^{i\vec{p} \cdot \vec{x}}}{\sqrt{2E_p}}}. \quad (137)$$

I get the Heisenberg conjugate momentum density operator using

$$\pi(\vec{x}) = \partial_t \phi(\vec{x}). \quad (138)$$

This equation holds for CFT and I really should prove that it also holds for QFT. But to save time I just take it from PS on faith.

$$\pi(\vec{x}) = \int \frac{d^3p}{(2\pi)^3} \frac{a_{\mathbf{p}} \partial_t e^{-i\vec{p} \cdot \vec{x}} + a_{\mathbf{p}}^\dagger \partial_t e^{i\vec{p} \cdot \vec{x}}}{\sqrt{2E_p}} \quad (139)$$

$$\begin{aligned} \partial_t e^{\pm i\vec{p} \cdot \vec{x}} &= \partial_t e^{\pm i(E_p t - \mathbf{p} \cdot \mathbf{r})} = \partial_t e^{\pm iE_p t} e^{-\pm i\mathbf{p} \cdot \mathbf{r}} = \\ \pm iE_p e^{\pm iE_p t} e^{-\pm i\mathbf{p} \cdot \mathbf{r}} &= \pm iE_p e^{\pm i(E_p t - \mathbf{p} \cdot \mathbf{r})} = \pm iE_p e^{\pm i\vec{p} \cdot \vec{x}} \end{aligned} \quad (140)$$

$$\boxed{\pi(\vec{x}) = \int \frac{d^3p}{(2\pi)^3} i \sqrt{\frac{E_p}{2}} (a_{\mathbf{p}} e^{i\vec{p} \cdot \vec{x}} - a_{\mathbf{p}}^\dagger e^{-i\vec{p} \cdot \vec{x}})}. \quad (141)$$

I derive the equation of motion for the Heisenberg field operator $\phi(\vec{x})$ using Eq. 125, the Heisenberg equation of motion. First I derive the identity,

$$[\mathcal{A}(\vec{x}), \mathcal{H}] = [e^{i\mathcal{H}t} \mathcal{A}(\mathbf{r}) e^{-i\mathcal{H}t}, \mathcal{H}] = e^{i\mathcal{H}t} [\mathcal{A}(\mathbf{r}), \mathcal{H}] e^{-i\mathcal{H}t} \quad (142)$$

$$\boxed{[\mathcal{A}(\vec{x}), \mathcal{H}] = e^{i\mathcal{H}t} [\mathcal{A}(\mathbf{r}), \mathcal{H}] e^{-i\mathcal{H}t}}. \quad (143)$$

Earlier (Eq. 38) I derived

$$[\phi(\mathbf{r}), \pi(\mathbf{q})] = i\delta^{(3)}(\mathbf{r} - \mathbf{q}). \quad (144)$$

I now derive

$$\begin{aligned} [\phi(\mathbf{r}), [\nabla\phi(\mathbf{q})]^2] &= [\phi(\mathbf{r}), \nabla\phi(\mathbf{q}) \cdot \nabla\phi(\mathbf{q})] = \\ &= [\phi(\mathbf{r}), \nabla_{\mathbf{q}}\phi(\mathbf{q})] \cdot \nabla_{\mathbf{q}}\phi(\mathbf{q}) + \nabla_{\mathbf{q}}\phi(\mathbf{q}) \cdot [\phi(\mathbf{r}), \nabla_{\mathbf{q}}\phi(\mathbf{q})] = \\ &= \cancel{[\nabla_{\mathbf{q}}[\phi(\mathbf{r}), \phi(\mathbf{q})]] \cdot \nabla_{\mathbf{q}}\phi(\mathbf{q}) + \nabla_{\mathbf{q}}\phi(\mathbf{q}) \cdot \nabla_{\mathbf{q}}[\phi(\mathbf{r}), \phi(\mathbf{q})]} = 0, \end{aligned} \quad (145)$$

$$\boxed{[\phi(\mathbf{r}), [\nabla\phi(\mathbf{q})]^2] = 0}. \quad (146)$$

I had to point out explicitly that ∇ acts on the coordinates $\mathbf{q}$, whereas previously this was implied.

Using these identities I derive two commutators.

$$\begin{aligned} [\phi(\mathbf{r}), \mathcal{H}] &= \\ &= \left[\phi(\mathbf{r}), \frac{1}{2} \int d^3q [\pi(\mathbf{q})^2 + [\nabla\phi(\mathbf{q})]^2 + m^2\phi(\mathbf{q})^2] \right] = \\ &= \frac{1}{2} \int d^3q [\phi(\mathbf{r}), \pi(\mathbf{q})^2 + [\nabla\phi(\mathbf{q})]^2 + m^2\phi(\mathbf{q})^2] = \\ &= \frac{1}{2} \int d^3q \left([\phi(\mathbf{r}), \pi(\mathbf{q})^2] + [\phi(\mathbf{r}), [\nabla\phi(\mathbf{q})]^2] + m^2[\phi(\mathbf{r}), \phi(\mathbf{q})^2] \right) = \\ &= \frac{1}{2} \int d^3q [\phi(\mathbf{r}), \pi(\mathbf{q})^2] = \\ &= \frac{1}{2} \int d^3q ([\phi(\mathbf{r}), \pi(\mathbf{q})]\pi(\mathbf{q}) + \pi(\mathbf{q})[\phi(\mathbf{r}), \pi(\mathbf{q})]) = \\ &= i \int d^3q \pi(\mathbf{q})\delta^{(3)}(\mathbf{r} - \mathbf{q}) = i\pi(\mathbf{r}) \end{aligned} \quad (147)$$

$$\boxed{[\phi(\mathbf{r}), \mathcal{H}] = i\pi(\mathbf{r})} \quad (148)$$

$$\begin{aligned} [\pi(\mathbf{r}), \mathcal{H}] &= \\ &= \frac{1}{2} \int d^3q [\pi(\mathbf{r}), \pi(\mathbf{q})^2 + [\nabla\phi(\mathbf{q})]^2 + m^2\phi(\mathbf{q})^2] = \\ &= \frac{1}{2} \int d^3q [\pi(\mathbf{r}), [\nabla\phi(\mathbf{q})]^2 + m^2\phi(\mathbf{q})^2] \end{aligned} \quad (149)$$

To go farther I need to integrate by parts using the first Green's identity

$$\int d^3r [\nabla\phi(\mathbf{r})]^2 = - \int d^3r \phi(\mathbf{r}) \nabla^2\phi(\mathbf{r}), \quad (150)$$

where I assume that $\phi(\infty) = 0$.

$$\begin{aligned} [\pi(\mathbf{r}), \mathcal{H}] &= \frac{1}{2} \int d^3q [\pi(\mathbf{r}), [m^2\phi(\mathbf{q})^2 - \phi(\mathbf{q})\nabla^2\phi(\mathbf{q})]] = \\ &= \frac{1}{2} \int d^3q [\pi(\mathbf{r}), \phi(\mathbf{q})(m^2 - \nabla_{\mathbf{q}}^2)\phi(\mathbf{q})] = \\ &= \frac{1}{2} \int d^3q \left([\pi(\mathbf{r}), \phi(\mathbf{q})](m^2 - \nabla_{\mathbf{q}}^2)\phi(\mathbf{q}) + \right. \\ &\quad \left. \phi(\mathbf{q})[\pi(\mathbf{r}), (m^2 - \nabla_{\mathbf{q}}^2)\phi(\mathbf{q})] \right) = \\ &= \frac{1}{2} \int d^3q \left[-i\delta^{(3)}(\mathbf{r} - \mathbf{q})(m^2 - \nabla_{\mathbf{q}}^2)\phi(\mathbf{q}) + \right. \\ &\quad \left. \phi(\mathbf{q})[\pi(\mathbf{r}), m^2 - \nabla_{\mathbf{q}}^2]\phi(\mathbf{q}) + \phi(\mathbf{q})(m^2 - \nabla_{\mathbf{q}}^2)[\pi(\mathbf{r}), \phi(\mathbf{q})] \right] = \\ &= \frac{1}{2} \int d^3q \left[-i\delta^{(3)}(\mathbf{r} - \mathbf{q})(m^2 - \nabla_{\mathbf{q}}^2)\phi(\mathbf{q}) \right. \\ &\quad \left. - i\phi(\mathbf{q})(m^2 - \nabla_{\mathbf{q}}^2)\delta^{(3)}(\mathbf{r} - \mathbf{q}) \right] \end{aligned} \quad (151)$$

$$\begin{aligned} 2i[\pi(\mathbf{r}), \mathcal{H}] &= \int d^3q \delta^{(3)}(\mathbf{r} - \mathbf{q})(m^2 - \nabla_{\mathbf{q}}^2)\phi(\mathbf{q}) + \\ &= \int d^3q \phi(\mathbf{q})(m^2 - \nabla_{\mathbf{q}}^2)\delta^{(3)}(\mathbf{r} - \mathbf{q}) = \\ &= (m^2 - \nabla_{\mathbf{r}}^2)\phi(\mathbf{r}) + m^2 \int d^3q \phi(\mathbf{q})\delta^{(3)}(\mathbf{r} - \mathbf{q}) - \\ &= \int d^3q \phi(\mathbf{q})\nabla_{\mathbf{q}}^2\delta^{(3)}(\mathbf{r} - \mathbf{q}) = \\ &= (m^2 - \nabla^2)\phi(\mathbf{r}) + m^2\phi(\mathbf{r}) - \int d^3q \phi(\mathbf{q})\nabla_{\mathbf{q}}^2\delta^{(3)}(\mathbf{r} - \mathbf{q}) \end{aligned} \quad (152)$$

I evaluate the remaining integral using Integration by parts (Green's second identity). The boundary terms are zero because ϕ and its derivatives are zero at infinity.

$$\int d^3q \phi(\mathbf{q})\nabla_{\mathbf{q}}^2\delta^{(3)}(\mathbf{r} - \mathbf{q}) = \int d^3q \delta^{(3)}(\mathbf{r} - \mathbf{q})\nabla_{\mathbf{q}}^2\phi(\mathbf{q}) = \nabla^2\phi(\mathbf{r}) \quad (153)$$

$$\begin{aligned} 2i[\pi(\mathbf{r}), \mathcal{H}] &= (m^2 - \nabla^2)\phi(\mathbf{r}) + m^2\phi(\mathbf{r}) - \nabla^2\phi(\mathbf{r}) = \\ &= (m^2 - \nabla^2)\phi(\mathbf{r}) + (m^2 - \nabla^2)\phi(\mathbf{r}) = 2(m^2 - \nabla^2)\phi(\mathbf{r}) \end{aligned} \quad (154)$$

$$\boxed{i[\pi(\mathbf{r}), \mathcal{H}] = (m^2 - \nabla^2)\phi(\mathbf{r})} \quad (155)$$

Now that I have these Schrödinger commutators I get their Heisenberg versions.

$$[\phi(\vec{x}), \mathcal{H}] = e^{i\mathcal{H}t}[\phi(\mathbf{r}), \mathcal{H}]e^{-i\mathcal{H}t} = ie^{i\mathcal{H}t}\pi(\mathbf{r})e^{-i\mathcal{H}t} = i\pi(\vec{x}) \quad (156)$$

$$\boxed{[\phi(\vec{x}), \mathcal{H}] = i\pi(\vec{x})} \quad (157)$$

$$\begin{aligned} [\pi(\vec{x}), \mathcal{H}] &= e^{i\mathcal{H}t} [\pi(\mathbf{r}), \mathcal{H}] e^{-i\mathcal{H}t} = -ie^{i\mathcal{H}t} i[\pi(\mathbf{r}), \mathcal{H}] e^{-i\mathcal{H}t} \\ &= -ie^{i\mathcal{H}t} (m^2 - \nabla^2) \phi(\mathbf{r}) e^{-i\mathcal{H}t} = i(\nabla^2 - m^2) e^{i\mathcal{H}t} \phi(\mathbf{r}) e^{-i\mathcal{H}t} = \\ &= i(\nabla^2 - m^2) \phi(\vec{x}) \end{aligned} \quad (158)$$

$$\boxed{[\pi(\vec{x}), \mathcal{H}] = i(\nabla^2 - m^2) \phi(\vec{x})} \quad (159)$$

Now I derive the equation of motion for $\phi(\vec{x})$ using Heisenberg's equation of motion, Eq. 125.

$$i\partial_t \phi(\vec{x}) = [\phi(\vec{x}), \mathcal{H}] \implies \boxed{\dot{\phi}(\vec{x}) = \pi(\vec{x})}. \quad (160)$$

This is the same relationship between ϕ and π I found in CFT. Pushing forward

$$i\partial_t \pi(\vec{x}) = [\pi(\vec{x}), \mathcal{H}] \quad (161)$$

$$\partial_t \pi(\vec{x}) = \partial_t \dot{\phi}(\vec{x}) = \ddot{\phi}(\vec{x}) \quad (162)$$

$$i\ddot{\phi}(\vec{x}) = [\pi(\vec{x}), \mathcal{H}] = i(\nabla^2 - m^2) \phi(\vec{x}) \quad (163)$$

$$\boxed{\ddot{\phi}(\vec{x}) = (\nabla^2 - m^2) \phi(\vec{x})} \quad (164)$$

This is the Klein-Gordon equation. So the equation of motion for ϕ the Heisenberg operator and ϕ the classical field is the same.

In relativity there is a deep connection between space and time, and so what I just did for time translation can be repeated for space translation. Consider the 4-vector

$$P^\mu = (\mathcal{H}, \mathbf{P}), \quad \mathbf{P} = \int d^3x \mathbf{K}, \quad (165)$$

where I define $\mathbf{K}$ in Eq. 35. Then the spacetime translation operator is

$$e^{i\vec{P} \cdot \vec{x}} \phi(\vec{0}) e^{-i\vec{P} \cdot \vec{x}} \quad (166)$$

where $\vec{0}$ is the spacetime origin.

4. LOCALITY

Consider the matrix element

$$D(\vec{x} - \vec{y}) = \langle 0 | \phi(\vec{x}) \phi(\vec{y}) | 0 \rangle. \quad (167)$$

Plugging in Eq. 137 yields

$$\begin{aligned} D(\vec{x} - \vec{y}) &= \int \frac{d^3p d^3k}{2(2\pi)^6 \sqrt{E_p E_k}} \\ &\langle 0 | (a_{\mathbf{p}} e^{-i\vec{p} \cdot \vec{x}} + a_{\mathbf{p}}^\dagger e^{i\vec{p} \cdot \vec{x}}) (a_{\mathbf{k}} e^{-i\vec{k} \cdot \vec{y}} + a_{\mathbf{k}}^\dagger e^{i\vec{k} \cdot \vec{y}}) | 0 \rangle, \end{aligned} \quad (168)$$

$$\begin{aligned} D(\vec{x} - \vec{y}) &= \int \frac{d^3p d^3k}{2(2\pi)^6 \sqrt{E_p E_k}} [\langle 0 | a_{\mathbf{p}} a_{\mathbf{k}} | 0 \rangle e^{-i(\vec{p} \cdot \vec{x} + \vec{k} \cdot \vec{y})} + \\ &\langle 0 | a_{\mathbf{p}} a_{\mathbf{k}}^\dagger | 0 \rangle e^{i(\vec{k} \cdot \vec{y} - \vec{p} \cdot \vec{x})} + \langle 0 | a_{\mathbf{p}}^\dagger a_{\mathbf{k}} | 0 \rangle e^{i(\vec{p} \cdot \vec{x} - \vec{k} \cdot \vec{y})} + \\ &\langle 0 | a_{\mathbf{p}}^\dagger a_{\mathbf{k}}^\dagger | 0 \rangle e^{i(\vec{p} \cdot \vec{x} + \vec{k} \cdot \vec{y})}]. \end{aligned} \quad (169)$$

I use $a_{\mathbf{p}} | 0 \rangle = \langle 0 | a_{\mathbf{p}}^\dagger = 0$ to annihilate terms

$$D(\vec{x} - \vec{y}) = \int \frac{d^3p d^3k}{2(2\pi)^6 \sqrt{E_p E_k}} \langle 0 | a_{\mathbf{p}} a_{\mathbf{k}}^\dagger | 0 \rangle e^{i(\vec{k} \cdot \vec{y} - \vec{p} \cdot \vec{x})}. \quad (170)$$

From nonrelativistic quantum mechanics (and the normalization convention in PS p. 22) I have

$$\langle 0 | a_{\mathbf{p}} a_{\mathbf{k}}^\dagger | 0 \rangle = \langle \mathbf{p} | \mathbf{k} \rangle = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{k}), \quad (171)$$

so

$$D(\vec{x} - \vec{y}) = \int \frac{d^3p d^3k}{2(2\pi)^3 \sqrt{E_p E_k}} \delta^{(3)}(\mathbf{p} - \mathbf{k}) e^{i(\vec{k} \cdot \vec{y} - \vec{p} \cdot \vec{x})}, \quad (172)$$

$$\boxed{2(2\pi)^3 D(\vec{x} - \vec{y}) = \int d^3p \frac{e^{-i\vec{p} \cdot (\vec{x} - \vec{y})}}{E_p}}, \quad (173)$$

where I used the fact that in relativity fixing the spatial momentum also fixes the energy so $\mathbf{p} = \mathbf{k} \implies E_p = E_k = \sqrt{p^2 + m^2} \implies \vec{p} = \vec{k}$. I won't prove it but this integral is Lorentz invariant.

I consider the case when the interval between spacetime events $\vec{x}$ and $\vec{y}$ is timelike. Then I can find a frame where they occur at the same point in space:

$$\begin{aligned} \vec{x} &= (x^0, \mathbf{r}), \quad \vec{y} = (y^0, \mathbf{r}), \\ \vec{x} - \vec{y} &= (x^0 - y^0, \mathbf{r} - \mathbf{r}) = (x^0 - y^0, 0, 0, 0). \end{aligned} \quad (174)$$

I use the shorthand $x^0 - y^0 = t$. Then

$$\begin{aligned} D(\vec{x} - \vec{y}) &= \int d^3p \frac{e^{-i\vec{p} \cdot (t, 0, 0, 0)}}{2(2\pi)^3 E_p} = \int d^3p \frac{e^{-ip^0 t}}{2(2\pi)^3 E_p} \\ &= \frac{1}{2(2\pi)^3} \int d^3p \frac{e^{-iE_p t}}{E_p} = \frac{1}{2(2\pi)^3} \int_0^{4\pi} d\Omega \int_0^\infty dp p^2 \frac{e^{-iE_p t}}{E_p} \\ &= \frac{4\pi}{2(2\pi)^3} \int_0^\infty dp p^2 \frac{e^{-iE_p t}}{E_p} = \frac{1}{(2\pi)^2} \int_0^\infty dp p^2 \frac{e^{-iE_p t}}{E_p}. \end{aligned} \quad (175)$$

I simplify this integral farther using substitution.

$$D(\vec{x} - \vec{y}) = \frac{1}{(2\pi)^2} \int_0^\infty dp p^2 \frac{\exp(-i\sqrt{p^2 + m^2} t)}{\sqrt{p^2 + m^2}} \quad (176)$$

$$\begin{aligned} E &= E(p) = \sqrt{p^2 + m^2}, \quad E(0) = m, \quad E(\infty) = \infty, \\ p^2 &= E^2 - m^2, \quad \frac{dE}{dp} = \frac{p}{\sqrt{p^2 + m^2}} \implies E dE = p dp \end{aligned} \quad (177)$$

$$4\pi^2 D(\vec{x} - \vec{y}) = \int_m^\infty dE \sqrt{E^2 - m^2} e^{-iEt}. \quad (178)$$

This is the exact result for timelike separations. I can't simplify it further.

If the interval between $\vec{x}$ and $\vec{y}$ is spacelike I can always find a frame where they are simultaneous

$$\vec{x} = (t, \mathbf{x}), \quad \vec{y} = (t, \mathbf{y}), \quad \vec{x} - \vec{y} = (0, \mathbf{r}), \quad \mathbf{r} = \mathbf{x} - \mathbf{y}. \quad (179)$$

Then

$$2(2\pi)^3 D(\vec{x} - \vec{y}) = \int d^3p \frac{e^{i\mathbf{p} \cdot \mathbf{r}}}{E_p}. \quad (180)$$

Without any loss of generality I set $\vec{y}$ to be my spacetime origin and align $\mathbf{r}$ along the z axis

$$D(\vec{x} - \vec{y}) = \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin\theta \int_0^\infty dp p^2 \frac{\exp(ipr \cos\theta)}{2(2\pi)^3 E_p}, \quad (181)$$

$$8\pi^2 D(\vec{x} - \vec{y}) = \int_0^\infty dp \frac{p^2}{E_p} \int_0^\pi d\theta \sin\theta \exp(ipr \cos\theta), \quad (182)$$

$$u = u(\theta) = \cos\theta, \quad u(0) = 1, \quad u(\pi) = -1, \quad -du = d\theta \sin\theta \quad (183)$$

$$8\pi^2 D(\vec{x} - \vec{y}) = \int_0^\infty dp \frac{p^2}{E_p} \int_{-1}^1 du e^{ipru}, \quad (184)$$

$$8\pi^2 D(\vec{x} - \vec{y}) = \int_0^\infty dp \frac{p^2}{E_p} I(pr), \quad (185)$$

$$\begin{aligned} I(pr) &= \int_{-1}^1 du e^{ipru} = \int_{-1}^1 du \cos(pru) + i \int_{-1}^1 du \sin(pru) \\ &= \int_{-1}^1 du \cos(pru) = \frac{2}{pr} \sin(pr) \end{aligned} \quad (186)$$

$$4\pi^2 r D(\vec{x} - \vec{y}) = \int_0^\infty dp \frac{p}{E_p} \sin(pr). \quad (187)$$

I prefer this form over the one in PS (p. 27). The key point is that this integral is not generally zero.

What is the physical meaning of $D(\vec{x} - \vec{y})$? First consider how the Schrödinger field operator acts on the vacuum state (PS p. 24)

$$\phi(\mathbf{x}) |0\rangle = |\mathbf{x}\rangle, \quad (188)$$

where $|\mathbf{x}\rangle$ is a point particle at position $\mathbf{x}$: I won't prove it, but this definition agrees with the position eigenstate in the nonrelativistic limit (up to a constant). In QFT though

$\mathbf{x}$ is not an operator and $|\mathbf{x}\rangle$ is not its eigenstate. Eq. 114 yields

$$(2\pi)^3 \mathcal{H} |0\rangle = \int d^3p \omega_p a_{\mathbf{p}}^\dagger a_{\mathbf{p}} |0\rangle = 0. \quad (189)$$

I generalize my notation to include time in the ket state vector

$$e^{-i\mathcal{H}t} |\Psi\rangle = |\Psi(t)\rangle = |\Psi; t\rangle. \quad (190)$$

For example the vacuum state is stationary $|0; t\rangle = |0\rangle$. Then

$$\begin{aligned} \phi(\vec{x}) |0\rangle &= e^{i\mathcal{H}t} \phi(\mathbf{x}) e^{-i\mathcal{H}t} |0\rangle = e^{i\mathcal{H}t} \phi(\mathbf{x}) e^0 |0\rangle = \\ &= e^{i\mathcal{H}t} \phi(\mathbf{x}) |0\rangle = e^{i\mathcal{H}t} |\mathbf{x}\rangle = |\mathbf{x}; -t\rangle = |\vec{x}\rangle. \end{aligned} \quad (191)$$

The right interpretation must be $|\mathbf{x}; -t\rangle = |\vec{x}\rangle$: an instantaneous point particle isolated at event $\vec{x}$. I expected $|\mathbf{x}; t\rangle = |\vec{x}\rangle$ but I don't see another possible interpretation. So I put aside my misgivings for the moment and push forward. Because ϕ is Hermitian and because $\phi(\vec{x}) |0\rangle = |\vec{x}\rangle$

$$D(\vec{x} - \vec{y}) = \langle 0 | \phi(\vec{x}) \phi(\vec{y}) | 0 \rangle = \langle \vec{x} | \vec{y} \rangle. \quad (192)$$

So $|D(\vec{x} - \vec{y})|^2 = |\langle \vec{x} | \vec{y} \rangle|^2$ is the probability for a point particle to move from one event to another. Since $D(\vec{x} - \vec{y})$ is nonzero for spacelike separation (Eq. 187) particles can move faster than light in QFT, just like in nonrelativistic quantum mechanics. But information cannot be shared faster than light so everything is still kosher.

I calculate the commutator

$$\begin{aligned} [\phi(\vec{x}), \phi(\vec{y})] &= \int \frac{d^3p d^3k}{2(2\pi)^6 \sqrt{E_p E_k}} \\ &[a_{\mathbf{p}} e^{-i\vec{p} \cdot \vec{x}} + a_{\mathbf{p}}^\dagger e^{i\vec{p} \cdot \vec{x}}, a_{\mathbf{k}} e^{-i\vec{k} \cdot \vec{y}} + a_{\mathbf{k}}^\dagger e^{i\vec{k} \cdot \vec{y}}], \end{aligned} \quad (193)$$

$$\begin{aligned} [\phi(\vec{x}), \phi(\vec{y})] &= \int \frac{d^3p d^3k}{2(2\pi)^6 \sqrt{E_p E_k}} ([a_{\mathbf{p}}, a_{\mathbf{k}}] e^{-i(\vec{p} \cdot \vec{x} + \vec{k} \cdot \vec{y})} \\ &+ [a_{\mathbf{p}}, a_{\mathbf{k}}^\dagger] e^{i(\vec{k} \cdot \vec{y} - \vec{p} \cdot \vec{x})} + [a_{\mathbf{p}}^\dagger, a_{\mathbf{k}}] e^{i(\vec{p} \cdot \vec{x} - \vec{k} \cdot \vec{y})} + \\ &\quad [a_{\mathbf{p}}^\dagger, a_{\mathbf{k}}^\dagger] e^{i(\vec{p} \cdot \vec{x} + \vec{k} \cdot \vec{y})}), \end{aligned} \quad (194)$$

$$\begin{aligned} [\phi(\vec{x}), \phi(\vec{y})] &= \int \frac{d^3p d^3k}{2(2\pi)^3 \sqrt{E_p E_k}} \delta^{(3)}(\mathbf{p} - \mathbf{k}) \\ &(e^{i(\vec{k} \cdot \vec{y} - \vec{p} \cdot \vec{x})} - e^{i(\vec{p} \cdot \vec{x} - \vec{k} \cdot \vec{y})}), \end{aligned} \quad (195)$$

$$[\phi(\vec{x}), \phi(\vec{y})] = \int \frac{d^3p}{2(2\pi)^3 E_p} (e^{-i\vec{p} \cdot (\vec{x} - \vec{y})} - e^{-i\vec{p} \cdot (\vec{y} - \vec{x})}), \quad (196)$$

$$[\phi(\vec{x}), \phi(\vec{y})] = D(\vec{x} - \vec{y}) - D(\vec{y} - \vec{x}). \quad (197)$$

When events $\vec{x}$ and $\vec{y}$ have a spacelike interval we can always find a frame where they are simultaneous $\vec{x} - \vec{y} = (0, \mathbf{x} - \mathbf{y}) = (0, \mathbf{r})$,

$$[\phi(\vec{x}), \phi(\vec{y})] = D[(0, \mathbf{r})] - D[(0, -\mathbf{r})], \quad (198)$$

and since each term is Lorentz invariant I can rotate my frame in the right hand term so that $-\mathbf{r} \rightarrow \mathbf{r}$ and

$$[\phi(\vec{x}), \phi(\vec{y})] = D[(0, \mathbf{r})] - D[(0, \mathbf{r})] = 0. \quad (199)$$

This means the information from a measurement cannot move faster than light. Particles can move faster than light but information cannot and if this commutator was not zero for spacelike separations sever problems would arise.

I derive a compact expression for $[\phi(\vec{x}), \phi(\vec{y})]$.

$$2(2\pi)^3 [\phi(\vec{x}), \phi(\vec{y})] = \int \frac{d^3p}{E_p} (e^{-i\vec{p} \cdot (\vec{x} - \vec{y})} - e^{-i\vec{p} \cdot (\vec{y} - \vec{x})}) \quad (200)$$

$$2(2\pi)^3 [\phi(\vec{x}), \phi(\vec{y})] = - \int \frac{d^3p}{E_p} (e^{i\vec{p} \cdot (\vec{x} - \vec{y})} - e^{-i\vec{p} \cdot (\vec{x} - \vec{y})}) \quad (201)$$

$$\vec{p} = (p^0, \mathbf{p}) = (E_p, \mathbf{p}), \quad \vec{q} = \vec{x} - \vec{y} = (t, \mathbf{q}), \\ \vec{x} = (x^0, \mathbf{x}), \quad \vec{y} = (y^0, \mathbf{y}) \quad (202)$$

$$2(2\pi)^3 [\phi(\vec{x}), \phi(\vec{y})] = - \int \frac{d^3p}{E_p} (e^{i\vec{p} \cdot \vec{q}} - e^{-i\vec{p} \cdot \vec{q}}) = \\ - \int \frac{d^3p}{E_p} [e^{i(E_p, \mathbf{p}) \cdot (t, \mathbf{q})} - e^{-i(E_p, \mathbf{p}) \cdot (t, \mathbf{q})}] = \\ - \int \frac{d^3p}{E_p} [e^{i(E_p t - \mathbf{p} \cdot \mathbf{q})} - e^{-i(E_p t - \mathbf{p} \cdot \mathbf{q})}] = \\ - \int \frac{d^3p}{E_p} (e^{iE_p t - i\mathbf{p} \cdot \mathbf{q}} - e^{-iE_p t + i\mathbf{p} \cdot \mathbf{q}}) \quad (203)$$

I substitute $\mathbf{p} \rightarrow -\mathbf{p}$ in one term.

$$2(2\pi)^3 [\phi(\vec{x}), \phi(\vec{y})] = - \int \frac{d^3p}{E_p} (e^{iE_p t + i\mathbf{p} \cdot \mathbf{q}} - e^{-iE_p t + i\mathbf{p} \cdot \mathbf{q}}) = \\ - \int \frac{d^3p}{E_p} (e^{iE_p t} - e^{-iE_p t}) e^{i\mathbf{p} \cdot \mathbf{q}} \quad (204)$$

$$(2\pi)^3 i [\phi(\vec{x}), \phi(\vec{y})] = \int d^3p \frac{\sin(E_p t)}{E_p} e^{i\mathbf{p} \cdot \mathbf{q}} \quad (205)$$

$$[\phi(\vec{x}), \phi(\vec{y})] = \int d^3p \frac{\sin[E_p(x^0 - y^0)]}{(2\pi)^3 i E_p} e^{i\mathbf{p} \cdot (\mathbf{x} - \mathbf{y})} \quad (206)$$

5. PROPAGATORS

Interestingly the commutator $[\phi(\vec{x}), \phi(\vec{y})]$ is a c-number. In QFT lingo c-number means not an operator, and more specifically a complex number or scalar. Then

$$\langle 0 | [\phi(\vec{x}), \phi(\vec{y})] | 0 \rangle = [\phi(\vec{x}), \phi(\vec{y})] = \int d^3p \frac{\sin(E_p t)}{(2\pi)^3 i E_p} e^{i\mathbf{p} \cdot \mathbf{q}}, \quad (207)$$

where I used the substitution $\mathbf{p} \rightarrow -\mathbf{p}$ and $\vec{q} = \vec{x} - \vec{y}$ and $q^0 = x^0 - y^0 = t$. The retarded propagator is

$$\mathcal{G}_R(\vec{x} - \vec{y}) = \Theta(x^0 - y^0) \langle 0 | [\phi(\vec{x}), \phi(\vec{y})] | 0 \rangle, \quad (208)$$

where Θ is the Heaviside step function. Alternatively

$$(2\pi)^4 \mathcal{G}_R(\vec{x} - \vec{y}) = i \int d^4p \frac{e^{i\vec{p} \cdot (\vec{x} - \vec{y})}}{\vec{p}^2 - m^2}, \quad (209)$$

where $\vec{p} \cdot \vec{p} = \vec{p}^2$. I use the script G because it is a Green's function of the Klein-Gordon equation, as I now prove. The derivation in PS (Eq. 2.56) is poorly done so I make it more explicit. First I specify which coordinates the d'Alembert operator acts on

$$(\square_{\vec{x}} + m^2) \mathcal{G}_R(\vec{x} - \vec{y}) = -i \delta^{(4)}(\vec{x} - \vec{y}). \quad (210)$$

Since the differential operator acts on $\vec{x}$, $\vec{y}$ is fixed. I can choose coordinates where $\vec{y} = \vec{0} = (0, 0, 0, 0)$ with no loss of generality because SR QFT is invariant under translation

$$(\square + m^2) \mathcal{G}_R(\vec{x}) = -i \delta^{(4)}(\vec{x}). \quad (211)$$

Where I dropped the subscript on $\square$ because there is no more ambiguity about which coordinates I act on.

$$(\square + m^2) \Theta(x^0) \langle 0 | [\phi(\vec{x}), \phi(\vec{0})] | 0 \rangle = -i \delta^{(4)}(\vec{x}). \quad (212)$$

I define the shorthand $f(\vec{x}) = \langle 0 | [\phi(\vec{x}), \phi(\vec{0})] | 0 \rangle$:

$$(\square + m^2) \Theta(x^0) f(\vec{x}) = -i \delta^{(4)}(\vec{x}). \quad (213)$$

$$\square \Theta(x^0) f(\vec{x}) + m^2 \mathcal{G}_R(\vec{x}) = -i \delta^{(4)}(\vec{x}) \quad (214)$$

$$\square \Theta(x^0) f(\vec{x}) + m^2 \mathcal{G}_R(\vec{x}) = -i \delta^{(4)}(\vec{x}) \quad (215)$$

Dirac delta functions are only well defined under the integral sign

$$\int d^4x [\square \Theta(x^0) f(\vec{x}) + m^2 \mathcal{G}_R(\vec{x})] = -i \int d^4x \delta^{(4)}(\vec{x}). \quad (216)$$

$$\int d^4x [\square \Theta(x^0) f(\vec{x}) + m^2 \mathcal{G}_R(\vec{x})] = -i \quad (217)$$

$$\int d^4x \square \Theta(x^0) f(\vec{x}) + m^2 \int d^4x \mathcal{G}_R(\vec{x}) = -i \quad (218)$$

I use $t = x^0$.

$$\int d^4x \square \Theta(t) f(\vec{x}) + m^2 \int d^4x \mathcal{G}_R(\vec{x}) = -i \quad (219)$$

$$\int d^4x \square \Theta(t) f(\vec{x}) + m^2 I = -i \quad (220)$$

$$I = \int d^4x \mathcal{G}_R(\vec{x}) = \int d^4x \Theta(t) f(\vec{x}) = \int_0^\infty dt \int d^3x f(\vec{x}) = \int_0^\infty dt \mathcal{A}(\vec{x}), \quad (221)$$

$$\mathcal{A}(\vec{x}) = \int d^3x f(\vec{x}) = \int d^3x \langle 0 | [\phi(\vec{x}), \phi(\vec{0})] | 0 \rangle = \int d^3x [\phi(\vec{x}), \phi(\vec{0})] \langle 0 | 0 \rangle = \int d^3x [\phi(\vec{x}), \phi(\vec{0})] \quad (222)$$

I use Eq. 196

$$[\phi(\vec{x}), \phi(\vec{0})] = \int \frac{d^3p}{2(2\pi)^3 E_p} (e^{-i\vec{p}\cdot\vec{x}} - e^{i\vec{p}\cdot\vec{x}}), \quad (223)$$

$$[\phi(\vec{x}), \phi(\vec{0})] = \int \frac{d^3p}{2(2\pi)^3 E_p} (e^{-iE_p t} e^{i\vec{p}\cdot\vec{x}} - e^{iE_p t} e^{-i\vec{p}\cdot\vec{x}}). \quad (224)$$

I substitute $-\mathbf{p} \rightarrow \mathbf{p}$ in one term

$$[\phi(\vec{x}), \phi(\vec{0})] = \int \frac{d^3p}{2(2\pi)^3 E_p} (e^{-iE_p t} - e^{iE_p t}) e^{i\vec{p}\cdot\vec{x}}, \quad (225)$$

$$[\phi(\vec{x}), \phi(\vec{0})] = \frac{1}{(2\pi)^3 i} \int d^3p \frac{\sin(E_p t)}{E_p} e^{i\vec{p}\cdot\vec{x}}. \quad (226)$$

$$\mathcal{A}(\vec{x}) = \frac{1}{(2\pi)^3 i} \int d^3p \frac{\sin(E_p t)}{E_p} \int d^3x e^{i\vec{p}\cdot\vec{x}} \quad (227)$$

$$\int d^3x e^{i\vec{p}\cdot\vec{x}} = (2\pi)^3 \delta^{(3)}(\mathbf{p}) \quad (228)$$

$$\mathcal{A}(\vec{x}) = \frac{1}{i} \int d^3p \frac{\sin(E_p t)}{E_p} \delta^{(3)}(\mathbf{p}) \quad (229)$$

$$\mathcal{A}(\vec{x}) = \frac{\sin(E_0 t)}{i E_0} \quad (230)$$

$$E_p = \sqrt{p^2 + m^2} \implies E_0 = m \quad (231)$$

$$\mathcal{A}(\vec{x}) = \frac{\sin(mt)}{im} \quad (232)$$

$$imI = \int_0^\infty dt \sin(mt) \quad (233)$$

$$x = x(t) = mt, \quad x(0) = 0, \quad x(\infty) = \infty, \quad \frac{dx}{m} = dt \quad (234)$$

$$im^2 I = \int_0^\infty dx \sin x \quad (235)$$

At first glance such an integral appears non-convergent but I explain how to solve it in HW. 3 Problem 2.

$$im^2 I = \lim_{\epsilon \rightarrow 0^+} \int_0^\infty dx e^{-\epsilon x} \sin x = \lim_{\epsilon \rightarrow 0^+} \frac{1}{1 + \epsilon^2} = 1 \quad (236)$$

$$I = \frac{1}{im^2} \quad (237)$$

I plug this back into Eq. 220

$$\int d^4x \square \Theta(t) f(\vec{x}) + \frac{1}{i} = -i, \quad (238)$$

$$\int d^4x \square \Theta(t) f(\vec{x}) = 0. \quad (239)$$

$$\int d^4x (\partial_t^2 - \nabla^2) \Theta(t) f(\vec{x}) = 0 \quad (240)$$

$$\int d^4x [\partial_t^2 \Theta(t) f(\vec{x}) - \Theta(t) \nabla^2 f(\vec{x})] = 0 \quad (241)$$

$$\int d^4x [\ddot{\Theta}(t) f(\vec{x}) + \Theta(t) \ddot{f}(\vec{x}) + 2\dot{\Theta}(t) \dot{f}(\vec{x}) - \Theta(t) \nabla^2 f(\vec{x})] = 0 \quad (242)$$

The integral is over all space and I assume that $\phi(\vec{x})$ and its derivatives vanish at the boundary. This implies that $f(\vec{x}) = \langle 0 | [\phi(\vec{x}), \phi(\vec{0})] | 0 \rangle$ and its derivatives vanish on the boundary as well, so I can integrate by parts without worrying about boundary terms.

$$\int d^4x [\Theta(t) \ddot{f}(\vec{x}) + \Theta(t) \ddot{f}(\vec{x}) - 2\Theta(t) \ddot{f}(\vec{x}) - \cancel{f(\vec{x}) \nabla^2 \Theta(t)}] = 0 \quad (243)$$

$$(2 - 2) \int d^4x \Theta(t) \ddot{f}(\vec{x}) = 0 \quad (244)$$

$$0 = 0 \quad (245)$$

This implies that Eq. 210 is true. QED.

5.1. Feynman Propagator

In addition to the retarded propagator there is the advanced propagator. But more important than either of these is the Feynman propagator which is half retarded and half advanced. This propagator is one of the building blocks of Feynman diagrams and deserves special attention. One way to write this propagator is

$$(2\pi)^4 \mathcal{G}_F(\vec{x} - \vec{y}) = i \lim_{\epsilon \rightarrow 0_+} \int d^4p \frac{e^{i\vec{p} \cdot (\vec{y} - \vec{x})}}{\vec{p}^2 - m^2 + i\epsilon}. \quad (246)$$

Starting from this expression I can derive another more useful version of $\mathcal{G}_F(\vec{x} - \vec{y})$, by integrating over p^0 . First I make the expression less cumbersome by dropping the limit symbol

$$(2\pi)^4 \mathcal{G}_F(\vec{x} - \vec{y}) = i \int d^4p \frac{e^{i\vec{p} \cdot (\vec{y} - \vec{x})}}{\vec{p}^2 - m^2 + i\epsilon}. \quad (247)$$

From now on ϵ implies the limit $\epsilon \rightarrow 0_+$. I use the shorthand $\vec{q} = \vec{x} - \vec{y} = (t, \mathbf{q})$

$$(2\pi)^4 \mathcal{G}_F(\vec{q}) = i \int d^4p \frac{e^{-i\vec{p} \cdot \vec{q}}}{\vec{p}^2 - m^2 + i\epsilon}. \quad (248)$$

When I was integrating over d^3p , $p^0 = E_p$ was the energy associated with 3-momentum $\mathbf{p}$. But now p^0 is an integration variable. I use $\vec{p} = (p^0, \mathbf{p})$

$$(2\pi)^4 \mathcal{G}_F(\vec{q}) = i \int d^3p \int_{-\infty}^{\infty} dp^0 \frac{\exp[-i(p^0, \mathbf{p}) \cdot \vec{q}]}{(p^0, \mathbf{p})^2 - m^2 + i\epsilon}. \quad (249)$$

Note that p^0 is the kinetic plus mass energy of a point particle with 4-momentum $\vec{p}$, and physically cannot be negative. But I integrate p^0 over all the reals. That's okay, it's not a physical energy. But I still prefer to use $E = p^0$ when integrating

$$(2\pi)^4 \mathcal{G}_F(\vec{q}) = i \int d^3p \int_{-\infty}^{\infty} dE \frac{\exp[-i(E, \mathbf{p}) \cdot \vec{q}]}{(E, \mathbf{p})^2 - m^2 + i\epsilon}. \quad (250)$$

$$(2\pi)^4 \mathcal{G}_F(\vec{q}) = i \int d^3p \int_{-\infty}^{\infty} dE \frac{e^{-i(Et - \mathbf{p} \cdot \mathbf{q})}}{E^2 - \mathbf{p}^2 - m^2 + i\epsilon} \quad (251)$$

$$(2\pi)^4 \mathcal{G}_F(\vec{q}) = i \int d^3p e^{i\mathbf{p} \cdot \mathbf{q}} \int_{-\infty}^{\infty} dE \frac{e^{-iEt}}{E^2 - p^2 - m^2 + i\epsilon} \quad (252)$$

$$E_p^2 = p^2 + m^2 \quad (253)$$

$$(2\pi)^4 \mathcal{G}_F(\vec{q}) = i \int d^3p e^{i\mathbf{p} \cdot \mathbf{q}} \int_{-\infty}^{\infty} dE \frac{e^{-iEt}}{E^2 - E_p^2 + i\epsilon} \quad (254)$$

$$(2\pi)^4 \mathcal{G}_F(\vec{q}) = i \int d^3p e^{i\mathbf{p} \cdot \mathbf{q}} I(t) \quad (255)$$

$$I(t) = \int_{-\infty}^{\infty} dE \frac{e^{-iEt}}{E^2 - E_p^2 + i\epsilon} \quad (256)$$

Since this integral must be solved by contour integration I use the traditional dummy variable x

$$I(t) = \int_{-\infty}^{\infty} dx \frac{e^{-ixt}}{x^2 - E_p^2 + i\epsilon}, \quad (257)$$

and find the poles $z_{\pm}$

$$z_{\pm}^2 - E_p^2 + i\epsilon = 0 \implies z_{\pm} = \pm \sqrt{E_p^2 - i\epsilon}. \quad (258)$$

Because $\epsilon \rightarrow 0_+$ these simplify further

$$\begin{aligned} z_{\pm} &= \pm E_p \sqrt{1 - i\frac{\epsilon}{E_p^2}} = \pm E_p \left(1 - \frac{i\epsilon}{2E_p^2}\right) \\ &= \pm \left(E_p - \frac{i\epsilon}{2E_p}\right). \end{aligned} \quad (259)$$

I can simplify farther by realizing that taking $\epsilon \rightarrow 0_+$ is equivalent to taking $\beta \rightarrow 0_+$ where

$$z_{\pm} = \pm(E_p - i\beta), \quad \beta = \frac{\epsilon}{2E_p}. \quad (260)$$

But I can rename β to ϵ and get the poles

$$z_{\pm} = \pm E_p \mp i\epsilon. \quad (261)$$

The lesson here is that positive constants can be absorbed into limiting variables like ϵ .

$$I(t) = \int_{-\infty}^{\infty} dx \frac{e^{-ixt}}{(x - z_+)(x - z_-)}. \quad (262)$$

Now we see that the $i\epsilon$ bumps the poles off the real axis, which prevents the integral from diverging. After integration we can safely take the limit $\epsilon \rightarrow 0_+$. There are two cases to consider: $t > 0$ and $t < 0$. The case $t = 0$ is an isolated point and does not have to be explicitly considered. The two other cases will account for this one as well.

$$I(t) = \Theta(t)I_+ + \Theta(-t)I_-, \quad (263)$$

where $I_{\pm}$ correspond to positive and negative t . I show the integration contours for $I_{\pm}$ in Fig. 1.

Now we integrate $I_{\pm}$

$$I_{\pm} = \int_{-\infty}^{\infty} dx f(x), \quad f(x) = \frac{e^{-ixt}}{(x - z_+)(x - z_-)}. \quad (264)$$

To get I_- I use the top contour in Fig. 1

$$\oint dz f(z) = \int_{-R}^R dx f(x) + \int_R^{-R} dz f(z), \quad R \rightarrow \infty, \quad (265)$$

where the red term is integrated along the red contour.

$$\int_R^{-R} dz f(z) = \int_R^{-R} dz \frac{e^{-izt}}{(z - z_+)(z - z_-)} \quad (266)$$

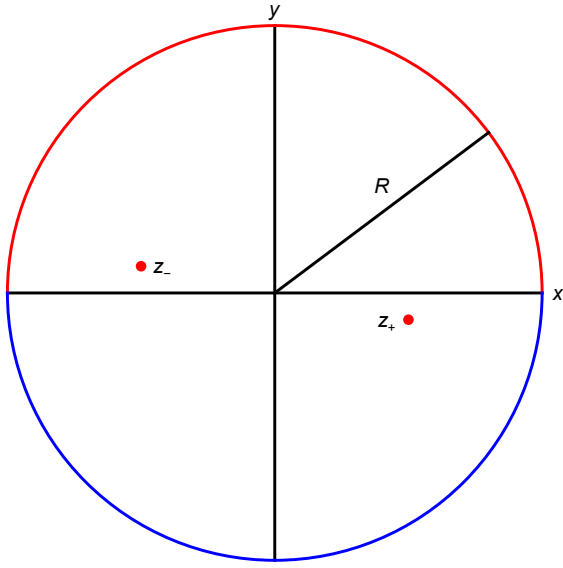


FIG. 1. Integration contour for I_+ (I_-) closed from the top (bottom). The limit $R \rightarrow \infty$ is assumed.

Along the red contour

$$z = Re^{i\phi}, \quad dz = iRe^{i\phi}d\phi. \quad (267)$$

$$\begin{aligned} \int_R^{-R} dz f(z) &= iR \int_0^\pi d\phi \frac{e^{i\phi} \exp(-iRe^{i\phi}t)}{(Re^{i\phi} - z_+)(Re^{i\phi} - z_-)} \\ &= \frac{i}{R} \int_0^\pi d\phi \frac{\exp(-iRe^{i\phi}t)}{e^{i\phi}} \\ &= \frac{i}{R} \int_0^\pi d\phi \exp[-i(Re^{i\phi}t + \phi)] \\ &= \frac{i}{R} \int_0^\pi d\phi \exp[-i[R(\cos\phi + i\sin\phi)t + \phi]] = \\ &= \frac{i}{R} \int_0^\pi d\phi \exp[Rt \sin\phi - i(Rt \cos\phi + \phi)] = \\ &= \frac{i}{R} \int_0^\pi d\phi e^{Rt \sin\phi} e^{-i(Rt \cos\phi + \phi)} \end{aligned} \quad (268)$$

So

$$\int_R^{-R} dz f(z) = \frac{i}{R} \int_0^\pi d\phi e^{Rt \sin\phi} g(\phi) \quad (269)$$

where

$$g(\phi) = e^{-i(Rt \cos\phi + \phi)}. \quad (270)$$

The function $g(\phi)$ oscillates between ± 1 , so it is of order 1

$$|g(\phi)| \sim 1. \quad (271)$$

For I_- , $t = -|t| < 0$ and $\sin\phi > 0 \quad \forall \phi \in (0, \pi)$, so

$$\int_R^{-R} dz f(z) = \frac{i}{R} \int_0^\pi d\phi e^{-R|t| \sin\phi} g(\phi). \quad (272)$$

In the $R \rightarrow \infty$ limit

$$\int_R^{-R} dz f(z) = \frac{i}{\infty} \int_0^\pi d\phi e^{-\infty} g(\phi), \quad (273)$$

$$\int_R^{-R} dz f(z) = 0. \quad (274)$$

$$\oint dz f(z) = \int_{-R}^R dx f(x) + \int_R^{-R} dz f(z) \quad (275)$$

$$\int_{-R}^R dx f(x) = \oint dz f(z) \quad (276)$$

$$\int_{-\infty}^\infty dx f(x) = \oint dz f(z) \quad (277)$$

$$I_- = \oint dz f(z) = \frac{2\pi i e^{-iz_-t}}{z_- - z_+} \quad (278)$$

$$I_- = \frac{\pi i e^{i(E_p - i\epsilon)t}}{i\epsilon - E_p} \quad (279)$$

I take $\epsilon \rightarrow 0_+$

$$I_- = \frac{\pi e^{iE_p t}}{iE_p}. \quad (280)$$

We repeat this procedure using the lower (blue) contour to get I_+ .

$$-\oint dz f(z) = \int_{-R}^R dx f(x) + \int_R^{-R} dz f(z), \quad (281)$$

where the minus sign comes from integrating clockwise.

$$\begin{aligned} \int_R^{-R} dz f(z) &= \int_R^{-R} dz \frac{e^{-izt}}{(z - z_+)(z - z_-)} = \\ &= iR \int_0^{-\pi} d\phi \frac{e^{i\phi} \exp(-iRe^{i\phi}t)}{(Re^{i\phi} - z_+)(Re^{i\phi} - z_-)} = \\ &= \frac{i}{R} \int_0^{-\pi} d\phi \exp[-i(Re^{i\phi}t + \phi)] = \\ &= \frac{i}{R} \int_0^{-\pi} d\phi \exp[Rt \sin\phi - i(Rt \cos\phi + \phi)] = \\ &= \frac{i}{R} \int_0^{-\pi} d\phi e^{Rt \sin\phi} e^{-i(Rt \cos\phi + \phi)} \end{aligned} \quad (282)$$

For I_+ , $t > 0$ and $\sin\phi < 0 \quad \forall \phi \in (-\pi, 0)$

$$\begin{aligned} \int_R^{-R} dz f(z) &= \frac{i}{R} \int_0^{-\pi} d\phi e^{-R|t| \sin\phi} e^{-i(Rt \cos\phi + \phi)} = \\ &= \frac{i}{\infty} \int_0^{-\pi} d\phi e^{-\infty} e^{-i(Rt \cos\phi + \phi)} = 0. \end{aligned} \quad (283)$$

$$-\oint dz f(z) = \int_{-R}^R dx f(x) + \cancel{\int_R^{-R} dz f(z)}, \quad (284)$$

$$\int_{-R}^R dx f(x) = -\oint dz f(z) \quad (285)$$

$$\int_{-\infty}^{\infty} dx f(x) = -\oint dz f(z) \quad (286)$$

$$I_+ = -\frac{2\pi i e^{-iz_+t}}{z_+ - z_-} = \frac{\pi i e^{-i(E_p - i\epsilon)t}}{i\epsilon - E_p} \quad (287)$$

$$I_+ = \frac{\pi e^{-iE_p t}}{iE_p} \quad (288)$$

I put the $I_{\pm}$ together

$$I(t) = \Theta(t)I_+ + \Theta(-t)I_-, \quad (289)$$

$$I(t) = \frac{\pi}{iE_p} [e^{-iE_p t} \Theta(t) + e^{iE_p t} \Theta(-t)]. \quad (290)$$

The Feynman propagator is

$$(2\pi)^4 \mathcal{G}_F(\vec{q}) = i \int d^3p e^{i\mathbf{p} \cdot \mathbf{q}} I(t), \quad (291)$$

$$(2\pi)^4 \mathcal{G}_F(\vec{q}) = \pi \int d^3p \frac{e^{i\mathbf{p} \cdot \mathbf{q}}}{E_p} [e^{-iE_p t} \Theta(t) + e^{iE_p t} \Theta(-t)]. \quad (292)$$

$$2(2\pi)^3 \mathcal{G}_F(\vec{q}) = \int d^3p \frac{e^{-i(E_p t - \mathbf{p} \cdot \mathbf{q})} \Theta(t) + e^{i(E_p t + \mathbf{p} \cdot \mathbf{q})} \Theta(-t)}{E_p} \quad (293)$$

I substitute $\mathbf{p} \rightarrow -\mathbf{p}$ in one of the terms.

$$2(2\pi)^3 \mathcal{G}_F(\vec{q}) = \int d^3p \frac{e^{-i(E_p t - \mathbf{p} \cdot \mathbf{q})} \Theta(t) + e^{i(E_p t - \mathbf{p} \cdot \mathbf{q})} \Theta(-t)}{E_p} \quad (294)$$

In Eq. 246 p^0 is an integration variable. But here $E_p = p^0$ is the on-shell energy of a free particle with momentum $\mathbf{p}$.

$$2(2\pi)^3 \mathcal{G}_F(\vec{q}) = \int d^3p \frac{e^{-i(E_p t - \mathbf{p} \cdot \mathbf{q})} \Theta(t) + e^{i(E_p t - \mathbf{p} \cdot \mathbf{q})} \Theta(-t)}{E_p} \quad (295)$$

$$2(2\pi)^3 \mathcal{G}_F(\vec{q}) = \int d^3p \frac{e^{-i(p^0 q^0 - \mathbf{p} \cdot \mathbf{q})} \Theta(t) + e^{i(p^0 q^0 - \mathbf{p} \cdot \mathbf{q})} \Theta(-t)}{E_p} \quad (296)$$

$$2(2\pi)^3 \mathcal{G}_F(\vec{q}) = \int d^3p \frac{e^{-i\vec{p} \cdot \vec{q}} \Theta(t) + e^{i\vec{p} \cdot \vec{q}} \Theta(-t)}{E_p} \quad (297)$$

Comparing this with Eq. 173 yields

$$\mathcal{G}_F(\vec{q}) = D(\vec{q})\Theta(t) + D(-\vec{q})\Theta(-t). \quad (298)$$

$$\mathcal{G}_F(\vec{x} - \vec{y}) = D(\vec{x} - \vec{y})\Theta(x^0 - y^0) + D(\vec{y} - \vec{x})\Theta(y^0 - x^0) \quad (299)$$

$$\mathcal{G}_F(\vec{x} - \vec{y}) = \langle 0 | \phi(\vec{x}) \phi(\vec{y}) | 0 \rangle \Theta(x^0 - y^0) + \langle 0 | \phi(\vec{y}) \phi(\vec{x}) | 0 \rangle \Theta(y^0 - x^0) \quad (300)$$

$$\mathcal{G}_F(\vec{x} - \vec{y}) = \langle 0 | \phi(\vec{x}) \phi(\vec{y}) \Theta(x^0 - y^0) + \phi(\vec{y}) \phi(\vec{x}) \Theta(y^0 - x^0) | 0 \rangle \quad (301)$$

I condense this expression using the time ordering symbol T which puts the latest operator to the left

$$\boxed{\mathcal{G}_F(\vec{x} - \vec{y}) = \langle 0 | T \phi(\vec{x}) \phi(\vec{y}) | 0 \rangle}. \quad (302)$$

The Feynman propagator is even

$$\mathcal{G}_F(-\vec{q}) = i \int \frac{d^4p}{(2\pi)^4} \frac{e^{i\vec{p} \cdot \vec{q}}}{\vec{p} \cdot \vec{p} - m^2 + i\epsilon} = i \int \frac{d^4p}{(2\pi)^4} \frac{e^{-i\vec{p} \cdot \vec{q}}}{\vec{p} \cdot \vec{p} - m^2 + i\epsilon} = \mathcal{G}_F(\vec{q}), \quad (303)$$

$$\mathcal{G}_F(\vec{x} - \vec{y}) = \mathcal{G}_F(\vec{y} - \vec{x}). \quad (304)$$

It is also a Green's function of the Klein-Gordon equation

$$\boxed{(\square_x + m^2) \mathcal{G}_F(\vec{x} - \vec{y}) = -i\delta^{(4)}(\vec{x} - \vec{y})}. \quad (305)$$

Part II

Interactions

6. SELF-INTERACTING FIELDS

The simplest possible interacting Lagrangians are

$$\mathcal{L}[\Phi] = \mathcal{L}_0[\Phi] + \mathcal{L}_{\text{int}}[\Phi], \quad (306)$$

where $\mathcal{L}_0[\Phi]$ is the free field Lagrangian from Eq. 1

$$2\mathcal{L}_0[\Phi] = \dot{\Phi}^2 - (\nabla\Phi)^2 - m^2\Phi^2 \quad (307)$$

and $\mathcal{L}_{\text{int}}[\Phi]$ is the interacting Lagrangian. There are many possible interactions. For example $\mathcal{L}_{\text{int}}[\Phi] \propto \Phi^3$ or Φ^4 . Since there is only one field, Φ , involved how is $\mathcal{L}_{\text{int}}[\Phi]$ an interaction? Because the pure momentum plane waves in the Fourier decomposition of $\Phi(\vec{x})$ are only independent if $V(\phi) \propto \phi^2$. With any other $V(\phi)$ the modes interact among themselves. In Sec. 12 I consider derivative interactions (where $\mathcal{L}_{\text{int}}$ is a functional of $\partial_\mu\Phi$) and interactions with other fields.

I use Φ for the interacting field to distinguish it from the free field ϕ . I use $|\Omega\rangle$ for the vacuum (ground) state of the interacting theory to distinguish it from the free theory vacuum $|0\rangle$. There is no exactly solvable interacting theory in 4D spacetime. My approach is to treat the interactions as small perturbations. This allows me to replace expressions with Φ and $|\Omega\rangle$ (which I don't know how to evaluate) with perturbative expressions composed of ϕ and $|0\rangle$ which I know how to evaluate.

I construct the interacting Hamiltonian for perturbation theory. Using Eq. 5 I derive the conjugate momentum

$$\pi(\vec{x}) = \frac{\partial\mathcal{L}}{\partial\dot{\Phi}} = \frac{\partial}{\partial\dot{\Phi}}(\mathcal{L}_0 + \mathcal{L}_{\text{int}}) = \frac{\partial\mathcal{L}_0}{\partial\dot{\Phi}} = \dot{\Phi}. \quad (308)$$

Obviously this will change if derivative interactions are involved, and that will change the Hamiltonian I use for perturbation theory. That's why derivative interactions have to be considered separately. I construct the Hamiltonian using Eq. 17

$$\mathcal{H} = \pi(\vec{x})\dot{\Phi} - \mathcal{L} = \dot{\Phi}^2 - \mathcal{L}_0 - \mathcal{L}_{\text{int}}, \quad (309)$$

$$\mathcal{H}[\Phi] = \mathcal{H}_0[\Phi] + \mathcal{V}[\Phi], \quad (310)$$

where

$$\mathcal{H}_0[\Phi] = \dot{\Phi}^2 - \mathcal{L}_0[\Phi] = \frac{\dot{\Phi}^2 + (\nabla\Phi)^2 + m^2\Phi^2}{2}, \quad (311)$$

is the free Hamiltonian from Eq. 18 and

$$\mathcal{V}[\Phi] = -\mathcal{L}_{\text{int}}[\Phi], \quad (312)$$

is the interacting Hamiltonian which I will treat as a perturbation. Actually these are the Hamiltonian densities. They are related to the Hamiltonian in the usual way (Eq. 19)

$$\mathcal{H} = \int d^3x \mathcal{H}, \quad \mathcal{H}_0 = \int d^3x \mathcal{H}_0, \quad \mathcal{V} = \int d^3x \mathcal{V}. \quad (313)$$

7. PERTURBATION THEORY

To solve scattering problems we need to use an old friend from nonrelativistic quantum mechanics: time-dependent perturbation theory in the interaction picture. I would love to re-derive this theory but I don't have time, so I will take the results without proving them whenever I can.

The Schrödinger picture interacting and non-interacting field operators and the Heisenberg picture interacting field operator at some fixed time t_0 are the same

$$\Phi(\mathbf{x}) = \phi(\mathbf{x}) = \Phi(t_0, \mathbf{x}) = \int d^3p \frac{a_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{r}} + a_{\mathbf{p}}^\dagger e^{-i\mathbf{p}\cdot\mathbf{r}}}{(2\pi)^3 \sqrt{2E_p}}. \quad (314)$$

The interaction picture interacting field operator, $\Phi_I(\vec{x})$, is just the free field Heisenberg operator

$$\Phi_I(\vec{x}) = e^{i\mathcal{H}_0(t-t_0)} \Phi(\mathbf{x}) e^{-i\mathcal{H}_0(t-t_0)} = \int d^3p \frac{a_{\mathbf{p}} e^{-i\vec{p}\cdot\vec{x}} + a_{\mathbf{p}}^\dagger e^{i\vec{p}\cdot\vec{x}}}{(2\pi)^3 \sqrt{2E_p}} = \phi(\vec{x}). \quad (315)$$

The interacting part of the Hamiltonian density is the perturbation. It is a function of spacetime and a functional of the Heisenberg interacting field operator $\Phi(\vec{x})$

$$\mathcal{V} = \mathcal{V}[\Phi](\vec{x}). \quad (316)$$

It is easy to get confused with functional notation (trust me on this, I studied DFT). So we will consider a concrete example as we go along developing the scattering perturbation theory

$$\mathcal{V}(\vec{x}) = \frac{g}{3!} \Phi(\vec{x})^3. \quad (317)$$

This is a Φ^3 interaction (obviously) and g is a coupling constant. For perturbation theory to work g must be sufficiently small. I give the perturbation in Eq. 313

$$\mathcal{V}(t) = \int d^3x \mathcal{V}(\vec{x}). \quad (318)$$

For the Φ^3 interaction this is

$$\mathcal{V}(t) = \frac{g}{3!} \int d^3x \Phi(\vec{x})^3 = \frac{g}{3!} \int d^3x \Phi(t, \mathbf{x})^3. \quad (319)$$

The interaction picture perturbation is

$$\mathcal{V}_I(t) = e^{i\mathcal{H}_0(t-t_0)} \mathcal{V}(t_0) e^{-i\mathcal{H}_0(t-t_0)}. \quad (320)$$

For the Φ^3 example Eq. 314 implies

$$\mathcal{V}(t_0) = \frac{g}{3!} \int d^3x \Phi(t_0, \mathbf{x})^3 = \frac{g}{3!} \int d^3x \phi(\mathbf{x})^3, \quad (321)$$

then using Eq. 315 yields

$$\begin{aligned}\mathcal{V}_1(t) &= e^{i\mathcal{H}_0(t-t_0)}\mathcal{V}(t_0)e^{-i\mathcal{H}_0(t-t_0)} = \\ &= \frac{g}{3!} \int d^3x e^{i\mathcal{H}_0(t-t_0)} \phi(\mathbf{x})^3 e^{-i\mathcal{H}_0(t-t_0)} = \\ &= \frac{g}{3!} \int d^3x e^{i\mathcal{H}_0(t-t_0)} \phi(\mathbf{x}) e^{-i\mathcal{H}_0(t-t_0)} \\ &\quad e^{i\mathcal{H}_0(t-t_0)} \phi(\mathbf{x}) e^{-i\mathcal{H}_0(t-t_0)} e^{i\mathcal{H}_0(t-t_0)} \phi(\mathbf{x}) e^{-i\mathcal{H}_0(t-t_0)} \\ &= \frac{g}{3!} \int d^3x \phi(\vec{x})^3. \quad (322)\end{aligned}$$

Combining this with Eqs. 312 & 317 implies

$$\mathcal{V}_1(t) = - \int d^3x \mathcal{L}_{\text{int}}[\phi]. \quad (323)$$

Though I won't prove it this result is general: it holds for all the self-interacting theories I am considering.

The interaction picture time evolution operator is

$$U(t, t_0) = T \exp \left[-i \int_{t_0}^t d\tau \mathcal{V}_1(\tau) \right]. \quad (324)$$

It connects the Heisenberg picture interacting and free fields

$$\Phi(t, \mathbf{x}) = U^\dagger(t, t_0) \phi(t, \mathbf{x}) U(t, t_0). \quad (325)$$

To actually use this operator we must expand it in a Dyson series

$$\begin{aligned}U(t, t_0) &= 1 - i \int_{t_0}^t dt_1 \mathcal{V}_1(t_1) + \\ &\quad \frac{(-i)^2}{2} \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 T[\mathcal{V}_1(t_1) \mathcal{V}_1(t_2)] + \mathcal{O}(3). \quad (326)\end{aligned}$$

In the scattering perturbation theory I am developing I need to connect the distant past, $t_0 = -\infty$, to the far future, $t = \infty$,

$$U(\infty, -\infty) = T \exp \left(i \int d^4x \mathcal{L}_{\text{int}}[\phi] \right), \quad (327)$$

where I used Eq. 323.

The true time evolution operator, $S(t, t_0)$, acts like

$$|\Psi(t)\rangle = S(t, t_0) |\Psi(t_0)\rangle, \quad \Phi(t, \mathbf{x}) = S^\dagger(t, t_0) \Phi(t_0, \mathbf{x}) S(t, t_0) \quad (328)$$

It is connected to the interaction picture time evolution operator by

$$S(t, t_0) = e^{-i\mathcal{H}_0(t-t_0)} U(t, t_0). \quad (329)$$

8. SCATTERING

In a scattering process the fields start out as free with no interactions. That means the particles inside of the fields

move like free particles with well defined momenta. Then the particles approach each other and interact. Then the particles (or different particles created in the interaction) scatter off to infinity and no longer interact. Again all of the fields are free and we end with particles with well defined momenta.

The free state in the far distant past is $|\mathcal{I}(-\infty)\rangle$. The free state in the far future is $|\mathcal{F}(\infty)\rangle$. If there was no scattering the initial state would have evolved into

$$|\mathcal{I}(\infty)\rangle = S(\infty, -\infty) |\mathcal{I}(-\infty)\rangle, \quad (330)$$

where S is the time evolution operator in Eq. 328. The scattering matrix (S-matrix) is the overlap between the final state and what the final state would have been without scattering

$$\langle \mathcal{F}(\infty) | \mathcal{I}(\infty) \rangle = \langle \mathcal{F}(\infty) | S(\infty, -\infty) | \mathcal{I}(-\infty) \rangle. \quad (331)$$

Once we have the S-matrix we can plug it into expressions for the scattering cross section and decay rate. These are experimental values particle physicists care most about. I personally care more about other things: for example predicting spontaneous emission. Since the initial and final particles have well defined momenta I can use those momenta to specify each state

$$|\mathcal{I}(-\infty)\rangle = |\mathbf{p}_1 \dots \mathbf{p}_M\rangle, \quad |\mathcal{F}(\infty)\rangle = |\mathbf{p}_{M+1} \dots \mathbf{p}_N\rangle, \quad (332)$$

where $M = 1$ for decay and 2 for collisions. A 3-way collision is very hard to set up so usually M is no greater than 2 .

The LSZ (Lehmann, Symanzik, and Zimmermann) reduction formula is a very convenient (exact) way to write the S-matrix. This is another thing that I will prove later if I need to. But for now to save time I just give the result

$$\begin{aligned}\langle \mathbf{p}_{M+1} \dots \mathbf{p}_N | S(\infty, -\infty) | \mathbf{p}_1 \dots \mathbf{p}_M \rangle &= i^N \\ &\int d^4x_1 e^{-i\vec{p}_1 \cdot \vec{x}_1} (\square_1 + m^2) \dots \int d^4x_M e^{-i\vec{p}_M \cdot \vec{x}_M} (\square_M + m^2) \\ &\int d^4x_{M+1} e^{i\vec{p}_{M+1} \cdot \vec{x}_{M+1}} (\square_{M+1} + m^2) \dots \\ &\int d^4x_N e^{i\vec{p}_N \cdot \vec{x}_N} (\square_N + m^2) C(\vec{x}_1 \dots \vec{x}_N) \quad (333)\end{aligned}$$

where

$$C(\vec{x}_1 \dots \vec{x}_N) = \langle \Omega | T[\Phi(\vec{x}_1) \dots \Phi(\vec{x}_N)] | \Omega \rangle, \quad (334)$$

is a correlation function.

9. WICK'S THEOREM

To predict measurable quantities like the cross section I need the scattering matrix. To get the scattering matrix I need to evaluate correlation functions. To evaluate correlation functions I need Wick's theorem. Wick's theorem is

$$T(\phi_1 \dots \phi_N) =: \phi_1 \dots \phi_N : + : \text{all contractions} :, \quad (335)$$

where I introduce the shorthand $\phi_1 = \phi(\vec{x}_1)$. A contraction (denoted by an over-bracket) of two free fields is their Feynman propagator

$$\overline{\phi_1 \phi_2} = \mathcal{G}_F(\vec{x}_1 - \vec{x}_2) = \mathcal{G}_F^{12} = \mathcal{G}_F^{21}, \quad (336)$$

where I used Eq. 304 (Feynman propagator is even). The normal ordered product (denoted by $::$) moves all creation/annihilation operators to the left/right. For example

$$: a_{\mathbf{p}} a_{\mathbf{p}}^\dagger := a_{\mathbf{p}}^\dagger a_{\mathbf{p}}. \quad (337)$$

The simplest example of Wick's theorem is

$$T(\phi_1 \phi_2) = : \phi_1 \phi_2 : + \overline{\phi_1 \phi_2} = : \phi_1 \phi_2 : + \mathcal{G}_F^{12}. \quad (338)$$

9.1. Free Vacuum Expectation Values

In practice Wick's theorem is mostly used to evaluate expectation values in the free vacuum, $\langle \mathcal{A} \rangle = \langle 0 | \mathcal{A} | 0 \rangle$. In the simplest case

$$\langle 0 | T(\phi_1 \phi_2) | 0 \rangle = \langle 0 | : \phi_1 \phi_2 : | 0 \rangle + \langle 0 | \mathcal{G}_F^{12} | 0 \rangle = \mathcal{G}_F^{12} \langle 0 | 0 \rangle = \mathcal{G}_F^{12} \quad (339)$$

where I used

$$\begin{aligned} \langle 0 | : \phi_1 \phi_2 : | 0 \rangle &\propto \int \frac{d^3 p d^3 k}{\sqrt{E_p E_k}} \\ \langle 0 | : (a_{\mathbf{p}} e^{-i\vec{p} \cdot \vec{x}_1} + a_{\mathbf{p}}^\dagger e^{i\vec{p} \cdot \vec{x}_1}) (a_{\mathbf{k}} e^{-i\vec{k} \cdot \vec{x}_2} + a_{\mathbf{k}}^\dagger e^{i\vec{k} \cdot \vec{x}_2}) : | 0 \rangle &= \\ \langle 0 | : a_{\mathbf{p}} a_{\mathbf{k}} : | 0 \rangle e^{-i(\vec{p} \cdot \vec{x}_1 + \vec{k} \cdot \vec{x}_2)} + \langle 0 | : a_{\mathbf{p}} a_{\mathbf{k}}^\dagger : | 0 \rangle e^{i(\vec{k} \cdot \vec{x}_2 - \vec{p} \cdot \vec{x}_1)} + \\ \langle 0 | : a_{\mathbf{p}}^\dagger a_{\mathbf{k}} : | 0 \rangle e^{i(\vec{p} \cdot \vec{x}_1 - \vec{k} \cdot \vec{x}_2)} + \langle 0 | : a_{\mathbf{p}}^\dagger a_{\mathbf{k}}^\dagger : | 0 \rangle e^{i(\vec{p} \cdot \vec{x}_1 + \vec{k} \cdot \vec{x}_2)} &= \\ \langle 0 | a_{\mathbf{p}} a_{\mathbf{k}} | 0 \rangle e^{-i(\vec{p} \cdot \vec{x}_1 + \vec{k} \cdot \vec{x}_2)} + \langle 0 | a_{\mathbf{k}}^\dagger a_{\mathbf{p}} | 0 \rangle e^{i(\vec{k} \cdot \vec{x}_2 - \vec{p} \cdot \vec{x}_1)} + \\ \langle 0 | a_{\mathbf{p}}^\dagger a_{\mathbf{k}} | 0 \rangle e^{i(\vec{p} \cdot \vec{x}_1 - \vec{k} \cdot \vec{x}_2)} + \langle 0 | a_{\mathbf{p}}^\dagger a_{\mathbf{k}}^\dagger | 0 \rangle e^{i(\vec{p} \cdot \vec{x}_1 + \vec{k} \cdot \vec{x}_2)} &= \\ \langle 0 | a_{\mathbf{k}}^\dagger a_{\mathbf{p}} | 0 \rangle e^{i(\vec{k} \cdot \vec{x}_2 - \vec{p} \cdot \vec{x}_1)} = 0. \end{aligned} \quad (340)$$

This shows that any term that is not fully contracted (not completely factored into Feynman propagators) will not contribute to a vacuum expectation value and can be ignored.

10. CORRELATION FUNCTIONS

We need to evaluate correlation functions using perturbation theory. To do that we put them into a form that is conducive for perturbative expansions

$$C(\vec{x}_1 \dots \vec{x}_N) = \langle \Omega | T(\Phi_1 \dots \Phi_N) | \Omega \rangle = \frac{\langle 0 | T(\phi_1 \dots \phi_N \mathcal{F}) | 0 \rangle}{\langle 0 | T \mathcal{F} | 0 \rangle}, \quad (341)$$

where

$$\mathcal{F} = \exp \left(i \int d^4 x \mathcal{L}_{\text{int}}[\phi] \right). \quad (342)$$

This is another thing that I state without proof.

10.1. Φ^3 Example

I will show how to expand the correlation function perturbatively for the simple interaction

$$\mathcal{L}_{\text{int}}[\Phi] = \frac{g}{3!} \Phi(\vec{x})^3 \quad (343)$$

and the simplest correlation function

$$C(\vec{x}_1, \vec{x}_2) = \langle \Omega | T(\Phi_1 \Phi_2) | \Omega \rangle = \frac{\langle 0 | T(\phi_1 \phi_2 \mathcal{F}) | 0 \rangle}{\langle 0 | T \mathcal{F} | 0 \rangle}. \quad (344)$$

Then

$$\mathcal{F} = \exp \left(\frac{ig}{3!} \int d^4 x \phi(\vec{x})^3 \right). \quad (345)$$

The coupling constant g must be small for perturbation theory to work, so I expand in g

$$\mathcal{F} = 1 + \frac{ig}{3!} \int d^4 x \phi(\vec{x})^3 - \frac{g^2}{72} \int d^4 x d^4 y \phi(\vec{x})^3 \phi(\vec{y})^3 + \mathcal{O}(g^3). \quad (346)$$

I expand the denominator of Eq. 344 to order g^2 .

$$\mathcal{D} = \langle 0 | T \mathcal{F} | 0 \rangle \quad (347)$$

$$\begin{aligned} \mathcal{D} &= \langle 0 | 0 \rangle + \frac{ig}{3!} \int d^4 x \langle 0 | T \phi(\vec{x})^3 | 0 \rangle - \\ &\quad \frac{g^2}{72} \int d^4 x d^4 y \langle 0 | T[\phi(\vec{x})^3 \phi(\vec{y})^3] | 0 \rangle \end{aligned} \quad (348)$$

I introduce the shorthand $\phi_x = \phi(\vec{x})$.

$$\mathcal{D} = 1 + \frac{ig}{3!} \int d^4 x \langle 0 | T \phi_x^3 | 0 \rangle - \frac{g^2}{72} \int d^4 x d^4 y \langle 0 | T(\phi_x^3 \phi_y^3) | 0 \rangle \quad (349)$$

There is no way to fully contract 3 (or any other odd number) fields so (Sec. 9.1)

$$\langle 0 | T \phi_x^3 | 0 \rangle = 0. \quad (350)$$

$$\mathcal{D} = 1 - \frac{g^2}{72} \int d^4 x d^4 y \langle 0 | T(\phi_x^3 \phi_y^3) | 0 \rangle \quad (351)$$

I use Wick's theorem

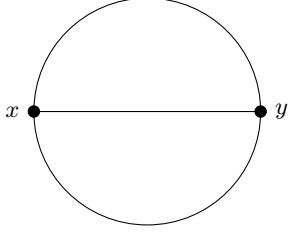
$$\langle 0 | T(\phi_x^3 \phi_y^3) | 0 \rangle = 9 \mathcal{G}_F^{xx} \mathcal{G}_F^{yy} \mathcal{G}_F^{xy} + 6 (\mathcal{G}_F^{xy})^3 \quad (352)$$

where I introduce the shorthand $\mathcal{G}_F(\vec{x} - \vec{y}) = \mathcal{G}_F^{xy} = \mathcal{G}_F^{yx}$.

$$\mathcal{D} = 1 - \frac{g^2}{24} \int d^4 x d^4 y [3 \mathcal{G}_F^{xx} \mathcal{G}_F^{yy} \mathcal{G}_F^{xy} + 2 (\mathcal{G}_F^{xy})^3] \quad (353)$$

I label these integrals

$$\begin{aligned} \mathfrak{Z}[0] &= g^2 \int d^4 x d^4 y \mathcal{G}_F^{xx} \mathcal{G}_F^{yy} \mathcal{G}_F^{xy}, \\ \mathfrak{Z}[1] &= g^2 \int d^4 x d^4 y (\mathcal{G}_F^{xy})^3, \end{aligned} \quad (354)$$

FIG. 2. Feynman diagram for $\mathfrak{Y}[0]$.FIG. 3. Feynman diagram for $\mathfrak{Y}[1]$.

so

$$\mathcal{D} = 1 - \frac{\mathfrak{Y}[0]}{8} - \frac{\mathfrak{Y}[1]}{12} \quad (355)$$

These integrals correspond to the diagrams in Figs. 2 & 3. Because they are not connected to any external vertex they are called bubble diagrams.

I work out the numerator of Eq. 344 to order g^2 .

$$\mathcal{N} = \langle 0|T(\phi_1\phi_2\mathcal{F})|0\rangle \quad (356)$$

$$\begin{aligned} \mathcal{N} &= \langle 0|T(\phi_1\phi_2)|0\rangle + \frac{ig}{3!} \int d^4x \langle 0|T(\phi_1\phi_2\phi_x^3)|0\rangle \\ &\quad - \frac{g^2}{72} \int d^4x d^4y \langle 0|T(\phi_1\phi_2\phi_x^3\phi_y^3)|0\rangle \end{aligned} \quad (357)$$

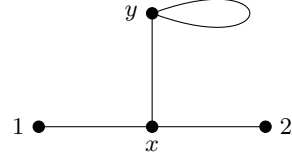
$$\mathcal{N} = \mathcal{G}_F^{12} - \frac{g^2}{72} \int d^4x d^4y \langle 0|T(\phi_1\phi_2\phi_x^3\phi_y^3)|0\rangle \quad (358)$$

$$\begin{aligned} \langle 0|T(\phi_1\phi_2\phi_x^3\phi_y^3)|0\rangle &= 9\mathcal{G}_F^{12}\mathcal{G}_F^{xx}\mathcal{G}_F^{yy}\mathcal{G}_F^{xy} + 6\mathcal{G}_F^{12}(\mathcal{G}_F^{xy})^3 \\ &+ 18\mathcal{G}_F^{1x}\mathcal{G}_F^{2x}\mathcal{G}_F^{xy}\mathcal{G}_F^{yy} + 9\mathcal{G}_F^{1x}\mathcal{G}_F^{2y}\mathcal{G}_F^{xx}\mathcal{G}_F^{yy} + 18\mathcal{G}_F^{1x}\mathcal{G}_F^{2y}(\mathcal{G}_F^{xy})^2 \\ &+ 18\mathcal{G}_F^{1y}\mathcal{G}_F^{2y}\mathcal{G}_F^{xx}\mathcal{G}_F^{xy} + 9\mathcal{G}_F^{1y}\mathcal{G}_F^{2x}\mathcal{G}_F^{xx}\mathcal{G}_F^{yy} + 18\mathcal{G}_F^{1y}\mathcal{G}_F^{2x}(\mathcal{G}_F^{xy})^2 \end{aligned} \quad (359)$$

$$\begin{aligned} \mathcal{N} &= \mathcal{G}_F^{12} - \frac{g^2}{24} \int d^4x d^4y [3\mathcal{G}_F^{12}\mathcal{G}_F^{xx}\mathcal{G}_F^{yy}\mathcal{G}_F^{xy} + 2\mathcal{G}_F^{12}(\mathcal{G}_F^{xy})^3 \\ &+ 6\mathcal{G}_F^{1x}\mathcal{G}_F^{2x}\mathcal{G}_F^{xy}\mathcal{G}_F^{yy} + 3\mathcal{G}_F^{1x}\mathcal{G}_F^{2y}\mathcal{G}_F^{xx}\mathcal{G}_F^{yy} + 6\mathcal{G}_F^{1x}\mathcal{G}_F^{2y}(\mathcal{G}_F^{xy})^2 \\ &+ 6\mathcal{G}_F^{1y}\mathcal{G}_F^{2y}\mathcal{G}_F^{xx}\mathcal{G}_F^{xy} + 3\mathcal{G}_F^{1y}\mathcal{G}_F^{2x}\mathcal{G}_F^{xx}\mathcal{G}_F^{yy} + 6\mathcal{G}_F^{1y}\mathcal{G}_F^{2x}(\mathcal{G}_F^{xy})^2] \end{aligned} \quad (360)$$

That's a lot of terms. To condense things I rename them.

$$\begin{aligned} \mathfrak{Y}[0] &= \mathcal{G}_F^{12} \\ \mathfrak{Y}[1] &= g^2 \int d^4x d^4y \mathcal{G}_F^{1x}\mathcal{G}_F^{2x}\mathcal{G}_F^{xy}\mathcal{G}_F^{yy} \\ \mathfrak{Y}[2] &= g^2 \int d^4x \mathcal{G}_F^{1x}\mathcal{G}_F^{xx} \int d^4y \mathcal{G}_F^{2y}\mathcal{G}_F^{yy} \\ \mathfrak{Y}[3] &= g^2 \int d^4x d^4y \mathcal{G}_F^{1x}\mathcal{G}_F^{2y}(\mathcal{G}_F^{xy})^2 \end{aligned} \quad (361)$$

FIG. 4. Feynman diagram for $\mathfrak{Y}[0]$.FIG. 5. Feynman diagram for $\mathfrak{Y}[1]$.

The new terms correspond to the Feynman diagrams in Figs. 4-7. In terms of these diagrams the numerator becomes

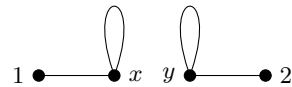
$$\begin{aligned} \mathcal{N} &= \mathfrak{Y}[0] - \left(\frac{\mathfrak{Y}[0]\mathfrak{Y}[0]}{8} + \frac{\mathfrak{Y}[0]\mathfrak{Y}[1]}{12} + \frac{\mathfrak{Y}[1]}{4} + \frac{\mathfrak{Y}[2]}{8} + \frac{\mathfrak{Y}[3]}{4} \right. \\ &\quad + \frac{g^2}{4} \int d^4x d^4y \mathcal{G}_F^{1y}\mathcal{G}_F^{2y}\mathcal{G}_F^{xx}\mathcal{G}_F^{xy} + \\ &\quad \left. \frac{g^2}{8} \int d^4x \mathcal{G}_F^{2x}\mathcal{G}_F^{xx} \int d^4y \mathcal{G}_F^{1y}\mathcal{G}_F^{yy} + \right. \\ &\quad \left. \frac{g^2}{4} \int d^4x d^4y \mathcal{G}_F^{1y}\mathcal{G}_F^{2x}(\mathcal{G}_F^{xy})^2 \right), \end{aligned} \quad (362)$$

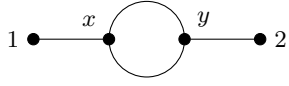
$$\begin{aligned} \mathcal{N} &= \mathfrak{Y}[0] - \left(\frac{\mathfrak{Y}[0]\mathfrak{Y}[0]}{8} + \frac{\mathfrak{Y}[0]\mathfrak{Y}[1]}{12} + \frac{\mathfrak{Y}[1]}{4} + \frac{\mathfrak{Y}[2]}{8} + \frac{\mathfrak{Y}[3]}{4} \right. \\ &\quad + \frac{g^2}{4} \int d^4x d^4y \mathcal{G}_F^{1x}\mathcal{G}_F^{2x}\mathcal{G}_F^{xy}\mathcal{G}_F^{yy} + \\ &\quad \left. \frac{g^2}{8} \int d^4x \mathcal{G}_F^{1x}\mathcal{G}_F^{xx} \int d^4y \mathcal{G}_F^{2y}\mathcal{G}_F^{yy} + \right. \\ &\quad \left. \frac{g^2}{4} \int d^4x d^4y \mathcal{G}_F^{1x}\mathcal{G}_F^{2y}(\mathcal{G}_F^{xy})^2 \right), \end{aligned} \quad (363)$$

$$\mathcal{N} = \mathfrak{Y}[0] - \frac{1}{2} \left(\frac{\mathfrak{Y}[0]\mathfrak{Y}[0]}{4} + \frac{\mathfrak{Y}[0]\mathfrak{Y}[1]}{6} + \mathfrak{Y}[1] + \frac{\mathfrak{Y}[2]}{2} + \mathfrak{Y}[3] \right). \quad (364)$$

Now that I have the numerator and denominator I expand the correlation function to order g^2 ,

$$C = \frac{\mathcal{N}}{\mathcal{D}} = \frac{\mathcal{N}^{(0)} + g^2\Delta\mathcal{N}}{\mathcal{D}^{(0)} + g^2\Delta\mathcal{D}} = \frac{\mathcal{N}^{(0)}}{\mathcal{D}^{(0)}} + \frac{g^2}{\mathcal{D}^{(0)}} \left(\Delta\mathcal{N} - \frac{\mathcal{N}^{(0)}}{\mathcal{D}^{(0)}}\Delta\mathcal{D} \right), \quad (365)$$

FIG. 6. Feynman diagram for $\mathfrak{Y}[2]$.


 FIG. 7. Feynman diagram for $\mathfrak{Y}[3]$.

where the superscript 0 denotes leading order and Δ denotes the lading correction. Plugging in the leading order terms $\mathcal{D}^{(0)} = 1$ & $\mathcal{N}^{(0)} = \mathfrak{Y}[0]$ from Eqs. 355 & 364 yields

$$C = \mathfrak{Y}[0] + g^2(\Delta\mathcal{N} - \mathfrak{Y}[0]\Delta\mathcal{D}). \quad (366)$$

Plugging in the leading (second) order corrections yields the final result

$$C = \mathfrak{Y}[0] + g^2\Delta\mathcal{N} - g^2\mathfrak{Y}[0]\Delta\mathcal{D} = \mathfrak{Y}[0] - \frac{1}{2} \left(\frac{\mathfrak{Y}[0]\mathfrak{Y}[0]}{4} - \frac{\mathfrak{Y}[0]\mathfrak{Y}[0]}{4} - \frac{\mathfrak{Y}[0]\mathfrak{Y}[1]}{6} + \frac{\mathfrak{Y}[0]\mathfrak{Y}[1]}{6} + \mathfrak{Y}[1] + \frac{\mathfrak{Y}[2]}{2} + \mathfrak{Y}[3] \right), \quad (367)$$

$$C(\vec{x}_1, \vec{x}_2) = \mathfrak{Y}[0] - \frac{1}{2} \left(\mathfrak{Y}[1] + \frac{\mathfrak{Y}[2]}{2} + \mathfrak{Y}[3] \right). \quad (368)$$

The bubble diagrams denoted by $\mathfrak{Y}$ canceled out. There is a beautiful proof that this must always happen. If I ever need to calculate another correlation function I won't do it the hard way by expanding everything and using Wick's theorem. I will just draw all the non-bubble diagrams. However I will have to nail down the rules for assigning coefficients to the diagrams first, which I have neglected for now to save time.

11. S-MATRIX ELEMENTS & THE FEYNMAN RULES

Now I have everything I need to calculate S-matrix elements. To convince myself of that I will calculate the S-matrix element for the scattering process in Sec. 10.1

$$\mathcal{S} = \langle \mathbf{p}_2 | S(\infty, -\infty) | \mathbf{p}_1 \rangle. \quad (369)$$

I did not mention it while calculating the correlation function but this is a 1 to 1 (or $\Phi \rightarrow \Phi$) scattering process. I use the LSZ reduction formula

$$\mathcal{S} = - \int d^4x_1 e^{-i\vec{p}_1 \cdot \vec{x}_1} (\square_1 + m^2) \int d^4x_2 e^{i\vec{p}_2 \cdot \vec{x}_2} (\square_2 + m^2) C(\vec{x}_1, \vec{x}_2). \quad (370)$$

I plug in Eq. 368.

$$\mathcal{S} = - \int d^4x_1 e^{-i\vec{p}_1 \cdot \vec{x}_1} (\square_1 + m^2) \int d^4x_2 e^{i\vec{p}_2 \cdot \vec{x}_2} (\square_2 + m^2) \left[\mathcal{G}_F^{12} - \frac{1}{2} \left(\mathfrak{Y}[1] + \frac{\mathfrak{Y}[2]}{2} + \mathfrak{Y}[3] \right) \right]. \quad (371)$$

I seek an expansion to order g^2

$$\mathcal{S} = \mathcal{S}^{(0)} + g^2\Delta\mathcal{S}. \quad (372)$$

I calculate the lowest order term using Eq. 305.

$$\mathcal{S}^{(0)} = - \int d^4x_1 e^{-i\vec{p}_1 \cdot \vec{x}_1} (\square_1 + m^2) \int d^4x_2 e^{i\vec{p}_2 \cdot \vec{x}_2} (\square_2 + m^2) \mathcal{G}_F^{12} \quad (373)$$

$$\mathcal{S}^{(0)} = - \int d^4x_1 e^{-i\vec{p}_1 \cdot \vec{x}_1} (\square_1 + m^2) \int d^4x_2 e^{i\vec{p}_2 \cdot \vec{x}_2} (\square_2 + m^2) \mathcal{G}_F(\vec{x}_1 - \vec{x}_2) \quad (374)$$

$$\mathcal{S}^{(0)} = - \int d^4x e^{-i\vec{p}_1 \cdot \vec{x}} (\square_x + m^2) \int d^4y e^{i\vec{p}_2 \cdot \vec{y}} (\square_y + m^2) \mathcal{G}_F(\vec{x} - \vec{y}) \quad (375)$$

$$\mathcal{S}^{(0)} = i \int d^4x d^4y e^{i(\vec{p}_2 \cdot \vec{y} - \vec{p}_1 \cdot \vec{x})} (\square_x + m^2) \delta^{(4)}(\vec{x} - \vec{y}) \quad (376)$$

$$\mathcal{S}^{(0)} = i \int d^4x d^4y e^{i(\vec{p}_2 \cdot \vec{y} - \vec{p}_1 \cdot \vec{x})} \square_x \delta^{(4)}(\vec{x} - \vec{y}) + im^2 \int d^4x d^4y e^{i(\vec{p}_2 \cdot \vec{y} - \vec{p}_1 \cdot \vec{x})} \delta^{(4)}(\vec{x} - \vec{y}) \quad (377)$$

I assume that I can integrate by parts without worrying about boundary terms: in other words I can move derivatives around with impunity.

$$\mathcal{S}^{(0)} = i \int d^4x d^4y \delta^{(4)}(\vec{x} - \vec{y}) e^{i\vec{p}_2 \cdot \vec{y}} \square_x e^{-i\vec{p}_1 \cdot \vec{x}} + im^2 \int d^4x e^{i(\vec{p}_2 - \vec{p}_1) \cdot \vec{x}} \quad (378)$$

$$\square e^{i\vec{p} \cdot \vec{x}} = (\partial_t^2 - \nabla^2) e^{i\vec{p} \cdot \vec{x}} = (\partial_t^2 - \nabla^2) e^{i(E, \mathbf{p}) \cdot (t, \mathbf{x})} = (\partial_t^2 - \nabla^2) e^{i(tE - \mathbf{p} \cdot \mathbf{x})} = (\partial_t^2 - \nabla^2) e^{itE} e^{-i\mathbf{p} \cdot \mathbf{x}} = e^{-i\mathbf{p} \cdot \mathbf{x}} \partial_t^2 e^{itE} - e^{itE} \nabla^2 e^{-i\mathbf{p} \cdot \mathbf{x}} = (p^2 - E^2) e^{i\vec{p} \cdot \vec{x}} = -m^2 e^{i\vec{p} \cdot \vec{x}} \quad (379)$$

$$\square e^{-i\vec{p} \cdot \vec{x}} = -m^2 e^{-i\vec{p} \cdot \vec{x}} \quad (380)$$

$$\mathcal{S}^{(0)} = -im^2 \int d^4x d^4y \delta^{(4)}(\vec{x} - \vec{y}) e^{i(\vec{p}_2 \cdot \vec{y} - \vec{p}_1 \cdot \vec{x})} + im^2 \int d^4x e^{i(\vec{p}_2 - \vec{p}_1) \cdot \vec{x}} \quad (381)$$

$$\mathcal{S}^{(0)} = im^2 \int d^4x e^{i(\vec{p}_2 - \vec{p}_1) \cdot \vec{x}} \quad \cancel{im^2 \int d^4x e^{i(\vec{p}_2 - \vec{p}_1) \cdot \vec{x}}} \quad (382)$$

$$\mathcal{S}^{(0)} = 0 \quad (383)$$

To lowest order no scattering happens and $S^{(0)}(\infty, -\infty)$ is the identity. So this implies that the initial and final momenta are different. But since this is 1 to 1 scattering that appears to violate momentum conservation. I will have to ask someone about this later.

I will follow Schwartz (Sec. 7.3) and only calculate one of the second order contributions. Maybe I will come back and do them all later.

$$g^2 \Delta \mathcal{S} = \int d^4x_1 e^{-i\vec{p}_1 \cdot \vec{x}_1} (\square_1 + m^2) \int d^4x_2 e^{i\vec{p}_2 \cdot \vec{x}_2} (\square_2 + m^2) \quad (384)$$

$$\frac{1}{2} \left(\mathfrak{G}[1] + \frac{\mathfrak{G}[2]}{2} + \mathfrak{G}[3] \right) = I[1] + I[2] + I[3]$$

$$2I[3] = \int d^4x_1 e^{-i\vec{p}_1 \cdot \vec{x}_1} (\square_1 + m^2) \int d^4x_2 e^{i\vec{p}_2 \cdot \vec{x}_2} (\square_2 + m^2) \quad (385)$$

$$\mathfrak{G}[3]$$

$$I[3] = \frac{g^2}{2} \int d^4x_1 e^{-i\vec{p}_1 \cdot \vec{x}_1} (\square_1 + m^2) \int d^4x_2 e^{i\vec{p}_2 \cdot \vec{x}_2} (\square_2 + m^2) \quad (386)$$

$$\int d^4x d^4y \mathcal{G}_F(\vec{x}_1 - \vec{x}) \mathcal{G}_F(\vec{x}_2 - \vec{y}) \mathcal{G}_F(\vec{x} - \vec{y})^2$$

$$(2\pi)^4 \mathcal{G}_F(\vec{x} - \vec{y}) = i \int d^4p \frac{e^{i\vec{p} \cdot (\vec{y} - \vec{x})}}{\vec{p}^2 - m^2 + i\epsilon} \quad (387)$$

For simplicity I follow Schwartz and only treat the $m = 0$ case. I also use the evenness of the Feynman propagator.

$$\mathcal{G}_F(\vec{x} - \vec{y}) = \frac{i}{(2\pi)^4} \int d^4p \frac{e^{i\vec{p} \cdot (\vec{x} - \vec{y})}}{\vec{p}^2 + i\epsilon} \quad (388)$$

$$(2\pi)^{16} I[3] = \frac{g^2}{2} \int d^4x_1 e^{-i\vec{p}_1 \cdot \vec{x}_1} \square_1 \int d^4x_2 e^{i\vec{p}_2 \cdot \vec{x}_2} \square_2 \quad (389)$$

$$\int d^4x d^4y \int d^4p d^4k d^4w d^4v$$

$$\frac{e^{i\vec{v} \cdot (\vec{x}_1 - \vec{x})}}{\vec{v}^2 + i\epsilon} \frac{e^{i\vec{w} \cdot (\vec{x}_2 - \vec{y})}}{\vec{w}^2 + i\epsilon} \frac{e^{i\vec{k} \cdot (\vec{x} - \vec{y})}}{\vec{k}^2 + i\epsilon} \frac{e^{i\vec{p} \cdot (\vec{x} - \vec{y})}}{\vec{p}^2 + i\epsilon}$$

$$(2\pi)^{16} I[3] = \frac{g^2}{2} \int d^4x_1 e^{-i\vec{p}_1 \cdot \vec{x}_1} \square_1 \int d^4x_2 e^{i\vec{p}_2 \cdot \vec{x}_2} \square_2 \quad (390)$$

$$\int d^4x d^4y \int d^4p d^4k d^4w d^4v$$

$$\frac{e^{i\vec{v} \cdot \vec{x}_1} e^{i\vec{w} \cdot \vec{x}_2} e^{i(\vec{k} + \vec{p} - \vec{v}) \cdot \vec{x}} e^{-i(\vec{p} + \vec{k} + \vec{w}) \cdot \vec{y}}}{(\vec{v}^2 + i\epsilon)(\vec{w}^2 + i\epsilon)(\vec{k}^2 + i\epsilon)(\vec{p}^2 + i\epsilon)}$$

$$(2\pi)^{16} I[3] = \frac{g^2}{2} \int d^4x_1 d^4x_2 e^{-i\vec{p}_1 \cdot \vec{x}_1} e^{i\vec{p}_2 \cdot \vec{x}_2} \quad (391)$$

$$\int d^4x d^4y \int d^4p d^4k d^4w d^4v$$

$$\frac{(\square_1 e^{i\vec{v} \cdot \vec{x}_1})(\square_2 e^{i\vec{w} \cdot \vec{x}_2}) e^{i(\vec{k} + \vec{p} - \vec{v}) \cdot \vec{x}} e^{-i(\vec{p} + \vec{k} + \vec{w}) \cdot \vec{y}}}{(\vec{v}^2 + i\epsilon)(\vec{w}^2 + i\epsilon)(\vec{k}^2 + i\epsilon)(\vec{p}^2 + i\epsilon)}$$

$$\square e^{\pm i\vec{p} \cdot \vec{x}} = p^\mu p_\mu e^{\pm i\vec{p} \cdot \vec{x}} = \vec{p}^2 e^{\pm i\vec{p} \cdot \vec{x}} \quad (392)$$

For a massless particle $\vec{p}^2 = 0$, but Schwartz ignores this fact and just uses $\vec{p}^2$ to demonstrate overall conservation of momentum. I follow his lead.

$$(2\pi)^{16} I[3] = \frac{g^2}{2} \int d^4x_1 d^4x_2 e^{-i\vec{p}_1 \cdot \vec{x}_1} e^{i\vec{p}_2 \cdot \vec{x}_2} \quad (393)$$

$$\int d^4x d^4y \int d^4p d^4k d^4w d^4v \vec{v}^2 \vec{w}^2$$

$$\frac{e^{i\vec{v} \cdot \vec{x}_1} e^{i\vec{w} \cdot \vec{x}_2} e^{i(\vec{k} + \vec{p} - \vec{v}) \cdot \vec{x}} e^{-i(\vec{p} + \vec{k} + \vec{w}) \cdot \vec{y}}}{(\vec{v}^2 + i\epsilon)(\vec{w}^2 + i\epsilon)(\vec{k}^2 + i\epsilon)(\vec{p}^2 + i\epsilon)}$$

$$(2\pi)^{16} I[3] = \frac{g^2}{2} \int d^4x_1 d^4x_2 e^{-i\vec{p}_1 \cdot \vec{x}_1} e^{i\vec{p}_2 \cdot \vec{x}_2} \quad (394)$$

$$\int d^4p d^4k d^4w d^4v \vec{v}^2 \vec{w}^2$$

$$\frac{e^{i\vec{v} \cdot \vec{x}_1} e^{i\vec{w} \cdot \vec{x}_2}}{(\vec{v}^2 + i\epsilon)(\vec{w}^2 + i\epsilon)(\vec{k}^2 + i\epsilon)(\vec{p}^2 + i\epsilon)}$$

$$\int d^4y e^{-i(\vec{p} + \vec{k} + \vec{w}) \cdot \vec{y}} \int d^4x e^{i(\vec{k} + \vec{p} - \vec{v}) \cdot \vec{x}}$$

$$(2\pi)^8 I[3] = \frac{g^2}{2} \int d^4x_1 d^4x_2 e^{-i\vec{p}_1 \cdot \vec{x}_1} e^{i\vec{p}_2 \cdot \vec{x}_2} \quad (395)$$

$$\int d^4p d^4k d^4w d^4v \vec{v}^2 \vec{w}^2$$

$$\frac{e^{i\vec{v} \cdot \vec{x}_1} e^{i\vec{w} \cdot \vec{x}_2} \delta^{(4)}[-(\vec{p} + \vec{k} + \vec{w})] \delta^{(4)}(\vec{k} + \vec{p} - \vec{v})}{(\vec{v}^2 + i\epsilon)(\vec{w}^2 + i\epsilon)(\vec{k}^2 + i\epsilon)(\vec{p}^2 + i\epsilon)}$$

$$(2\pi)^8 I[3] = \frac{g^2}{2} \int d^4x_1 d^4x_2 e^{-i\vec{p}_1 \cdot \vec{x}_1} e^{i\vec{p}_2 \cdot \vec{x}_2} \quad (396)$$

$$\int d^4p d^4k d^4w d^4v \vec{v}^2 \vec{w}^2$$

$$\frac{e^{i\vec{v} \cdot \vec{x}_1} e^{i\vec{w} \cdot \vec{x}_2} \delta^{(4)}(\vec{p} + \vec{k} + \vec{w}) \delta^{(4)}(\vec{k} + \vec{p} - \vec{v})}{(\vec{v}^2 + i\epsilon)(\vec{w}^2 + i\epsilon)(\vec{k}^2 + i\epsilon)(\vec{p}^2 + i\epsilon)}$$

$$(2\pi)^8 I[3] = \frac{g^2}{2} \int d^4x_1 d^4x_2 e^{-i\vec{p}_1 \cdot \vec{x}_1} e^{i\vec{p}_2 \cdot \vec{x}_2} \quad (397)$$

$$\int d^4p d^4k \frac{(\vec{k} + \vec{p})^4 e^{i(\vec{k} + \vec{p}) \cdot \vec{x}_1} e^{-i(\vec{k} + \vec{p}) \cdot \vec{x}_2}}{[(\vec{k} + \vec{p})^2 + i\epsilon]^2 (\vec{k}^2 + i\epsilon)(\vec{p}^2 + i\epsilon)}$$

$$(2\pi)^8 I[3] = \frac{g^2}{2} \int \frac{d^4 p d^4 k (\vec{k} + \vec{p})^4}{[(\vec{k} + \vec{p})^2 + i\epsilon]^2 (\vec{k}^2 + i\epsilon) (\vec{p}^2 + i\epsilon)} \int d^4 x_1 e^{i(\vec{k} + \vec{p} - \vec{p}_1) \cdot \vec{x}_1} \int d^4 x_2 e^{i(\vec{p}_2 - \vec{k} - \vec{p}) \cdot \vec{x}_2} \quad (398)$$

$$I[3] = \frac{g^2}{2} \int \frac{d^4 p d^4 k (\vec{k} + \vec{p})^4}{[(\vec{k} + \vec{p})^2 + i\epsilon]^2 (\vec{k}^2 + i\epsilon) (\vec{p}^2 + i\epsilon)} \delta^{(4)}(\vec{k} + \vec{p} - \vec{p}_1) \delta^{(4)}(\vec{p}_2 - \vec{k} - \vec{p}) \quad (399)$$

$$I[3] = \frac{g^2 \delta^{(4)}(\vec{p}_2 - \vec{p}_1) (\vec{p}_1)^4}{2[(\vec{p}_1)^2 + i\epsilon]^2} \int \frac{d^4 p}{[(\vec{p}_1 - \vec{p})^2 + i\epsilon] (\vec{p}^2 + i\epsilon)} \quad (400)$$

Because $\epsilon \rightarrow 0_+$ I can cancel out the $(\vec{p}_1)^2$. And since the incoming scalar boson is massless this is actually $0/0 = 1$. This is a good example of how careful you have to be with 0 , ∞ , and the order of limits in QFT. It is not as forgiving as quantum mechanics.

$$I[3] = \frac{g^2 (\vec{p}_1)^4}{2(\vec{p}_1)^4} \delta^{(4)}(\vec{p}_2 - \vec{p}_1) \int \frac{d^4 p}{[(\vec{p}_1 - \vec{p})^2 + i\epsilon] (\vec{p}^2 + i\epsilon)} \quad (401)$$

$$I[3] = \frac{g^2}{2} \delta^{(4)}(\vec{p}_2 - \vec{p}_1) \int \frac{d^4 p}{[(\vec{p}_1 - \vec{p})^2 + i\epsilon] (\vec{p}^2 + i\epsilon)} \quad (402)$$

This is identical to what Schwartz got in Eq. 7.84. This took a lot of work. And it was only one term. I didn't even do the other two. This is where Feynman diagrams come in. We can speed things up dramatically by drawing pictures.

11.1. Momentum Space Feynman Rules

I can rewrite the S-matrix elements as

$$\mathcal{S} = I + (2\pi)^4 i \delta^{(4)}(\vec{P}_{\text{in}} - \vec{P}_{\text{out}}) \mathcal{M}, \quad (403)$$

where I is the identity matrix and $\vec{P}_{\text{in}}/\vec{P}_{\text{out}}$ is the total momentum of all the initial/final particles, so the delta function conserves momentum. The $\mathcal{M}$ is so important that it is called the matrix element in QFT. It is the nontrivial part of $\mathcal{S}$. Those famous Feynman diagrams that you see everywhere: those are momentum space Feynman diagrams that are used to compute $\mathcal{M}$ using these Feynman rules (which convert diagrams into integrals):

1. Internal lines (virtual particles not connected to external points, i.e. real particles) are propagators

$$\frac{i}{\vec{p}^2 - m^2 + i\epsilon}.$$

2. Each term in the interaction Lagrangian corresponds to a vertex. Each vertex in a diagram corresponds to a factor of the coupling constant times i in the integral.

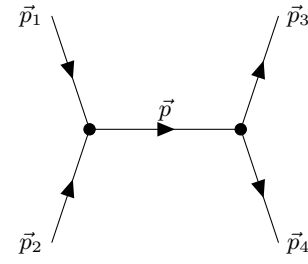


FIG. 8. Feynman diagram for $\mathfrak{S}[s]$.

3. Momentum is conserved at each vertex, so add a delta function enforcing conservation of momentum at that vertex in the integral: $\delta^{(4)}(\vec{p}_{\text{in}} - \vec{p}_{\text{out}})$.
4. Integrate over all internal 4-momenta $\int d^4 p$.
5. Add a factor of $(2\pi)^{-4}$ for every independent loop momentum. An independent loop momentum is one that remains in the integral after the delta functions enforcing momentum conservation are evaluated.
6. Work out the symmetry factor for loops. I show how to do this in Sec. 17.
7. Sum over all diagrams:

$$\delta^{(4)}(\vec{P}_{\text{in}} - \vec{P}_{\text{out}}) i \mathcal{M} = \text{Sum of all Feynman diagrams.}$$

To see how these work let's do an example.

11.2. Example: Φ^3 2 to 2

This example comes from Schwartz Sec. 7.4. The interaction Lagrangian is

$$\mathcal{L}_{\text{int}} = \frac{g^3}{3!} \Phi^3. \quad (404)$$

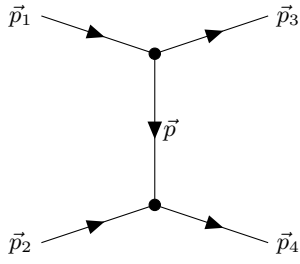
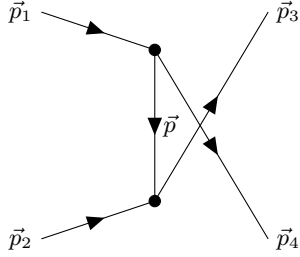
Consider 2 to 2 ($\Phi\Phi \rightarrow \Phi\Phi$) scattering. So two scalar boson particles enter and two leave. The matrix element has three terms

$$\delta^{(4)}(\vec{p}_1 + \vec{p}_2 - \vec{p}_3 - \vec{p}_4) i \mathcal{M} = \mathfrak{S}[s] + \mathfrak{S}[t] + \mathfrak{S}[u], \quad (405)$$

where each term corresponds to a diagram in Figs. 8-10, and each diagram corresponds to an integral. I translate the diagrams into integrals using the Feynman rules.

$$\mathfrak{S}[s] = i(i g)^2 \int d^4 p \frac{\delta^{(4)}(\vec{p}_1 + \vec{p}_2 - \vec{p}) \delta^{(4)}(\vec{p} - \vec{p}_3 - \vec{p}_4)}{\vec{p}^2 - m^2 + i\epsilon} \quad (406)$$

$$\mathfrak{S}[s] = -i g^2 \frac{\delta^{(4)}(\vec{p}_1 + \vec{p}_2 - \vec{p}_3 - \vec{p}_4)}{(\vec{p}_1 + \vec{p}_2)^2 - m^2 + i\epsilon} \quad (407)$$

FIG. 9. Feynman diagram for $\mathfrak{V}[t]$.FIG. 10. Feynman diagram for $\mathfrak{V}[u]$. Notice that the crossing between $\vec{p}_3$ and $\vec{p}_4$ does not have a dot. That means it is not a vertex.

I can condense this, and the other terms, by introducing Mandelstam variables, which are a nice shorthand for 1 to 3 decays and 2 to 2 scattering

$$\begin{aligned} s &= (\vec{p}_1 + \vec{p}_2)^2 = (\vec{p}_3 + \vec{p}_4)^2, \\ t &= (\vec{p}_1 - \vec{p}_3)^2 = (\vec{p}_2 - \vec{p}_4)^2, \\ u &= (\vec{p}_1 - \vec{p}_4)^2 = (\vec{p}_2 - \vec{p}_3)^2. \end{aligned} \quad (408)$$

$$\mathfrak{V}[s] = -ig^2 \frac{\delta^{(4)}(\vec{p}_1 + \vec{p}_2 - \vec{p}_3 - \vec{p}_4)}{s - m^2} \quad (409)$$

$$\mathfrak{V}[t] = i(ig)^2 \int d^4p \frac{\delta^{(4)}(\vec{p}_1 - \vec{p}_3 - \vec{p}) \delta^{(4)}(\vec{p}_2 + \vec{p} - \vec{p}_4)}{\vec{p}^2 - m^2 + i\epsilon} \quad (410)$$

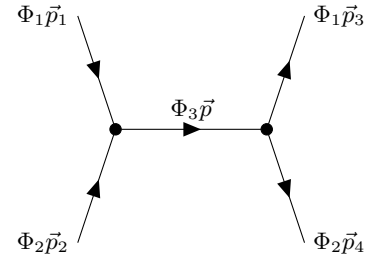
$$\mathfrak{V}[t] = -ig^2 \frac{\delta^{(4)}(\vec{p}_2 + \vec{p}_1 - \vec{p}_3 - \vec{p}_4)}{(\vec{p}_1 - \vec{p}_3)^2 - m^2 + i\epsilon} \quad (411)$$

$$\mathfrak{V}[t] = -ig^2 \frac{\delta^{(4)}(\vec{p}_1 + \vec{p}_2 - \vec{p}_3 - \vec{p}_4)}{t - m^2} \quad (412)$$

$$\mathfrak{V}[u] = i(ig)^2 \int d^4p \frac{\delta^{(4)}(\vec{p}_1 - \vec{p}_4 - \vec{p}) \delta^{(4)}(\vec{p}_2 + \vec{p} - \vec{p}_3)}{\vec{p}^2 - m^2 + i\epsilon} \quad (413)$$

$$\mathfrak{V}[u] = -ig^2 \frac{\delta^{(4)}(\vec{p}_2 + \vec{p}_1 - \vec{p}_4 - \vec{p}_3)}{(\vec{p}_1 - \vec{p}_4)^2 - m^2 + i\epsilon} \quad (414)$$

$$\mathfrak{V}[u] = -ig^2 \frac{\delta^{(4)}(\vec{p}_1 + \vec{p}_2 - \vec{p}_3 - \vec{p}_4)}{u - m^2} \quad (415)$$

FIG. 11. Sole Feynman diagram for leading correction to $\Phi_1 \Phi_2 \rightarrow \Phi_1 \Phi_2$ scattering.

I plug $\mathfrak{V}[s]$, $\mathfrak{V}[t]$, and $\mathfrak{V}[u]$ into Eq. 405.

$$\begin{aligned} \delta^{(4)}(\vec{p}_1 + \vec{p}_2 - \vec{p}_3 - \vec{p}_4) i\mathcal{M} &= -g^2 \frac{i\delta^{(4)}(\vec{p}_1 + \vec{p}_2 - \vec{p}_3 - \vec{p}_4)}{s - m^2} \\ &- g^2 \frac{i\delta^{(4)}(\vec{p}_1 + \vec{p}_2 - \vec{p}_3 - \vec{p}_4)}{t - m^2} - g^2 \frac{i\delta^{(4)}(\vec{p}_1 + \vec{p}_2 - \vec{p}_3 - \vec{p}_4)}{u - m^2} \end{aligned} \quad (416)$$

$$\mathcal{M} = -g^2 \left(\frac{1}{s - m^2} + \frac{1}{t - m^2} + \frac{1}{u - m^2} \right). \quad (417)$$

Wow! That was easy. Remember how much work going through LSZ and the correlation function was. The diagrams really speed things up.

12. DERIVATIVE INTERACTIONS & MULTIPLE SCALAR FIELDS

This section follows Sec. 7.4.2 of Schwartz. Consider the interaction

$$\mathcal{L}_{\text{int}} = \lambda \eta^{\mu\nu} \Phi_1 (\partial_\mu \Phi_2) (\partial_\nu \Phi_3), \quad (418)$$

where Φ_1 , Φ_2 , and Φ_3 are different real scalar fields. Note how the normalization of the coupling constant is different when 3 different fields are involved.

Let's consider $\Phi_1 \Phi_2 \rightarrow \Phi_1 \Phi_2$ scattering, whose leading correction is described by the single diagram in Fig. 11. There is also a new Feynman rule for derivative interactions:

1. We call fields whose derivatives appear in $\mathcal{L}_{\text{int}}$ D-fields. If a D-field with momentum $\vec{p}$ enters a vertex (annihilation) it contributes a factor $-i\vec{p}$. If a D-field with momentum $\vec{k}$ leaves a vertex (creation) it contributes a factor $i\vec{k}$. The final answer must be a scalar so take appropriate dot products of the 4-momenta. Second derivatives, $\square\Phi$, entering or leaving a vertex with momentum $\vec{p}$ contribute $-\vec{p} \cdot \vec{p}$.

$$\vec{P}_i = \vec{p}_1 + \vec{p}_2, \quad \vec{P}_f = \vec{p}_3 + \vec{p}_4 \quad (419)$$

I believe Schwartz has assumed the Φ_3 field is massless (without explicitly stating it) so I do the same.

$$\delta^{(4)}(\vec{P}_i - \vec{P}_f) i\mathcal{M} = i(i\lambda)^2 \int d^4p \frac{\delta^{(4)}(\vec{p}_1 + \vec{p}_2 - \vec{p}) \delta^{(4)}(\vec{p} - \vec{p}_3 - \vec{p}_4)}{\vec{p}^2 + i\epsilon} (\vec{p}_2 \cdot \vec{p}) (\vec{p}_4 \cdot \vec{p}) \quad (420)$$

$$\delta^{(4)}(\vec{P}_i - \vec{P}_f) \mathcal{M} = -\lambda^2 \int d^4p \frac{\delta^{(4)}(\vec{P}_i - \vec{p}) \delta^{(4)}(\vec{p} - \vec{P}_f)}{\vec{p}^2 + i\epsilon} (\vec{p}_2 \cdot \vec{p}) (\vec{p}_4 \cdot \vec{p}) \quad (421)$$

$$\delta^{(4)}(\vec{P}_i - \vec{P}_f) \mathcal{M} = -\lambda^2 \frac{\delta^{(4)}(\vec{P}_i - \vec{P}_f)}{(\vec{P}_i)^2} (\vec{p}_2 \cdot \vec{P}_i) (\vec{p}_4 \cdot \vec{P}_i) \quad (422)$$

$$\mathcal{M} = -\lambda^2 \frac{(\vec{p}_2 \cdot \vec{P}_i) (\vec{p}_4 \cdot \vec{P}_i)}{(\vec{P}_i)^2} \quad (423)$$

$$\mathcal{M} = -\lambda^2 \frac{\vec{p}_2 \cdot \vec{p}_1 + (\vec{p}_2)^2}{(\vec{p}_1 + \vec{p}_2)^2} (\vec{p}_4 \cdot \vec{P}_i) \quad (424)$$

Conservation of momentum means $\vec{P} = \vec{P}_i = \vec{P}_f = \vec{p}_3 + \vec{p}_4$.

$$\mathcal{M} = -\lambda^2 \frac{\vec{p}_2 \cdot \vec{p}_1 + (\vec{p}_2)^2}{(\vec{p}_1 + \vec{p}_2)^2} (\vec{p}_4 \cdot \vec{P}_f) \quad (425)$$

$$\boxed{\mathcal{M} = -\lambda^2 \frac{\vec{p}_2 \cdot \vec{p}_1 + (\vec{p}_2)^2}{\vec{P}^2} [\vec{p}_4 \cdot \vec{p}_3 + (\vec{p}_4)^2]} \quad (426)$$

The answer should not change if we integrate by parts and rewrite the interaction Lagrangian as

$$\begin{aligned} \mathcal{L}_{\text{int}} &= -\lambda \Phi_3 \eta^{\mu\nu} \partial_\nu (\Phi_1 \partial_\mu \Phi_2) = \\ &= -\lambda \Phi_3 [\eta^{\mu\nu} (\partial_\nu \Phi_1) (\partial_\mu \Phi_2) + \Phi_1 \eta^{\mu\nu} \partial_\nu \partial_\mu \Phi_2] = \\ &= -\lambda \Phi_3 [\eta^{\mu\nu} (\partial_\nu \Phi_1) (\partial_\mu \Phi_2) + \Phi_1 \square \Phi_2]. \end{aligned} \quad (427)$$

This is good practice for handling more complicated derivative interactions. Now there are 2 types of vertices which yield 4 diagrams (Figs. 12-15) which yield 4 terms.

$$\delta^{(4)}(\vec{P}_i - \vec{P}_f) i\mathcal{M} = \mathfrak{V}[1] + \mathfrak{V}[2] + \mathfrak{V}[3] + \mathfrak{V}[4]. \quad (428)$$

Note that the sign changed when I integrated by parts. The coupling at each vertex yields a factor. The coupling changed sign because of the integration by parts but this has no effect on two vertex diagrams because the minus cancels. I denote the $\Phi_3(\partial\Phi_1)(\partial\Phi_2)$ vertex with a circle and the $\Phi_3\Phi_1\square\Phi_2$ vertex with a square. I solve the integral in each diagram

$$\begin{aligned} \mathfrak{V}[1] &= i(i\lambda)^2 (-i\vec{p}_1) \cdot (-i\vec{p}_2) (i\vec{p}_3) \cdot (i\vec{p}_4) \\ &= \int d^4p \frac{\delta^{(4)}(\vec{p}_1 + \vec{p}_2 - \vec{p})}{\vec{p}^2 + i\epsilon} \delta^{(4)}(\vec{p} - \vec{p}_3 - \vec{p}_4) = \\ &= -i \frac{\lambda^2}{\vec{P}^2} \delta^{(4)}(\vec{P}_i - \vec{P}_f) (\vec{p}_1 \cdot \vec{p}_2) (\vec{p}_3 \cdot \vec{p}_4) \end{aligned} \quad (429)$$

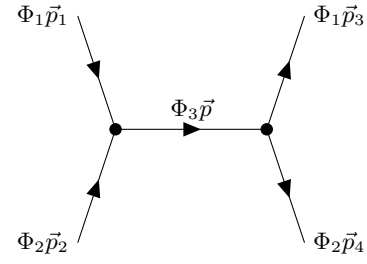


FIG. 12. Feynman diagram $\mathfrak{V}[1]$.

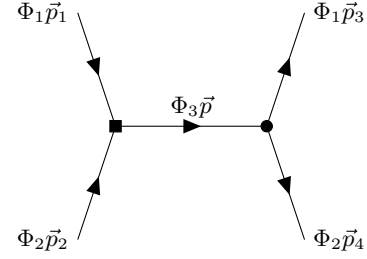


FIG. 13. Feynman diagram $\mathfrak{V}[2]$.

$$\begin{aligned} \mathfrak{V}[2] &= i(i\lambda)^2 (i\vec{p}_3) \cdot (i\vec{p}_4) [-(\vec{p}_2)^2] \\ &= \int d^4p \frac{\delta^{(4)}(\vec{P}_i - \vec{p}) \delta^{(4)}(\vec{p} - \vec{P}_f)}{\vec{p}^2 + i\epsilon} = \\ &= -i \frac{\lambda^2}{\vec{P}^2} \delta^{(4)}(\vec{P}_i - \vec{P}_f) (\vec{p}_3 \cdot \vec{p}_4) (\vec{p}_2)^2 \end{aligned} \quad (430)$$

$$\begin{aligned} \mathfrak{V}[3] &= i(i\lambda)^2 [-(\vec{p}_4)^2] (-i\vec{p}_1) \cdot (-i\vec{p}_2) \\ &= \int d^4p \frac{\delta^{(4)}(\vec{P}_i - \vec{p}) \delta^{(4)}(\vec{p} - \vec{P}_f)}{\vec{p}^2 + i\epsilon} = \\ &= -i \frac{\lambda^2}{\vec{P}^2} (\vec{p}_4)^2 \delta^{(4)}(\vec{P}_i - \vec{P}_f) (\vec{p}_1 \cdot \vec{p}_2) \end{aligned} \quad (431)$$

$$\begin{aligned} \mathfrak{V}[4] &= i(i\lambda)^2 [-(\vec{p}_2)^2] [-(\vec{p}_4)^2] \\ &= \int d^4p \frac{\delta^{(4)}(\vec{P}_i - \vec{p}) \delta^{(4)}(\vec{p} - \vec{P}_f)}{\vec{p}^2 + i\epsilon} = \\ &= -i \frac{\lambda^2}{\vec{P}^2} (\vec{p}_2)^2 (\vec{p}_4)^2 \delta^{(4)}(\vec{P}_i - \vec{P}_f) \end{aligned} \quad (432)$$

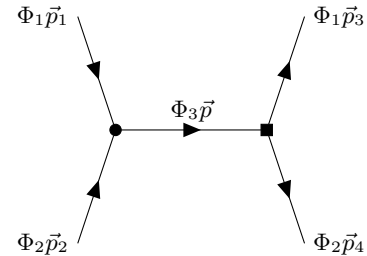
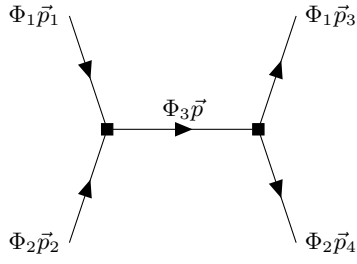


FIG. 14. Feynman diagram $\mathfrak{V}[3]$.


 FIG. 15. Feynman diagram $\mathfrak{F}[4]$.

I add all the diagrams together.

$$\mathcal{M} = -\lambda^2 \frac{\vec{p}_2 \cdot \vec{p}_1 + (\vec{p}_2)^2}{\vec{P}^2} [\vec{p}_4 \cdot \vec{p}_3 + (\vec{p}_4)^2] \quad (433)$$

This is exactly the same as Eq. 426, which it must be because integration by parts does not affect the action.

Part III

Path Integrals

13. GAUSSIAN INTEGRALS

In the path integral formulation multidimensional Gaussian integrals come up frequently. So in this section I will present a general formula for dealing with them. Consider the 1D real Gaussian integral

$$I = \int_{-\infty}^{\infty} dp \exp\left(Jp - \frac{a}{2}p^2\right), \quad a > 0. \quad (434)$$

Completing the square yields

$$I = \int_{-\infty}^{\infty} dp \exp\left[\frac{J^2}{2a} - \frac{a}{2}\left(p - \frac{J}{a}\right)^2\right] = \exp\left(\frac{J^2}{2a}\right) \int_{-\infty}^{\infty} dp \exp\left[-\frac{a}{2}\left(p - \frac{J}{a}\right)^2\right]. \quad (435)$$

Shifting the dummy variable $p \rightarrow p - J/a$ does not affect the value of the integrand

$$I = \exp\left(\frac{J^2}{2a}\right) \int_{-\infty}^{\infty} dp e^{-ap^2/2}. \quad (436)$$

I substitute

$$x = \sqrt{\frac{a}{2}}p, \quad x(\pm\infty) = \pm\infty, \quad \sqrt{\frac{2}{a}}dx = dp, \quad (437)$$

$$I = \sqrt{\frac{2}{a}} \exp\left(\frac{J^2}{2a}\right) \int_{-\infty}^{\infty} dx e^{-x^2}. \quad (438)$$

I have proved the famous result

$$\int_{-\infty}^{\infty} dx e^{-x^2} = \sqrt{\pi}, \quad (439)$$

many times and I will not do it again here. The final result is

$$\boxed{\int_{-\infty}^{\infty} dp \exp\left(Jp - \frac{a}{2}p^2\right) = \sqrt{\frac{2\pi}{a}} \exp\left(\frac{J^2}{2a}\right)}. \quad (440)$$

A good exercise would be to generalize this result to complex J , p , and a and see if there is any constraint on a , like it must be real.

The generalization to n dimensions is

$$\boxed{\int d\vec{p} \exp\left(\vec{J}^\dagger \vec{p} - \frac{\vec{p}^\dagger \mathbf{A} \vec{p}}{2}\right) = \sqrt{\frac{(2\pi)^n}{\det \mathbf{A}}} \exp\left(\frac{\vec{J}^\dagger \mathbf{A}^{-1} \vec{J}}{2}\right)}. \quad (441)$$

This result comes from Schwartz Eq. 14.7. I made a Mathematica script to test it for the 2D real case. Another fun

exercise would be to prove it analytically for the 2D complex case. In the real case I found that $\mathbf{A}$ must be symmetric. I wonder if Schwartz left implicit the assumption that $\mathbf{A}$ must be Hermitian? An analytic derivation would tell me. If n is continuous infinity $\vec{p}$ and $\vec{J}$ become functions, $\mathbf{A}$ becomes a differential operator $\mathcal{A}$, and the integral becomes a path integral

$$\int \mathcal{D}p(y) \exp\left[\int dx \left(J^*(x)p(x) - \frac{p^*(x)\mathcal{A}p(x)}{2}\right)\right] = \mathcal{B} \exp\left[\frac{1}{2} \int dx dy J^*(x)\mathcal{G}(x-y)J(y)\right]. \quad (442)$$

The constant $\mathcal{B}$ is nonsensical. For one thing $(2\pi)^n$ is infinite if n is infinite. Moreover $\det \mathcal{A}$ is the product of the eigenvalues of $\mathcal{A}$. If $\mathcal{A} = \square$ the eigenvalues are every real number. That includes 0 so maybe $\det \mathcal{A} = 0$ or maybe it is infinity. Luckily in path integral applications $\mathcal{B}$ cancels out of observables. $\mathcal{G}$ is the inverse of $\mathcal{A}$: $\mathcal{G} = \mathcal{A}^{-1} = 1/\mathcal{A}$. The inverse of a number is defined by $a/a = 1$. The inverse of a matrix is defined by $\mathbf{A}\mathbf{A}^{-1} = I$ where I is the identity matrix. The inverse of a differential operator satisfies

$$\mathcal{A}_x \mathcal{G}(x-y) = \delta(x-y). \quad (443)$$

So $\mathcal{G}$ is a Green's function. The generalization of Ref. 442 to functions of several variables is obvious.

14. NONRELATIVISTIC PATH INTEGRAL

Before formulating the path integral in QFT let's review it in nonrelativistic 1D quantum mechanics with the Hamiltonian

$$\mathcal{H}(t) = \frac{p^2}{2m} + V(t, x). \quad (444)$$

The initial state $|x_i, t_i\rangle$ is a position eigenstate at time t_i . The final state is $|x_f, t_f\rangle$. I want the amplitude for a particle to start at the initial state and end at the final state. Actually that is not quite what I want. What I want is the amplitude between the final state and the particle's state if it had stayed put (just like in Sec. 8)

$$A = \langle x_f, t_f | x_i, t_f \rangle = \langle x_f | x_i \rangle. \quad (445)$$

If the Hamiltonian was time independent the time evolution operator would be $e^{-i\mathcal{H}\Delta t}$ and

$$A = \langle x_f, t_f | x_i, t_f \rangle = \langle x_f, t_f | e^{-i(t_f-t_i)\mathcal{H}} | x_i, t_i \rangle. \quad (446)$$

This S-matrix element isn't that difficult to calculate.

But since $\mathcal{H}(t)$ is a function of time I need to break the amplitude into infinitesimal time slices over which $\mathcal{H}(t)$ is roughly constant. Let $U(t, t_0)$ be the actual time evolution operator and set $t_i = 0$ and $t_f = t$.

$$A = \langle x_f, t | x_i, t \rangle = \langle x_f, t | U(t, 0) | x_i, 0 \rangle. \quad (447)$$

Break the time evolution operator into infinitesimal pieces

$$U(t, 0) = U(t, \delta t)U(\delta t, 0) = \prod_{n=0}^{N-1} U[(n+1)\delta t, n\delta t], \quad (448)$$

I take $t_n = n\delta t$ and $t = (N+1)\delta t = t_{N+1}$

$$U(t, 0) = \prod_{n=0}^N U(t_{n+1}, t_n). \quad (449)$$

In each time slice the Hamiltonian is approximately constant

$$U(t_{n+1}, t_n) = e^{-i(t_{n+1}-t_n)\mathcal{H}(t_n)} = e^{-i\delta t\mathcal{H}(t_n)}, \quad (450)$$

$$U(t, 0) = \prod_{n=0}^N e^{-i\delta t\mathcal{H}(t_n)}. \quad (451)$$

I use the closure relation

$$U(t, 0) = \left[\prod_{n=1}^N e^{-i\delta t\mathcal{H}(t_n)} \right] \int_{-\infty}^{\infty} dx_1 |x_1\rangle \langle x_1| e^{-i\delta t\mathcal{H}(0)} \quad (452)$$

$$U(t, 0) = \prod_{n=0}^N \int_{-\infty}^{\infty} dx_{n+1} |x_{n+1}\rangle \langle x_{n+1}| e^{-i\delta t\mathcal{H}(t_n)} = \int dx_1 \dots dx_N e^{-i\delta t\mathcal{H}(t_N)} \prod_{n=0}^{N-1} |x_{n+1}\rangle \langle x_{n+1}| e^{-i\delta t\mathcal{H}(t_n)}. \quad (453)$$

I plug into the amplitude

$$A = \int dx_1 \dots dx_N \langle x_f, t | e^{-i\delta t\mathcal{H}(t_N)} | x_N \rangle \langle x_N | \dots | x_2 \rangle \langle x_2 | e^{-i\delta t\mathcal{H}(1)} | x_1 \rangle \langle x_1 | e^{-i\delta t\mathcal{H}(0)} | x_i, 0 \rangle. \quad (454)$$

To get a more compact expression take $|x_{N+1}\rangle = |x_f\rangle$ and $|x_0\rangle = |x_i, 0\rangle$,

$$A = \int dx_1 \dots dx_N \langle x_{N+1} | e^{-i\delta t\mathcal{H}(t_N)} | x_N \rangle \dots \langle x_1 | e^{-i\delta t\mathcal{H}(0)} | x_0 \rangle. \quad (455)$$

$$A = \int dx_1 \dots dx_N \prod_{n=0}^N M_n. \quad (456)$$

where

$$M_n = \langle x_{n+1} | e^{-i\delta t\mathcal{H}(t_n)} | x_n \rangle. \quad (457)$$

By the way time evolution operators do not commute at different times which implies a particular order in which the product symbol generates new factors: later times appear on the left.

I evaluate the M_n using the momentum closure relation. I follow Schwartz's convention for momentum eigenstates

$$\langle p | x \rangle = e^{-ipx}. \quad (458)$$

Usually I do not mark out operators with a hat because context is enough to differentiate operators and their corresponding eigenvalues (classical quantities) but this is one case where explicitly marking operators is good

$$M_n = \langle x_{n+1} | \exp \left[-i\delta t \left(\frac{\hat{p}^2}{2m} + V(t, \hat{x}) \right) \right] | x_n \rangle. \quad (459)$$

$$M_n = \int_{-\infty}^{\infty} dp \langle x_{n+1} | p \rangle \langle p | \exp \left[-i\delta t \left(\frac{\hat{p}^2}{2m} + V(t, \hat{x}) \right) \right] | x_n \rangle. \quad (460)$$

I use $\hat{x} | x \rangle = x | x \rangle$ and $\langle p | \hat{p} = \langle p | p$

$$M_n = e^{-i\delta t V(t, x_n)} \int_{-\infty}^{\infty} dp \exp \left(-i \frac{p^2 \delta t}{2m} \right) \langle x_{n+1} | p \rangle \langle p | x_n \rangle. \quad (461)$$

$$M_n = e^{-i\delta t V(t, x_n)} \int_{-\infty}^{\infty} dp \exp \left(-i \frac{p^2 \delta t}{2m} \right) e^{ip(x_{n+1} - x_n)} \quad (462)$$

$$M_n = e^{-i\delta t V(t, x_n)} \int_{-\infty}^{\infty} dp \exp \left[i(x_{n+1} - x_n)p - \frac{i\delta t}{2m} p^2 \right] \quad (463)$$

This is a 1D Gaussian integral which I evaluate with Eq. 440

$$M_n = b e^{-i\delta t V(t, x_n)} \exp \left[\frac{im}{2\delta t} (x_{n+1} - x_n)^2 \right] \quad (464)$$

where

$$b = \sqrt{\frac{2\pi m}{i\delta t}}. \quad (465)$$

$$M_n = b \exp \left[i\delta t \left(\frac{(x_{n+1} - x_n)^2}{2\delta t^2} m - V(t, x_n) \right) \right] \quad (466)$$

Each x_n must be integrated over all space, so there is no reason for x_{n+1} and x_n to be close together. And yet they were put in by using the closure relation on adjacent time slices, so they are the position of the particle before and after the infinitesimal time interval δt . It is reasonable to assume their difference is infinitesimal $\delta x_n = x_{n+1} - x_n$.

$$M_n = b \exp \left[i\delta t \left(\frac{\delta x_n^2}{2\delta t^2} m - V(t, x_n) \right) \right] \quad (467)$$

Clearly

$$\frac{\delta x}{\delta t} = \frac{dx}{dt} = \dot{x}. \quad (468)$$

$$M_n = b \exp \left[i \delta t \left(\frac{m}{2} \dot{x}_n^2 - V(t, x_n) \right) \right] \quad (469)$$

We recognize the Lagrangian

$$\mathcal{L}(\dot{x}, x) = \frac{m}{2} \dot{x}_n^2 - V(t, x). \quad (470)$$

$$M_n = b e^{i \mathcal{L}(\dot{x}_n, x_n) \delta t} \quad (471)$$

I plug into Eq. 456

$$A = b^{N+1} \int dx_1 \dots dx_N \prod_{n=0}^N e^{i \mathcal{L}(\dot{x}_n, x_n) \delta t}. \quad (472)$$

Remember that δt is infinitesimal so the product is over an infinite number of factors and $B = b^{N+1} = b^\infty$ is infinite. Luckily this infinite constant always cancels out when we evaluate observables.

$$A = B \int dx_1 \dots dx_N \prod_{n=0}^N e^{i \mathcal{L}(\dot{x}_n, x_n) \delta t} \quad (473)$$

This expression is the path integral, which we express with the more compact notation

$$A = B \int_{x(t_i)=x_i}^{x(t_f)=x_f} \mathcal{D}x(t) \exp \left[i \int_{t_i}^{t_f} dt \mathcal{L}(\dot{x}, x) \right]. \quad (474)$$

It is an integral over all classical paths $x(t)$. Notice that they are not operators (which is why I explicitly marked operators with hats). This interpretation of Eq. 473 is not obvious and for the moment I have to take it on faith. I don't have the time to go through the argument why this is true. Pay attention to the bounds of integration which constrain the particle between fixed points in space and time. We recognize the action

$$S[x] = \int_{t_i}^{t_f} dt \mathcal{L}(\dot{x}, x), \quad (475)$$

and use the even more compact expression

$$\langle x_f | x_i \rangle = B \int_{x(t_i)=x_i}^{x(t_f)=x_f} \mathcal{D}x(t) e^{i S[x]}. \quad (476)$$

Presumably this is the simplest correlation function in non-relativistic quantum mechanics. There are also higher order correlation functions expressed as path integrals and all observables can be expressed with correlation functions in the path integral formulation. But I won't get into that because my main interest is QFT path integrals.

15. QFT PATH INTEGRALS

Just like with path integrals in quantum mechanics (Sec. 14) I distinguish between operators and classical fields by

putting hats on operators. The analogues of $\hat{x} |x\rangle = x |x\rangle$ and $\hat{p} |p\rangle = p |p\rangle$ in QFT are

$$\hat{\phi}(\mathbf{x}) |\phi(\mathbf{x})\rangle = \phi(\mathbf{x}) |\phi(\mathbf{x})\rangle, \quad \hat{\pi}(\mathbf{x}) |\pi(\mathbf{x})\rangle = \pi(\mathbf{x}) |\pi(\mathbf{x})\rangle. \quad (477)$$

Note that I am working in the Schrödinger picture (Sec. 2). So far we have only seen vacuum states like $|0\rangle$. $|\phi(\mathbf{x})\rangle$ is something different: it corresponds to the field having a well defined classical value everywhere in space at some particular time. Just like $|x\rangle$ is a quantum state with a well defined classical position $|\phi(\mathbf{x})\rangle$ is a field with a well defined classical profile.

The QFT analogue of $\langle p | x \rangle = e^{-ipx}$ is

$$\langle \pi(\mathbf{x}) | \phi(\mathbf{x}) \rangle = \exp \left[-i \int d^3x \phi(\mathbf{x}) \pi(\mathbf{x}) \right]. \quad (478)$$

The analogue of $\langle y | x \rangle = \delta(x - y)$ is

$$\begin{aligned} \langle \phi_1(\mathbf{x}) | \phi(\mathbf{x}) \rangle &= \int \mathcal{D}\pi(\mathbf{x}) \langle \phi_1(\mathbf{x}) | \pi(\mathbf{x}) \rangle \langle \pi(\mathbf{x}) | \phi(\mathbf{x}) \rangle = \\ &= \int \mathcal{D}\pi(\mathbf{x}) \exp \left[i \int d^3x \pi(\mathbf{x}) [\phi_1(\mathbf{x}) - \phi(\mathbf{x})] \right]. \end{aligned} \quad (479)$$

Note the completeness relation. It will be fun to prove these identities when I have more time.

The classical Lagrangian density is

$$\mathcal{L} = \frac{\dot{\Phi}(\vec{x})^2}{2} - \frac{[\nabla \Phi(\vec{x})]^2}{2} - V[\Phi](\vec{x}). \quad (480)$$

The potential energy functional V might have spacetime dependence independent of $\Phi(\vec{x})$: e.g. $e^t \Phi(\vec{x})^2$. It might also include interactions. Remember we are using Φ for interacting fields and ϕ for free fields. Also remember that the free and interacting Schrödinger picture operators are identical (Eq. 314). We combine the gradient and potential energies into a generalized potential

$$\mathcal{V} = \frac{[\nabla \Phi(\vec{x})]^2}{2} + V[\Phi](\vec{x}), \quad (481)$$

so

$$\mathcal{L} = \frac{\dot{\Phi}(\vec{x})^2}{2} - \mathcal{V}(\vec{x}), \quad \mathcal{H} = \frac{\pi(\vec{x})^2}{2} + \mathcal{V}(\vec{x}). \quad (482)$$

The Hamiltonian density operator is

$$\hat{\mathcal{H}} = \frac{\hat{\pi}(\mathbf{x})^2}{2} + \mathcal{V}[\hat{\phi}(\mathbf{x})]. \quad (483)$$

I am still working in the Schrödinger picture and the Hamiltonian density might have intrinsic time dependence. The Hamiltonian is

$$\hat{\mathcal{H}}(t) = \int d^3x \hat{\mathcal{H}}. \quad (484)$$

Let $U(t, t_0)$ be the Schrödinger time evolution operator. I seek the amplitude

$$\langle \Omega, t_f | \Omega, t_f \rangle = \langle \Omega, t_f | U(t_f, t_i) | \Omega, t_i \rangle. \quad (485)$$

I am working from Schwartz but he calculates $\langle 0, t_f | 0, t_i \rangle$ in Eq. 14.26. A similar thing happened in Sec. 14. Also there is more tension between our treatments: 0 should only be used for the free vacuum and I am confused about whether Schwartz is working in the Heisenberg or Schrödinger pictures. I think the correct way in this case is the Schrödinger picture (for one thing we both associate states with time). It's very likely that I got something wrong, but that's okay: these are my notes not a publication. I will continue as best I can and revise later. For the moment though I stop here. I will finish this calculation later when I have more time. Because of the conceptual difficulties and time crunch I move on to describe only what I need in the immediate moment for calculations: the expression for correlation functions.

The two point correlation function is given by the path integrals

$$\langle \Omega | T \Phi(\vec{x}_1) \Phi(\vec{x}_2) | \Omega \rangle = \frac{\int \mathcal{D}\Phi(\vec{x}) e^{iS[\Phi]} \Phi(\vec{x}_1) \Phi(\vec{x}_2)}{\int \mathcal{D}\Phi(\vec{x}) e^{iS[\Phi]}}, \quad (486)$$

where

$$S[\Phi] = \int d^4x \mathcal{L}. \quad (487)$$

I again drop the hats, because it is clear that fields in path integrals are classical and fields in vacuum matrix elements are operators. The generalization to the N point correlation function is obvious.

16. GENERATING FUNCTIONAL

The generating functional is

$$Z[J] = \int \mathcal{D}\Phi(\vec{y}) \exp \left[iS[\Phi] + i \int d^4x J(\vec{x}) \Phi(\vec{x}) \right]. \quad (488)$$

Then

$$Z[0] = \int \mathcal{D}\Phi(\vec{x}) e^{iS[\Phi]}. \quad (489)$$

Consider functional derivatives with respect to J . From DFT I know how to take the functional derivative of a local functional. Putting that DFT background to work I get

$$\begin{aligned} \frac{\delta J(\vec{y})}{\delta J(\vec{x})} &= \frac{\delta}{\delta J(\vec{x})} \int d^4x \delta^{(4)}(\vec{x} - \vec{y}) J(\vec{x}) = \\ \frac{\partial}{\partial J(\vec{x})} \delta^{(4)}(\vec{x} - \vec{y}) J(\vec{x}) &= \delta^{(4)}(\vec{x} - \vec{y}) \frac{\partial J(\vec{x})}{\partial J(\vec{x})} = \delta^{(4)}(\vec{x} - \vec{y}). \end{aligned} \quad (490)$$

$$\begin{aligned} \frac{\delta}{\delta J(\vec{x})} \int d^4x J(\vec{x}) \Phi(\vec{x}) &= \frac{\partial}{\partial J(\vec{x})} J(\vec{x}) \Phi(\vec{x}) = \\ \Phi(\vec{x}) \frac{\partial J(\vec{x})}{\partial J(\vec{x})} &= \Phi(\vec{x}) \end{aligned} \quad (491)$$

Now consider

$$\frac{\delta Z[J]}{\delta J(\vec{x}_1)} = \frac{\delta}{\delta J(\vec{x}_1)} \int \mathcal{D}\Phi(\vec{y}) \exp \left[iS[\Phi] + i \int d^4x J(\vec{x}) \Phi(\vec{x}) \right]. \quad (492)$$

I did not learn how to take the functional derivative of a path integral in DFT class. I know the final answer from Schwartz Eq. 14.46, but I am not completely sure about the steps in this derivation. I will make a guess and fix it later if I am wrong.

$$\frac{\delta Z[J]}{\delta J(\vec{x}_1)} = \int \mathcal{D}\Phi(\vec{y}) \frac{\delta}{\delta J(\vec{x}_1)} \exp \left[iS[\Phi] + i \int d^4x J(\vec{x}) \Phi(\vec{x}) \right] \quad (493)$$

I use the chain rule

$$\begin{aligned} \frac{\delta Z[J]}{\delta J(\vec{x}_1)} &= \int \mathcal{D}\Phi(\vec{y}) \exp \left[iS[\Phi] + i \int d^4x J(\vec{x}) \Phi(\vec{x}) \right] \\ &\quad \left[i \frac{\delta S[\Phi]}{\delta J(\vec{x}_1)} + i \frac{\delta}{\delta J(\vec{x}_1)} \int d^4x J(\vec{x}) \Phi(\vec{x}) \right], \end{aligned} \quad (494)$$

$$\begin{aligned} -i \frac{\delta Z[J]}{\delta J(\vec{x}_1)} &= \int \mathcal{D}\Phi(\vec{y}) \exp \left[iS[\Phi] + i \int d^4x J(\vec{x}) \Phi(\vec{x}) \right] \\ &\quad \frac{\delta}{\delta J(\vec{x}_1)} \int d^4x J(\vec{x}) \Phi(\vec{x}), \end{aligned} \quad (495)$$

$$\begin{aligned} -i \frac{\delta Z[J]}{\delta J(\vec{x}_1)} &= \int \mathcal{D}\Phi(\vec{y}) \exp \left[iS[\Phi] + i \int d^4x J(\vec{x}) \Phi(\vec{x}) \right] \\ &\quad \Phi(\vec{x}_1). \end{aligned} \quad (496)$$

Then

$$\left. -i \frac{\delta Z[J]}{\delta J(\vec{x}_1)} \right|_{J=0} = \int \mathcal{D}\Phi(\vec{y}) e^{iS[\Phi]} \Phi(\vec{x}_1). \quad (497)$$

Dividing by $Z[0]$ and comparing with Eq. 486 I get

$$\begin{aligned} - \frac{i}{Z[0]} \left. \frac{\delta Z[J]}{\delta J(\vec{x}_1)} \right|_{J=0} &= \frac{1}{Z[0]} \int \mathcal{D}\Phi(\vec{y}) e^{iS[\Phi]} \Phi(\vec{x}_1) = \\ &\quad \langle \Omega | \Phi(\vec{x}_1) | \Omega \rangle. \end{aligned} \quad (498)$$

The two point function is

$$\langle \Omega | T \Phi(\vec{x}_1) \Phi(\vec{x}_2) | \Omega \rangle = - \frac{1}{Z[0]} \left. \frac{\delta^2 Z[J]}{\delta J(\vec{x}_1) \delta J(\vec{x}_2)} \right|_{J=0}. \quad (499)$$

The generalization to higher order correlation functions is obvious.

16.1. Example: Free Field

The free field Lagrangian is $2\mathcal{L} = -\phi(\square + m^2)\phi$. The free field generating functional is

$$Z[J] = \int \mathcal{D}\phi(\vec{y}) \exp \left[i \int d^4x [\mathcal{L} + J(\vec{x})\phi(\vec{x})] \right], \quad (500)$$

$$Z[J] = \int \mathcal{D}\phi(\vec{y}) \exp \left[i \int d^4x \left(J(\vec{x})\phi(\vec{x}) - \frac{\phi(\vec{x})(\square + m^2)\phi(\vec{x})}{2} \right) \right]. \quad (501)$$

This is a Gaussian integral I can evaluate analytically. I distribute i

$$Z[J] = \int \mathcal{D}\phi(\vec{y}) \exp \left[\int d^4x \left(iJ(\vec{x})\phi(\vec{x}) - \frac{\phi(\vec{x})i(\square + m^2)\phi(\vec{x})}{2} \right) \right]. \quad (502)$$

I define $\mathcal{J} = -iJ$ and $\mathcal{A} = i(\square + m^2)$

$$Z[J] = \int \mathcal{D}\phi(\vec{y}) \exp \left[\int d^4x \left(\mathcal{J}^*(\vec{x})\phi(\vec{x}) - \frac{\phi(\vec{x})\mathcal{A}\phi(\vec{x})}{2} \right) \right]. \quad (503)$$

Comparing to Eq. 442 I get

$$Z[J] = \mathcal{B} \exp \left[\frac{1}{2} \int d^4x d^4y \mathcal{J}^*(\vec{x})\mathcal{G}(\vec{x} - \vec{y})\mathcal{J}(\vec{y}) \right], \quad (504)$$

$$Z[J] = \mathcal{B} \exp \left[\frac{1}{2} \int d^4x d^4y J(\vec{x})\mathcal{G}(\vec{x} - \vec{y})J(\vec{y}) \right], \quad (505)$$

The constant $\mathcal{B}$ is probably infinite (see Sec. 13) but fortunately drops out of all observables. The Green's function is

$$\mathcal{G} = \frac{1}{\mathcal{A}} = \frac{1}{i(\square + m^2)} = -\frac{i}{\square + m^2}. \quad (506)$$

Defining

$$\mathcal{G} = \frac{1}{\square + m^2}, \quad (507)$$

yields $\mathcal{G} = -i\mathcal{G}$,

$$Z[J] = \mathcal{B} \exp \left[-\frac{i}{2} \int d^4x d^4y J(\vec{x})\mathcal{G}(\vec{x} - \vec{y})J(\vec{y}) \right]. \quad (508)$$

This is an exact closed-form expression for $Z[J]$. It completely defines the free field theory, just like the partition

function gives us everything in statistical mechanics. Life would be good if we could get such a thing for every QFT (we can't). The last loose end is the Green's function, which satisfies

$$(\square_x + m^2)\mathcal{G}(\vec{x} - \vec{y}) = \delta^{(4)}(\vec{x} - \vec{y}). \quad (509)$$

$\mathcal{G}$ is a propagator

$$(2\pi)^4 \mathcal{G}(\vec{x} - \vec{y}) = - \int d^4p \frac{e^{i\vec{p} \cdot (\vec{x} - \vec{y})}}{p^2 - m^2}. \quad (510)$$

I won't prove this, because I already analyzed propagators very similar to this one in Sec. 5. I just quote the result from Schwartz Eq. (14.53). Unlike the propagators I studied in Sec. 5 this one does not have a subscript like R for retarded or F for Feynman. Why? Because it is unbounded, meaning I did not put in a boundary condition. To get sensible physical results I actually do need to do that, and Schwartz Sec. 14.4 discusses how this is done. This is another one of those things I will do when I have the time. These notes are for a class (Physics 523) that only touches on path integrals and focuses on loops. That's why this chapter on path integrals as been sparser than the rest of the notes.

I compute the 2 point function using Eq. 499

$$\langle 0|T\phi_1\phi_2|0\rangle = -\frac{1}{Z[0]} \frac{\delta^2 Z[J]}{\delta J_1 \delta J_2} \Big|_{J=0} \quad (511)$$

where I use the shorthand $f(\vec{x}_n) = f_n$.

$$\frac{\delta Z}{\delta J_2} = \mathcal{B} \frac{\delta}{\delta J_2} \exp \left[-\frac{i}{2} \int d^4x d^4y J(\vec{x})\mathcal{G}(\vec{x} - \vec{y})J(\vec{y}) \right] \quad (512)$$

I do not know how to take this functional derivative but I try it anyway assuming that functional and ordinary derivatives treat exponentials the the same way and that the chain rule still applies. I use the shorthand

$$H[J] = \int d^4x d^4y J(\vec{x})\mathcal{G}(\vec{x} - \vec{y})J(\vec{y}). \quad (513)$$

$$\frac{\delta Z}{\delta J_2} = \mathcal{B} \frac{\delta}{\delta J_2} e^{-iH[J]/2} = -\mathcal{B} \frac{i}{2} e^{-iH[J]/2} \frac{\delta H}{\delta J_2} \quad (514)$$

$$\frac{\delta^2 Z}{\delta J_1 \delta J_2} = -\mathcal{B} \frac{i}{2} \frac{\delta}{\delta J_1} e^{-iH/2} \frac{\delta H}{\delta J_2} \quad (515)$$

I assume functional derivatives respect the product rule.

$$\frac{\delta^2 Z}{\delta J_1 \delta J_2} = -\mathcal{B} \frac{i}{2} \left(\frac{\delta H}{\delta J_2} \frac{\delta}{\delta J_1} e^{-iH/2} + e^{-iH/2} \frac{\delta^2 H}{\delta J_1 \delta J_2} \right) \quad (516)$$

$$\frac{\delta^2 Z}{\delta J_1 \delta J_2} = -\mathcal{B} \frac{i}{2} e^{-iH/2} \left(\frac{\delta^2 H}{\delta J_1 \delta J_2} - \frac{i}{2} \frac{\delta H}{\delta J_1} \frac{\delta H}{\delta J_2} \right) \quad (517)$$

$$\left. \frac{\delta^2 Z}{\delta J_1 \delta J_2} \right|_{J=0} = -\mathcal{B} \frac{i}{2} e^{-iH[0]/2} \left[\frac{\delta^2 H}{\delta J_1 \delta J_2} - \frac{i}{2} \frac{\delta H}{\delta J_1} \frac{\delta H}{\delta J_2} \right]_{J=0} \quad (518)$$

$$H[0] = 0 \quad (519)$$

$$\left. \frac{\delta^2 Z}{\delta J_1 \delta J_2} \right|_{J=0} = -\mathcal{B} \frac{i}{2} \left[\frac{\delta^2 H}{\delta J_1 \delta J_2} - \frac{i}{2} \frac{\delta H}{\delta J_1} \frac{\delta H}{\delta J_2} \right]_{J=0} \quad (520)$$

The non-local functional $H[J]$ has the same form as the Hartree energy in DFT. Using my DFT background (see Eqs. 2.16 & 2.17 in Kieron Burke's DFT textbook) I get

$$\frac{\delta H}{\delta J_1} = 2 \int d^4x \mathcal{G}(\vec{x} - \vec{x}_1) J(\vec{x}) \implies \left. \frac{\delta H}{\delta J_1} \right|_{J=0} = 0. \quad (521)$$

$$\left. \frac{\delta^2 Z}{\delta J_1 \delta J_2} \right|_{J=0} = -i\mathcal{B} \left[\frac{\delta^2 H}{\delta J_1 \delta J_2} - \frac{i}{2} \frac{\delta H}{\delta J_1} \frac{\delta H}{\delta J_2} \right]_{J=0} \quad (522)$$

$$\left. \frac{\delta^2 Z}{\delta J_1 \delta J_2} \right|_{J=0} = -i\mathcal{B} \left. \frac{\delta^2 H}{\delta J_1 \delta J_2} \right|_{J=0} \quad (523)$$

$$\frac{\delta^2 H}{\delta J_1 \delta J_2} = \frac{\delta}{\delta J_1} \frac{\delta H}{\delta J_2} = 2 \frac{\delta}{\delta J_1} \int d^4x \mathcal{G}(\vec{x} - \vec{x}_2) J(\vec{x}) = 2 \mathcal{G}(\vec{x}_1 - \vec{x}_2) \quad (524)$$

$$\left. \frac{\delta^2 Z}{\delta J_1 \delta J_2} \right|_{J=0} = -i\mathcal{B} \mathcal{G}(\vec{x}_1 - \vec{x}_2) \quad (525)$$

$$Z[0] = \mathcal{B} \quad (526)$$

The final result is

$$\langle 0 | T \phi_1 \phi_2 | 0 \rangle = i \mathcal{G}(\vec{x}_1 - \vec{x}_2). \quad (527)$$

This matches Eq. 14.55 in Schwartz. Like I said above putting in the boundary conditions rigorously is tough. I introduce a simple shortcut here: just replace $i\mathcal{G} \rightarrow \mathcal{G}_F$ where $\mathcal{G}_F$ is the Feynman propagator in Eq. 246. So the properly bounded final result is

$$\boxed{\langle 0 | T \phi(\vec{x}_1) \phi(\vec{x}_2) | 0 \rangle = \mathcal{G}_F(\vec{x}_1 - \vec{x}_2)}. \quad (528)$$

This matches my result in Eq. 302.

16.2. Example Φ^3 Theory

The Lagrangian is

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_1. \quad (529)$$

The free and interacting parts are

$$2\mathcal{L}_0 = -\Phi(\square + m^2)\Phi, \quad \mathcal{L}_1 = \frac{g}{3!} \Phi^3. \quad (530)$$

The action is

$$S = \int d^4x \mathcal{L} = S_0 + S_1, \quad (531)$$

$$S_0 = \int d^4x \mathcal{L}_0, \quad S_1 = \int d^4x \mathcal{L}_1. \quad (532)$$

The generating functional is

$$Z[J] = \int \mathcal{D}\Phi(\vec{y}) \exp \left[iS + i \int d^4x J(\vec{x}) \Phi(\vec{x}) \right]. \quad (533)$$

$$I[J] = \int d^4x J(\vec{x}) \Phi(\vec{x}) \quad (534)$$

$$Z = \int \mathcal{D}\Phi(\vec{y}) e^{i(S+I)} \quad (535)$$

$$Z = \int \mathcal{D}\Phi(\vec{y}) e^{i(S_0+S_1+I)} \quad (536)$$

$$Z = \int \mathcal{D}\Phi(\vec{y}) e^{i(S_0+I)} e^{iS_1} \quad (537)$$

I expand perturbatively assuming the interaction is weak

$$e^{iS_1} = 1 + iS_1 - \frac{S_1^2}{2} + \mathcal{O}. \quad (538)$$

$$e^{iS_1} = 1 + g \frac{i}{6} \int d^4x \Phi(\vec{x})^3 - \frac{g^2}{72} \left[\int d^4x \Phi(\vec{x})^3 \right]^2 \quad (539)$$

$$e^{iS_1} = 1 + g \frac{i}{6} \int d^4x \Phi(\vec{x})^3 - \frac{g^2}{72} \int d^4x d^4y \Phi(\vec{x})^3 \Phi(\vec{y})^3 \quad (540)$$

$$Z = \int \mathcal{D}\Phi(\vec{y}) e^{i(S_0+I)} + g \frac{i}{6} \int \mathcal{D}\Phi(\vec{y}) e^{i(S_0+I)} \int d^4x \Phi(\vec{x})^3 - \frac{g^2}{72} \int \mathcal{D}\Phi(\vec{y}) e^{i(S_0+I)} \int d^4x d^4z \Phi(\vec{x})^3 \Phi(\vec{z})^3 \quad (541)$$

The free generating functional is

$$Z_0[J] = \int \mathcal{D}\Phi(\vec{y}) e^{i(S_0+I)}. \quad (542)$$

$$Z = Z_0 + g \frac{i}{6} \int \mathcal{D}\Phi(\vec{y}) e^{i(S_0+I)} \int d^4x \Phi(\vec{x})^3 - \frac{g^2}{72} \int \mathcal{D}\Phi(\vec{y}) e^{i(S_0+I)} \int d^4x d^4z \Phi(\vec{x})^3 \Phi(\vec{z})^3 \quad (543)$$

Now comes a trick: I commute the functional (path) and regular integrals.

$$Z = Z_0 + g \frac{i}{6} \int d^4x \int \mathcal{D}\Phi(\vec{y}) e^{i(S_0+I)} \Phi(\vec{x})^3 - \frac{g^2}{72} \int d^4x d^4z \int \mathcal{D}\Phi(\vec{y}) e^{i(S_0+I)} \Phi(\vec{x})^3 \Phi(\vec{z})^3 \quad (544)$$

Next another trick: functional differentiation under the functional integral sign! Eq. 496 yields

$$-i \frac{\delta Z_0}{\delta J(\vec{x})} = \int \mathcal{D}\Phi(\vec{y}) e^{i(S_0+I)} \Phi(\vec{x}), \quad (545)$$

$$(-i)^n \frac{\delta^n Z_0}{\delta J(\vec{x})^n} = \int \mathcal{D}\Phi(\vec{y}) e^{i(S_0+I)} \Phi(\vec{x})^n, \quad (546)$$

$$Z = Z_0 - \frac{g}{6} \int d^4x \frac{\delta^3 Z_0}{\delta J(\vec{x})^3} + \frac{g^2}{72} \int d^4x d^4z \frac{\delta^6 Z_0}{\delta J(\vec{x})^3 \delta J(\vec{z})^3}. \quad (547)$$

I use the shorthand $f(\vec{x}) = f_x$

$$Z = Z_0 - \frac{g}{6} \int d^4x \frac{\delta^3 Z_0}{\delta J_x^3} + \frac{g^2}{72} \int d^4x d^4z \frac{\delta^6 Z_0}{\delta J_x^3 \delta J_z^3}. \quad (548)$$

Now I have the generating function to the needed order. I will use it to compute the 2 point correlation function $C = C(\vec{x}_1, \vec{x}_2)$ where

$$C = \langle \Omega | T \Phi_1 \Phi_2 | \Omega \rangle = -\frac{1}{Z[0]} \frac{\delta^2 Z}{\delta J_1 \delta J_2} \Big|_{J=0} \quad (549)$$

$$C = -\frac{1}{Z[0]} \left[\frac{\delta^2 Z_0}{\delta J_1 \delta J_2} - \frac{g}{6} \int d^4x \frac{\delta^5 Z_0}{\delta J_1 \delta J_2 \delta J_x^3} + \frac{g^2}{72} \int d^4x d^4y \frac{\delta^8 Z_0}{\delta J_1 \delta J_2 \delta J_x^3 \delta J_y^3} \right]_{J=0} \quad (550)$$

I expand $Z[0]$ perturbatively using Eq. 548.

$$Z[0] = Z_0[0] - \frac{g}{6} \int d^4x \frac{\delta^3 Z_0}{\delta J_x^3} \Big|_{J=0} + \frac{g^2}{72} \int d^4x d^4z \frac{\delta^6 Z_0}{\delta J_x^3 \delta J_z^3} \Big|_{J=0} \quad (551)$$

I found $Z_0[J]$ analytically in Eq. 508

$$Z_0[J] = \mathcal{B} e^{-iH[J]/2}, \quad Z_0[0] = \mathcal{B}, \quad (552)$$

where I give $H[J]$ in Eq. 513.

$$Z[0] = \mathcal{B} - \frac{g}{6} \int d^4x \frac{\delta^3 Z_0}{\delta J_x^3} \Big|_{J=0} + \frac{g^2}{72} \int d^4x d^4z \frac{\delta^6 Z_0}{\delta J_x^3 \delta J_z^3} \Big|_{J=0} \quad (553)$$

$$\frac{\delta^3 Z_0}{\delta J_x^3} = \mathcal{B} \frac{\delta^3}{\delta J_x^3} e^{-iH[J]/2} \quad (554)$$

$$\frac{\delta^3 Z_0}{\delta J_x^3} = \frac{\mathcal{B}}{2} e^{-iH[J]/2} \left[\frac{i}{4} \left(\frac{\delta H}{\delta J_x} \right)^3 - \frac{3}{2} \frac{\delta H}{\delta J_x} \frac{\delta^2 H}{\delta J_x^2} - \cancel{\frac{i \delta^3 H}{\delta J_x^3}} \right] \quad (555)$$

$$\frac{\delta^3 Z_0}{\delta J_x^3} = \frac{\mathcal{B}}{2} e^{-iH[J]/2} \left[\frac{i}{4} \left(\frac{\delta H}{\delta J_x} \right)^3 - \frac{3}{2} \frac{\delta H}{\delta J_x} \frac{\delta^2 H}{\delta J_x^2} \right] \quad (556)$$

$$\frac{\delta^3 Z_0}{\delta J_x^3} \Big|_{J=0} = \frac{\mathcal{B}}{2} \left[\frac{i}{4} \left(\frac{\delta H}{\delta J_x} \right)^3 - \frac{3}{2} \frac{\delta H}{\delta J_x} \frac{\delta^2 H}{\delta J_x^2} \right]_{J=0} \quad (557)$$

Eq. 521 yields

$$\frac{\delta^3 Z_0}{\delta J_x^3} \Big|_{J=0} = \frac{\mathcal{B}}{2} \left[\frac{i}{4} \left(\frac{\delta H}{\delta J_x} \right)^3 - \frac{3}{2} \frac{\delta H}{\delta J_x} \frac{\delta^2 H}{\delta J_x^2} \right]_{J=0} = 0. \quad (558)$$

$$Z[0] = \mathcal{B} - \frac{g}{6} \int d^4x \frac{\delta^3 Z_0}{\delta J_x^3} \Big|_{J=0} + \frac{g^2}{72} \int d^4x d^4z \frac{\delta^6 Z_0}{\delta J_x^3 \delta J_z^3} \Big|_{J=0} \quad (559)$$

$$Z[0] = \mathcal{B} + \frac{g^2}{72} \int d^4x d^4y \frac{\delta^6 Z_0}{\delta J_x^3 \delta J_y^3} \Big|_{J=0} \quad (560)$$

The derivatives are starting to have an obscenely large number of terms. Luckily we see that only second derivatives will survive. This is the analogue of only full contractions contributing to correlation functions in canonical quantization (Sec. 9.1).

$$\frac{\delta^6 Z_0}{\delta J_x^3 \delta J_y^3} \Big|_{J=0} = \frac{3}{4} i \mathcal{B} \delta_{xy} H \left[\delta_{xy} H^2 + \frac{3}{2} \delta_{xx} H \delta_{yy} H \right] \quad (561)$$

where none of the surviving terms depend on J and I use the shorthand

$$\frac{\delta}{\delta J_x} = \delta_x. \quad (562)$$

$$\delta_y H = 2 \int d^4x \mathcal{G}(\vec{x} - \vec{y}) J(\vec{x}) \quad (563)$$

$$\delta_{xy} H = 2 \mathcal{G}(\vec{x} - \vec{y}) = 2 \mathcal{G}^{xy} \quad (564)$$

Where we use the shorthand $\mathcal{G}(\vec{x} - \vec{y}) = \mathcal{G}^{xy}$.

$$\begin{aligned} \frac{\delta_{yy} H}{2} &= \frac{\delta}{\delta J(\vec{y})} \int d^4x \mathcal{G}(\vec{x} - \vec{y}) J(\vec{x}) = \\ &\int d^4z \delta^{(4)}(\vec{z} - \vec{y}) \frac{\delta}{\delta J(\vec{y})} \int d^4x \mathcal{G}(\vec{x} - \vec{z}) J(\vec{x}) = \\ &\int d^4z \delta^{(4)}(\vec{z} - \vec{y}) \mathcal{G}(\vec{y} - \vec{z}) = \mathcal{G}(\vec{y} - \vec{y}) = \\ &\mathcal{G}(0) = \mathcal{G}^{yy} = \mathcal{G}^{xx} \end{aligned} \quad (565)$$

$$\left. \frac{\delta^6 Z_0}{\delta J_x^3 \delta J_y^3} \right|_{J=0} = 3i\mathcal{B}\mathcal{G}^{xy}[2(\mathcal{G}^{xy})^2 + 3\mathcal{G}^{xx}\mathcal{G}^{yy}] \quad (566)$$

I calculated this correlation function in the canonical commutation formalism in Sec. 10.1. I do not write $\mathcal{G}^{xx}\mathcal{G}^{yy} = \mathcal{G}(0)^2$ because I want my expressions in both calculations to be identical.

$$Z[0] = \mathcal{B} + \frac{g^2}{24}i\mathcal{B} \int d^4x d^4y \mathcal{G}^{xy}[2(\mathcal{G}^{xy})^2 + 3\mathcal{G}^{xx}\mathcal{G}^{yy}] \quad (567)$$

These propagators are unbounded. I turn them into Feynman propagators by hand like I discussed in Sec. 16.1: $\mathcal{G} \rightarrow -i\mathcal{G}_F$.

$$Z[0] = \mathcal{B} - \frac{g^2}{24}\mathcal{B} \int d^4x d^4y \mathcal{G}_F^{xy}[2(\mathcal{G}_F^{xy})^2 + 3\mathcal{G}_F^{xx}\mathcal{G}_F^{yy}] \quad (568)$$

$$Z[0] = \mathcal{B} - \frac{\mathcal{B}}{12}g^2 \int d^4x d^4y (\mathcal{G}_F^{xy})^3 - \frac{\mathcal{B}}{8}g^2 \int d^4x d^4y \mathcal{G}_F^{xy}\mathcal{G}_F^{xx}\mathcal{G}_F^{yy} \quad (569)$$

I label these integrals with the same labels I used the first time I calculated this correlation function (Eq. 354).

$$\frac{Z[0]}{\mathcal{B}} = 1 - \frac{\mathfrak{V}[0]}{8} - \frac{\mathfrak{V}[1]}{12} \quad (570)$$

I calculate the remaining derivatives in C (Eq. 550).

$$\begin{aligned} \frac{\delta^2 Z_0[J]}{\delta J_1 \delta J_2} &= \mathcal{B} \frac{\delta^2}{\delta J_1 \delta J_2} e^{-iH[J]/2} = \\ &- \frac{\mathcal{B}}{2} e^{-iH[J]/2} \left(i \frac{\delta^2 H}{\delta J_1 \delta J_2} + \frac{1}{2} \frac{\delta H}{\delta J_1} \frac{\delta H}{\delta J_2} \right) \end{aligned} \quad (571)$$

I simplify using shorthands

$$\frac{\delta^2 Z_0[J]}{\delta J_1 \delta J_2} = -\frac{\mathcal{B}}{2} e^{-iH[J]/2} \left(i\delta_{12}H + \frac{\delta_1 H \delta_2 H}{2} \right), \quad (572)$$

by this point the meaning of my shorthands is obvious.

$$\left. \frac{\delta^2 Z_0}{\delta J_1 \delta J_2} \right|_{J=0} = -\frac{\mathcal{B}}{2} \left[i\delta_{12}H + \frac{\delta_1 H \delta_2 H}{2} \right]_{J=0} \quad (573)$$

$$\left. \frac{\delta^2 Z_0}{\delta J_1 \delta J_2} \right|_{J=0} = -\frac{\mathcal{B}}{2} i\delta_{12}H_{J=0} \quad (574)$$

$$\delta_{12}H = 2\mathcal{G}^{12} \quad (575)$$

$$\left. \frac{\delta^2 Z_0}{\delta J_1 \delta J_2} \right|_{J=0} = -\mathcal{B}i\mathcal{G}^{12} \quad (576)$$

I swap unbounded with Feynman propagators.

$$\left. \frac{\delta^2 Z_0}{\delta J_1 \delta J_2} \right|_{J=0} = -\mathcal{B}\mathcal{G}_F^{12} \quad (577)$$

I relabel using the labels of Sec. 10.1 (Eqs. 354 & 361).

$$\left. \frac{\delta^2 Z_0}{\delta J_1 \delta J_2} \right|_{J=0} = -\mathcal{B}\mathfrak{V}[0] \quad (578)$$

$$\left. \frac{\delta^5 Z_0}{\delta J_1 \delta J_2 \delta J_x^3} \right|_{J=0} = 0 \quad (579)$$

$$\begin{aligned} \frac{g^2}{\mathcal{B}} \int d^4x d^4y \frac{\delta^8 Z_0}{\delta J_1 \delta J_2 \delta J_x^3 \delta J_y^3} \Big|_{J=0} = \\ 18(2\mathfrak{V}[1] + 2\mathfrak{V}[3] + \mathfrak{V}[2]) + 3(2\mathfrak{V}[1] + 3\mathfrak{V}[0])\mathfrak{V}[0] \end{aligned} \quad (580)$$

For these last two terms I did the algebra in Mathematica because the number of terms is astronomical. Just remember that the Feynman propagator is even.

Plugging Eqs. 578, 579, & 580 into Eq. 550 yields

$$C = \frac{\mathcal{B}}{Z[0]} \left[\mathfrak{V}[0] - \frac{6(2\mathfrak{V}[1] + 2\mathfrak{V}[3] + \mathfrak{V}[2]) + (2\mathfrak{V}[1] + 3\mathfrak{V}[0])\mathfrak{V}[0]}{24} \right]. \quad (581)$$

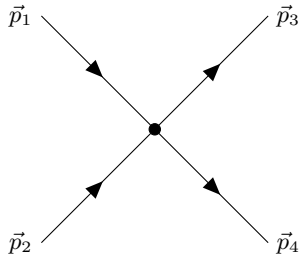
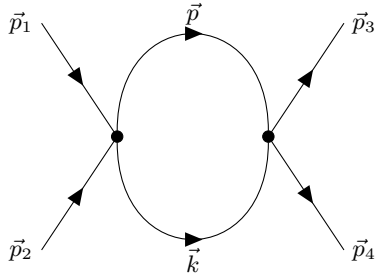
I plug in Eq. 570

$$C = \left(1 - \frac{\mathfrak{V}[0]}{8} - \frac{\mathfrak{V}[1]}{12} \right)^{-1} \left[\mathfrak{V}[0] - \frac{6(2\mathfrak{V}[1] + 2\mathfrak{V}[3] + \mathfrak{V}[2]) + (2\mathfrak{V}[1] + 3\mathfrak{V}[0])\mathfrak{V}[0]}{24} \right]. \quad (582)$$

I expand to order g^2

$$C(\vec{x}_1, \vec{x}_2) = \mathfrak{V}[0] - \frac{1}{2} \left(\mathfrak{V}[1] + \frac{\mathfrak{V}[2]}{2} + \mathfrak{V}[3] \right). \quad (583)$$

This is exactly what I got with canonical quantization in Eq. 368. Both formulations of QFT give the same answer but I found path integrals easier to use.

FIG. 16. $\star$ FIG. 17. $\mathfrak{F}$

Part IV

Loops

17. SIMPLE LOOP EXAMPLE

To get a feel for loops, divergences, and renormalization let's look at a simple example from Schwartz Sec. 15.4. Consider a real massless scalar field with Lagrangian

$$\mathcal{L} = -\frac{\Phi \square \Phi}{2} - \frac{\lambda}{4!} \Phi^4. \quad (584)$$

Let's look at $\Phi\Phi \rightarrow \Phi\Phi$ scattering. To lowest order there is only one diagram $\star$ (Fig. 16)

$$\star = -i\lambda \delta^{(4)}(\vec{P} - \vec{K}), \quad (585)$$

where $\vec{P} = \vec{p}_1 + \vec{p}_2$ and $\vec{K} = \vec{p}_3 + \vec{p}_4$. The leading correction comes at second order and has three diagrams (s , t , and u channels). But I follow Schwartz and only look at the s diagram $\mathfrak{F}$ (Fig. 17). The momentum space Feynman rules from Sec. 11.1 yield

$$\mathfrak{F} = \frac{(-i\lambda)^2 S}{(2\pi)^{4n}} \int d^4k d^4p \frac{i}{\vec{p}^2 + i\epsilon} \frac{i}{\vec{k}^2 + i\epsilon} \delta^{(4)}(\vec{p}_1 + \vec{p}_2 - \vec{p} - \vec{k}) \delta^{(4)}(\vec{p} + \vec{k} - \vec{p}_3 - \vec{p}_4), \quad (586)$$

where n is the number of independent loop momenta and S is the symmetry factor. To do the integral I have to determine both of these quantities. To keep the procedure transparent I did not simplify this integral: I want it to be obvious how to put together all of the pieces from the

Feynman rules. Now I evaluate it.

$$\mathfrak{F} = \frac{\lambda^2 S}{(2\pi)^{4n}} \int d^4k d^4p \frac{\delta^{(4)}(\vec{P} - \vec{p} - \vec{k})}{(\vec{p}^2 + i\epsilon)(\vec{k}^2 + i\epsilon)} \delta^{(4)}(\vec{p} + \vec{k} - \vec{K}) \quad (587)$$

$$\mathfrak{F} = \frac{\lambda^2 S}{(2\pi)^{4n}} \delta^{(4)}(\vec{P} - \vec{K}) \int \frac{d^4p}{(\vec{p}^2 + i\epsilon)[(\vec{P} - \vec{p})^2 + i\epsilon]} \quad (588)$$

I have evaluated the vertex delta functions and got an overall delta function enforcing momentum conservation for the scattering process, as expected. But one loop momentum remains in the integral. Therefore there is one independent loop momentum: $n = 1$.

$$\mathfrak{F} = \frac{\lambda^2 S}{(2\pi)^4} \delta^{(4)}(\vec{P} - \vec{K}) \int \frac{d^4p}{(\vec{p}^2 + i\epsilon)[(\vec{P} - \vec{p})^2 + i\epsilon]} \quad (589)$$

The symmetry factor ultimately comes from Wick's theorem (Sec. 9) contractions. To find it put the integral into the schematic form $\langle 12|\Phi\Phi\Phi\Phi|\Phi\Phi\Phi\Phi|34 \rangle$, where 1234 represent the external particles and each vertex contributes Φ^4 , because the interaction is Φ^4 . Then draw contractions that represent the connections in the Feynman diagram (Fig. 18). When connecting particle 2 to a vertex I have 2 vertices to choose from. In each vertex there are 4 fields to connect to. So there are a total of 8 ways to make this contraction. Particle 1 has to be connected to the same vertex as particle 2, and there are 3 ways to do this. Now the logic for associating numbers with contractions is obvious. The symmetry factor is

$$S = \frac{8 \times 4 \times 3^2 \times 2}{(4!)^2 2!} = \frac{1}{2}. \quad (590)$$

The numerator comes from multiplying all the numbers in Fig. 18. In the denominator the $2!$ comes from permuting the 2 four particle vertices and the two $4!$ factors come from permuting the fields in each vertex. The rule for computing symmetry factors is now obvious. The result is

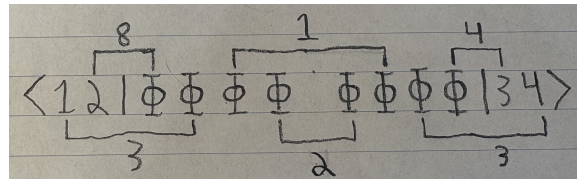


FIG. 18. Determining the symmetry factor.

$$\mathfrak{F} = \frac{\lambda^2 \delta^{(4)}(\vec{P} - \vec{K})}{2(2\pi)^4} \int \frac{d^4p}{(\vec{p}^2 + i\epsilon)[(\vec{P} - \vec{p})^2 + i\epsilon]}. \quad (591)$$

This integral has the implied limit $\epsilon \rightarrow 0_+$. I believe it is proper to take this limit after solving the integral. But I follow Schwartz's lead and take it now

$$\mathfrak{Y} = \frac{\lambda^2}{2(2\pi)^4} \delta^{(4)}(\vec{P} - \vec{K}) \int \frac{d^4 p}{\vec{p}^2 (\vec{P} - \vec{p})^2}. \quad (592)$$

This integral is over all of 4-momentum space and it obviously diverges as $\vec{p}$ becomes large

$$\mathfrak{Y} \sim \int_{-\infty}^{\infty} \frac{dp}{p}. \quad (593)$$

I define

$$\mathfrak{Y} = \delta^{(4)}(\vec{P} - \vec{K}) \mathfrak{Q}, \quad (594)$$

so

$$\mathfrak{Q} = \frac{\lambda^2}{2(2\pi)^4} \int \frac{d^4 p}{\vec{p}^2 (\vec{P} - \vec{p})^2}. \quad (595)$$

The Mandelstam variable s is

$$s = \vec{P}^2 = \vec{K}^2. \quad (596)$$

We cure the divergence with a trick.

$$\frac{\partial \mathfrak{Q}}{\partial P^\mu} = \frac{\partial \mathfrak{Q}}{\partial s} \frac{\partial s}{\partial P^\mu} \quad (597)$$

$$\begin{aligned} \frac{\partial s}{\partial P^\mu} &= \frac{\partial}{\partial P^\mu} P_\beta P^\beta = \frac{\partial}{\partial P^\mu} \eta_{\alpha\beta} P^\alpha P^\beta = \eta_{\alpha\beta} \frac{\partial}{\partial P^\mu} P^\alpha P^\beta \\ &= \eta_{\alpha\beta} P^\beta \frac{\partial P^\alpha}{\partial P^\mu} + \eta_{\alpha\beta} P^\alpha \frac{\partial P^\beta}{\partial P^\mu} \end{aligned} \quad (598)$$

$$\frac{\partial P^\alpha}{\partial P^\mu} = \delta_\mu^\alpha \quad (599)$$

$$\frac{\partial s}{\partial P^\mu} = P_\alpha \delta_\mu^\alpha + P_\beta \delta_\mu^\beta = 2P_\mu \quad (600)$$

$$2P_\mu \frac{\partial \mathfrak{Q}}{\partial s} = \frac{\partial \mathfrak{Q}}{\partial P^\mu} \quad (601)$$

$$2P^\mu P_\mu \frac{\partial \mathfrak{Q}}{\partial s} = P^\mu \frac{\partial \mathfrak{Q}}{\partial P^\mu} \quad (602)$$

$$2s \frac{\partial \mathfrak{Q}}{\partial s} = P^\mu \frac{\partial \mathfrak{Q}}{\partial P^\mu} \quad (603)$$

$$\frac{\partial \mathfrak{Q}}{\partial s} = \frac{P^\mu}{2s} \frac{\partial \mathfrak{Q}}{\partial P^\mu} \quad (604)$$

$$\frac{\partial \mathfrak{Q}}{\partial P^\mu} = \frac{\lambda^2}{2(2\pi)^4} \int \frac{d^4 p}{\vec{p}^2} \frac{\partial}{\partial P^\mu} (\vec{P} - \vec{p})^{-2} \quad (605)$$

$$\begin{aligned} \frac{\partial}{\partial P^\mu} (\vec{P} - \vec{p})^{-2} &= \frac{\partial}{\partial P^\mu} (\vec{P}^2 + \vec{p}^2 - 2\vec{P} \cdot \vec{p})^{-1} = \\ &= -(\vec{P}^2 + \vec{p}^2 - 2\vec{P} \cdot \vec{p})^{-2} \frac{\partial}{\partial P^\mu} (\vec{P}^2 + \vec{p}^2 - 2\vec{P} \cdot \vec{p}) = \\ &= -\frac{1}{(\vec{P} - \vec{p})^4} \frac{\partial}{\partial P^\mu} (\vec{P}^2 - 2\vec{P} \cdot \vec{p}) \end{aligned} \quad (606)$$

$$\frac{\partial \mathfrak{Q}}{\partial P^\mu} = -\frac{\lambda^2}{2(2\pi)^4} \int \frac{d^4 p}{\vec{p}^2 (\vec{P} - \vec{p})^4} \frac{\partial}{\partial P^\mu} (\vec{P}^2 - 2\vec{P} \cdot \vec{p}) \quad (607)$$

$$\begin{aligned} \frac{\partial}{\partial P^\mu} (\vec{P}^2 - 2\vec{P} \cdot \vec{p}) &= \frac{\partial \vec{P}^2}{\partial P^\mu} - 2 \frac{\partial}{\partial P^\mu} \vec{P} \cdot \vec{p} = \frac{\partial s}{\partial P^\mu} - 2p_\beta \frac{\partial P^\beta}{\partial P^\mu} \\ &= 2P_\mu - 2p_\mu \end{aligned} \quad (608)$$

$$\frac{\partial \mathfrak{Q}}{\partial P^\mu} = -\frac{\lambda^2}{(2\pi)^4} \int d^4 p \frac{P_\mu - p_\mu}{\vec{p}^2 (\vec{P} - \vec{p})^4} \quad (609)$$

$$\frac{\partial \mathfrak{Q}}{\partial s} = -\frac{\lambda^2}{2(2\pi)^4 s} \int d^4 p \frac{P^\mu P_\mu - P^\mu p_\mu}{\vec{p}^2 (\vec{P} - \vec{p})^4} \quad (610)$$

$$\frac{\partial \mathfrak{Q}}{\partial s} = -\frac{\lambda^2}{2(2\pi)^4 s} \int d^4 p \frac{s - \vec{P} \cdot \vec{p}}{\vec{p}^2 (\vec{P} - \vec{p})^4} \quad (611)$$

Now the integral is convergent. I rewrite it as

$$I = -i \frac{\partial \mathfrak{Q}}{\partial s} = \frac{i\lambda^2}{2(2\pi)^4 s} \int d^4 p \frac{s - \vec{P} \cdot \vec{p}}{\vec{p}^2 (\vec{P} - \vec{p})^4}. \quad (612)$$

Schwartz does not explain how to evaluate it, he simply gives the answer. That is what I will do. I might return to this and do the integral later but the important part was making it convergent.

$$I = -\frac{\lambda^2}{32\pi^2 s}. \quad (613)$$

What happened to the i ? How did this integral become real? Maybe this is a consequence of taking the $\epsilon \rightarrow 0_+$ limit prematurely. I will just go with it and blindly follow Schwartz. Using this result I get $\mathfrak{Q}$.

$$\frac{\partial \mathfrak{Q}}{\partial s} = iI \quad (614)$$

$$\frac{\partial \mathfrak{Q}}{\partial s} = -\frac{\lambda^2 i}{32\pi^2 s} \quad (615)$$

$$\mathfrak{Q} = -\frac{\lambda^2 i}{32\pi^2} \int \frac{ds}{s} \quad (616)$$

$$\mathfrak{Q} = -\frac{\lambda^2 i}{32\pi^2} \ln s + C \quad (617)$$

C is an integration constant. Because $\mathfrak{C}$ is divergent C must be infinite. This is progress though since we have extracted the divergent piece, and separated it from the finite part. Like integral surgery. It looks like Schwartz has some typos at this step but following him as best we can (making sure to correct his mistakes) we set

$$C = \frac{\lambda^2 i}{32\pi^2} \ln \Lambda^2. \quad (618)$$

The infinite part is isolated in Λ . Then

$$\mathfrak{C} = -\frac{\lambda^2 i}{32\pi^2} \ln s + \frac{\lambda^2 i}{32\pi^2} \ln \Lambda^2, \quad (619)$$

$$\boxed{\mathfrak{C} = -\frac{\lambda^2 i}{32\pi^2} \ln \left(\frac{s}{\Lambda^2} \right)}. \quad (620)$$

The mass dimension of s is 2: $[s] = 2$. Since the argument of a logarithm must be dimensionless $[\Lambda] = 1$. Plugging into Eq. 594 yields the diagram

$$\boxed{\mathfrak{G} = -\delta^{(4)}(\vec{P} - \vec{K}) \frac{\lambda^2 i}{32\pi^2} \ln \left(\frac{s}{\Lambda^2} \right)}. \quad (621)$$

Now I get the amplitude to one loop order (with the t and u channels artificially set to 0 to simplify the math).

$$\delta^{(4)}(\vec{P} - \vec{K}) i\mathcal{M} = \mathfrak{X} + \mathfrak{G} \quad (622)$$

$$\cancel{\delta^{(4)}(\vec{P} - \vec{K})} i\mathcal{M} = -\cancel{\delta^{(4)}(\vec{P} - \vec{K})} \left[\lambda + \frac{\lambda^2}{32\pi^2} \ln \left(\frac{s}{\Lambda^2} \right) \right] \quad (623)$$

$$\boxed{\mathcal{M} = -\lambda - \frac{\lambda^2}{32\pi^2} \ln \left(\frac{s}{\Lambda^2} \right)} \quad (624)$$

This is exactly what Schwartz got in Eq. (15.63). So that is how the divergent part in loop integral/amplitude can be isolated.

17.1. Renormalization

Let's use the amplitude in Eq. 624 to show how renormalization works. Notice that $\mathcal{M} = \mathcal{M}(s)$ is infinite, but the difference

$$\begin{aligned} \mathcal{M}(s_1) - \mathcal{M}(s_2) &= -\cancel{\lambda} - \frac{\lambda^2}{32\pi^2} \ln \left(\frac{s_1}{\Lambda^2} \right) + \cancel{\lambda} + \frac{\lambda^2}{32\pi^2} \ln \left(\frac{s_2}{\Lambda^2} \right) \\ &= \frac{\lambda^2}{32\pi^2} \left[\ln \left(\frac{s_2}{\Lambda^2} \right) - \ln \left(\frac{s_1}{\Lambda^2} \right) \right] = \frac{\lambda^2}{32\pi^2} \ln \left(\frac{s_2 \Lambda^2}{s_1 \Lambda^2} \right) = \\ &= \frac{\lambda^2}{32\pi^2} \ln \left(\frac{s_2}{s_1} \right), \end{aligned} \quad (625)$$

is finite because the divergent Λ cancels.

The coupling constant λ is the strength of the interaction. We do not measure it directly. We measure scattering cross sections. So let us define the renormalized coupling λ_R as

$$\lambda_R = -\mathcal{M}(s_0), \quad (626)$$

where s_0 is some standard value of s we pick. To one loop order

$$\lambda_R = \lambda + \frac{\lambda^2}{32\pi^2} \ln \left(\frac{s_0}{\Lambda^2} \right). \quad (627)$$

Because λ_R is observable it must be finite. The infinity has gone into the bare coupling λ . We expand λ as a perturbative series in λ_R

$$\lambda = \lambda^{(0)} + \Delta\lambda^{(2)} + \mathcal{O}(3) = \lambda_R + a\lambda_R^2 + \mathcal{O}(\lambda_R^3). \quad (628)$$

To lowest order

$$\boxed{\lambda^{(0)} = \lambda_R}. \quad (629)$$

I use Eq. 627 to get the first correction.

$$\lambda_R = \lambda^{(1)} + \beta \frac{\lambda^{(1)2}}{32\pi^2} \ln \left(\frac{s_0}{\Lambda^2} \right) \quad (630)$$

The β is a parameter to track the order. At the end of a calculation it is set to 1.

$$\lambda_R = \lambda^{(0)} + \beta\Delta\lambda^{(1)} + \beta \frac{(\lambda^{(0)} + \beta\Delta\lambda^{(1)})^2}{32\pi^2} \ln \left(\frac{s_0}{\Lambda^2} \right) \quad (631)$$

I only expand to first order in β .

$$\lambda_R = \lambda^{(0)} + \beta\Delta\lambda^{(1)} + \beta \frac{\lambda^{(0)2}}{32\pi^2} \ln \left(\frac{s_0}{\Lambda^2} \right) \quad (632)$$

$$\cancel{\lambda_R} = \cancel{\lambda_R} + \beta\Delta\lambda^{(1)} + \beta \frac{\lambda_R^2}{32\pi^2} \ln \left(\frac{s_0}{\Lambda^2} \right) \quad (633)$$

$$0 = \beta\Delta\lambda^{(1)} + \beta \frac{\lambda_R^2}{32\pi^2} \ln \left(\frac{s_0}{\Lambda^2} \right) \quad (634)$$

The first correction is

$$\boxed{\Delta\lambda^{(1)} = -\frac{\lambda_R^2}{32\pi^2} \ln \left(\frac{s_0}{\Lambda^2} \right)}. \quad (635)$$

One would think that a perturbative correction would be small. But this is infinity big. That's QFT for you.

I use the expansion in λ and Eq. 624 to get a finite perturbative expansion for the amplitude. First I do lowest order.

$$\mathcal{M}^{(0)} = -\lambda^{(0)} \quad (636)$$

$$\boxed{\mathcal{M}^{(0)} = -\lambda_R} \quad (637)$$

Now the leading correction.

$$\mathcal{M}^{(1)}(s) = \mathcal{M}^{(0)} + \beta \Delta \mathcal{M}^{(1)}(s) \quad (638)$$

$$\mathcal{M}^{(1)}(s) = -\lambda^{(1)} - \beta \frac{\lambda^{(0)2}}{32\pi^2} \ln\left(\frac{s}{\Lambda^2}\right) \quad (639)$$

$$\mathcal{M}^{(1)}(s) = -\lambda^{(0)} - \beta \Delta \lambda^{(1)} - \beta \frac{\lambda^{(0)2}}{32\pi^2} \ln\left(\frac{s}{\Lambda^2}\right) \quad (640)$$

$$\mathcal{M}^{(1)}(s) = -\lambda_R - \beta \Delta \lambda^{(1)} - \beta \frac{\lambda_R^2}{32\pi^2} \ln\left(\frac{s}{\Lambda^2}\right) \quad (641)$$

$$\mathcal{M}^{(0)} + \beta \Delta \mathcal{M}^{(1)}(s) = -\lambda_R - \beta \Delta \lambda^{(1)} - \beta \frac{\lambda_R^2}{32\pi^2} \ln\left(\frac{s}{\Lambda^2}\right) \quad (642)$$

$$\cancel{\text{X}}_R + \beta \Delta \mathcal{M}^{(1)}(s) = \cancel{\text{X}}_R - \beta \Delta \lambda^{(1)} - \beta \frac{\lambda_R^2}{32\pi^2} \ln\left(\frac{s}{\Lambda^2}\right) \quad (643)$$

$$\beta \Delta \mathcal{M}^{(1)}(s) = -\beta \Delta \lambda^{(1)} - \beta \frac{\lambda_R^2}{32\pi^2} \ln\left(\frac{s}{\Lambda^2}\right) \quad (644)$$

$$\Delta \mathcal{M}^{(1)}(s) = -\Delta \lambda^{(1)} - \frac{\lambda_R^2}{32\pi^2} \ln\left(\frac{s}{\Lambda^2}\right) \quad (645)$$

$$\Delta \mathcal{M}^{(1)}(s) = \frac{\lambda_R^2}{32\pi^2} \ln\left(\frac{s_0}{\Lambda^2}\right) - \frac{\lambda_R^2}{32\pi^2} \ln\left(\frac{s}{\Lambda^2}\right) \quad (646)$$

$$\Delta \mathcal{M}^{(1)}(s) = \frac{\lambda_R^2}{32\pi^2} \ln\left(\frac{s_0}{\Lambda^2} \frac{\Lambda^2}{s}\right) \quad (647)$$

$$\boxed{\Delta \mathcal{M}^{(1)}(s) = \frac{\lambda_R^2}{32\pi^2} \ln\left(\frac{s_0}{s}\right)} \quad (648)$$

This correction is finite and small, like the perturbative correction to a real measurable quantity must be. This is renormalization. There is not much to it.

17.2. Renormalized Perturbation Theory

My calculation in Sec. 17.1 was done using physical/on-shell perturbation theory. Renormalized perturbation theory is another approach that gives the same answer, but is usually easier. We replace the Lagrangian (Eq. 584) with

$$\mathcal{L} = -\frac{\Phi \square \Phi}{2} - \frac{\lambda_R}{4!} \Phi^4 - \frac{\delta \lambda}{4!} \Phi^4. \quad (649)$$

The counterterm $\delta \lambda$ is infinite, but formally of order λ_R^2 . I can immediately write down the amplitude using my result in Eq. 624

$$\mathcal{M}(s) = -\lambda_R - \delta \lambda - \frac{\lambda_R^2}{32\pi^2} \ln\left(\frac{s}{\Lambda^2}\right). \quad (650)$$

Basically $\lambda \rightarrow \lambda_R$ and to order λ_R^2 , the counterterm acts like the leading order interaction. Now we need to set the counterterm, by enforcing our condition $\mathcal{M}(s_0) = -\lambda_R$.

$$\mathcal{M}(s_0) = -\lambda_R - \delta \lambda - \frac{\lambda_R^2}{32\pi^2} \ln\left(\frac{s_0}{\Lambda^2}\right) \quad (651)$$

$$\cancel{\text{X}}_R = \cancel{\text{X}}_R - \delta \lambda - \frac{\lambda_R^2}{32\pi^2} \ln\left(\frac{s_0}{\Lambda^2}\right) \quad (652)$$

$$\boxed{\delta \lambda = \frac{\lambda_R^2}{32\pi^2} \ln\left(\frac{\Lambda^2}{s_0}\right)} \quad (653)$$

Plugging this back into Eq. 650 yields

$$\boxed{\mathcal{M}(s) = -\lambda_R - \frac{\lambda_R^2}{32\pi^2} \ln\left(\frac{s}{s_0}\right)}. \quad (654)$$

This is exactly what I got in Sec. 17.1, but with considerably less effort.

18. WICK ROTATIONS

I follow Schwartz Sec. B.2. One-loop integrals often reduce to the form

$$I = \int \frac{d^4 p}{(\vec{p}^2 - \Delta + i\epsilon)^n}, \quad n > \frac{1}{2}, \quad \Delta > 0, \quad (655)$$

where Δ does not depend on $\vec{p}$. Wick rotation also works for $\Delta < 0$ but following Schwartz I won't prove that here. This is another one of those things I can do when I have more time. I plug $\vec{p} = (p^0, \mathbf{p}) = (E, \mathbf{p})$ into I .

$$I = \int \frac{d^4 p}{[(E, \mathbf{p})^2 - \Delta + i\epsilon]^n} = \int \frac{d^4 p}{(E^2 - \mathbf{p}^2 - \Delta + i\epsilon)^n} \quad (656)$$

$$p = |\mathbf{p}| \quad (657)$$

$$I = \int \frac{d^4 p}{(E^2 - p^2 - \Delta + i\epsilon)^n} \quad (658)$$

I find the poles for $E = p^0$.

$$E^2 - p^2 - \Delta + i\epsilon = 0 \implies E_{\pm} = \pm \sqrt{p^2 + \Delta - i\epsilon} \quad (659)$$

Because of the implied limit $\epsilon \rightarrow 0_+$ these simplify

$$E_{\pm} = \pm \sqrt{p^2 + \Delta} \mp \frac{i\epsilon}{2\sqrt{p^2 + \Delta}}. \quad (660)$$

Because $\epsilon \rightarrow 0_+$ it can absorb positive constants

$$E_{\pm} = \pm \sqrt{p^2 + \Delta} \mp i\epsilon. \quad (661)$$

$$I = \int_{-\infty}^{\infty} dE \int \frac{d^3p}{(E - E_+)^n (E - E_-)^n} \quad (662)$$

Consider the complex E plane in Fig. 19. I is integrated along the blue contour. Defining

$$f(E) = \int \frac{d^3p}{(E - E_+)^n (E - E_-)^n} = \int \frac{d^3p}{(\vec{p}^2 - \Delta + i\epsilon)^n}, \quad (663)$$

we have

$$I = \int_{-R}^R dE f(E). \quad (664)$$

where the limit $R \rightarrow \infty$ is implied. The U and L in Fig. 19 represent two closed loop contours. The upper loop is

$$\oint_U dE f(E) = \int_0^R dE f(E) + \int_R^{iR} dE f(E) + \int_{iR}^0 dE f(E). \quad (665)$$

The lower loop is

$$\oint_L dE f(E) = \int_{-R}^{-iR} dE f(E) + \int_{-iR}^0 dE f(E) + \int_0^{-R} dE f(E). \quad (666)$$

The closed loops enclose no poles so

$$\oint_U dE f(E) = \oint_L dE f(E) = 0. \quad (667)$$

Now let's do the integrals along the black quarter circles.

$$Q = \int_R^{iR} dE f(E). \quad (668)$$

Along this contour $E = Re^{i\phi}$ and $dE = iRe^{i\phi}d\phi$.

$$Q = iR \int_0^{\pi/2} d\phi e^{i\phi} f(Re^{i\phi}) \quad (669)$$

$$\begin{aligned} f(Re^{i\phi}) &= \int \frac{d^3p}{(Re^{i\phi} - E_+)^n (Re^{i\phi} - E_-)^n} = \\ &= \int \frac{d^3p}{[Re^{i\phi} - E_+(p)]^n [Re^{i\phi} - E_-(p)]^n} = \\ &= \frac{1}{R^{2n} e^{2ni\phi}} \int d^3p \left(1 - \frac{E_+}{Re^{i\phi}}\right)^{-n} \left(1 - \frac{E_-}{Re^{i\phi}}\right)^{-n} \end{aligned} \quad (670)$$

I define

$$h(E) = \int d^3p \left(1 - \frac{E_+}{E}\right)^{-n} \left(1 - \frac{E_-}{E}\right)^{-n}. \quad (671)$$

To kill the integrals along the quarter circle contours I must assume that $h(Re^{i\phi})$ is finite or diverges slower than R^{-2n} . Then

$$f(Re^{i\phi}) = \frac{h(Re^{i\phi})}{R^{2n} e^{2ni\phi}}, \quad (672)$$

$$Q = iR \int_0^{\pi/2} d\phi e^{i\phi} \frac{h(Re^{i\phi})}{R^{2n} e^{2ni\phi}} = \frac{i}{R^{2n-1}} \int_0^{\pi/2} d\phi \frac{h(Re^{i\phi})}{e^{(2n-1)i\phi}} \quad (673)$$

Because $n > 1/2$ (Eq. 655) and $h(Re^{i\phi})$ is small enough (as I assumed above)

$$Q = \int_R^{iR} dE f(E) = 0. \quad (674)$$

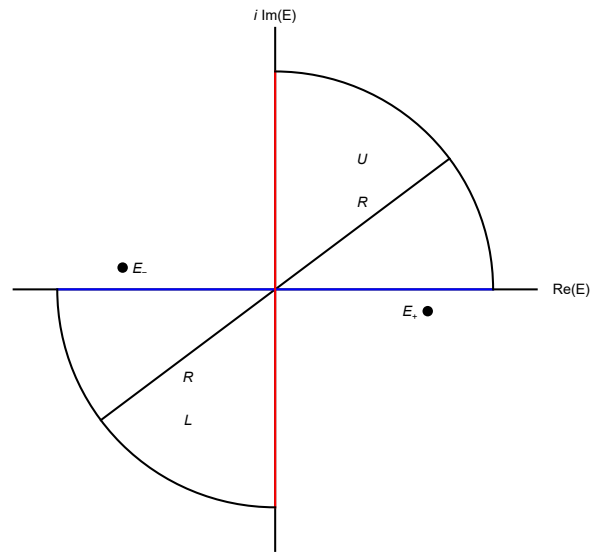


FIG. 19. Complex E plane for Eq. 662. The limit $R \rightarrow \infty$ is implied.

I confirm that the other quarter circle contour integral cancels.

$$\begin{aligned} \int_{-R}^{-iR} dE f(E) &= iR \int_{\pi}^{3\pi/2} d\phi e^{i\phi} f(Re^{i\phi}) = \\ &= \frac{i}{R^{2n-1}} \int_{\pi}^{3\pi/2} d\phi \frac{h(Re^{i\phi})}{e^{(2n-1)i\phi}} = 0. \end{aligned} \quad (675)$$

$$\int_{-R}^{-iR} dE f(E) = 0 \quad (676)$$

I combine Eqs. 665-667, 674, & 676.

$$\begin{aligned} \int_{-iR}^0 dE f(E) + \int_0^{-R} dE f(E) &= \\ &= \int_0^R dE f(E) + \int_{iR}^0 dE f(E) \end{aligned} \quad (677)$$

$$\int_{-R}^R dE f(E) = \int_{-iR}^{iR} dE f(E) \quad (678)$$

$$\int_{-\infty}^{\infty} dE f(E) = \int_{-i\infty}^{i\infty} dE f(E) \quad (679)$$

So there is a connection between integration along the red and blue contours. I plug in Eq. 663 to flesh it out.

$$\int_{-\infty}^{\infty} dE \int \frac{d^3 p}{(\vec{p}^2 - \Delta + i\epsilon)^n} = \int_{-i\infty}^{i\infty} dE \int \frac{d^3 p}{(\vec{p}^2 - \Delta + i\epsilon)^n} \quad (680)$$

We don't need ϵ along the red contour. We needed it to knock the poles off the blue contour but now we can take $\epsilon \rightarrow 0_+$ in the red.

$$\int_{-\infty}^{\infty} dE \int \frac{d^3 p}{(\vec{p}^2 - \Delta + i\epsilon)^n} = \int_{-i\infty}^{i\infty} dE \int \frac{d^3 p}{(\vec{p}^2 - \Delta)^n} \quad (681)$$

$$\int \frac{d^4 p}{(\vec{p}^2 - \Delta + i\epsilon)^n} = \int_{-i\infty}^{i\infty} dE \int \frac{d^3 p}{[(E, \mathbf{p})^2 - \Delta]^n} \quad (682)$$

I substitute $E = iY \implies Y = -iE$.

$$\int \frac{d^4 p}{(\vec{p}^2 - \Delta + i\epsilon)^n} = i \int_{-\infty}^{\infty} dY \int \frac{d^3 p}{[(iY, \mathbf{p})^2 - \Delta]^n} \quad (683)$$

$$\int \frac{d^4 p}{(\vec{p}^2 - \Delta + i\epsilon)^n} = i \int_{-\infty}^{\infty} dY \int \frac{d^3 p}{[(iY)^2 - \mathbf{p}^2 - \Delta]^n} \quad (684)$$

$$\int \frac{d^4 p}{(\vec{p}^2 - \Delta + i\epsilon)^n} = i \int_{-\infty}^{\infty} dY \int \frac{d^3 p}{(-Y^2 - \mathbf{p}^2 - \Delta)^n} \quad (685)$$

$$\int \frac{d^4 p}{(\vec{p}^2 - \Delta + i\epsilon)^n} = i \int_{-\infty}^{\infty} dY \int \frac{d^3 p}{[-(Y^2 + \mathbf{p}^2) - \Delta]^n} \quad (686)$$

I relabel the dummy variable Y to p^0 and drop the color coding.

$$\int \frac{d^4 p}{(\vec{p}^2 - \Delta + i\epsilon)^n} = i \int_{-\infty}^{\infty} dp^0 \int \frac{d^3 p}{[-[(p^0)^2 + \mathbf{p}^2] - \Delta]^n} \quad (687)$$

The regular 4-momentum is $\vec{p} = (p^0, \mathbf{p})$. I define the Euclidean 4-momentum $\vec{p}_E = (p^0, \mathbf{p})$. So both 4-momenta have the same components. But to square the Euclidean momentum I need to use the Euclidean metric

$$\vec{p}_E^2 = \delta_{\mu\nu} p_E^\mu p_E^\nu = \delta_{\mu\nu} p^\mu p^\nu = (p^0)^2 + \mathbf{p}^2. \quad (688)$$

It is crucial to understand that the two momenta have different metrics. Using the Euclidean momentum I simplify Eq. 687.

$$\int \frac{d^4 p}{(\vec{p}^2 - \Delta + i\epsilon)^n} = i \int_{-\infty}^{\infty} dp^0 \int \frac{d^3 p}{(-\vec{p}_E^2 - \Delta)^n}, \quad (689)$$

$$\int \frac{d^4 p}{(\vec{p}^2 - \Delta + i\epsilon)^n} = i \int \frac{d^4 p_E}{(-\vec{p}_E^2 - \Delta)^n} \quad (690)$$

For 3-vectors I use $p = |\mathbf{p}|$. The Euclidean 4-momentum is just a 3-vector with an extra spatial (not time) dimension. Therefore it is natural to define $p_E = \sqrt{\vec{p}_E^2} = |\vec{p}_E|$. With this definition

$$\int \frac{d^4 p}{(\vec{p}^2 - \Delta + i\epsilon)^n} = i \int \frac{d^4 p_E}{(-p_E^2 - \Delta)^n}. \quad (691)$$

This identity is beautiful, and we can simplify it further by exploiting spherical symmetry. The surface area of a sphere with radius R is $4\pi R^2$. A volume integral over all space of a spherically symmetric function simplifies

$$\int d^3 x f(r) = 4\pi \int_0^\infty dr r^2 f(r). \quad (692)$$

Let's scale this up to a 4D Euclidean space. The surface volume of a hyper-sphere is $2\pi^2 R^3$. Spherically symmetric integrals reduce to

$$\int d^4 x f(r) = 2\pi^2 \int_0^\infty dr r^3 f(r). \quad (693)$$

Applying this identity to Eq. 691 yields

$$\int \frac{d^4 p}{(\vec{p}^2 - \Delta + i\epsilon)^n} = 2\pi^2 i \int_0^\infty dp_E \frac{p_E^3}{(-p_E^2 - \Delta)^n}. \quad (694)$$

Some useful generalizations (which I might prove later are)

$$\begin{aligned} \int d^4 p \frac{f(\vec{p}^2)}{(\vec{p}^2 - \Delta + i\epsilon)^n} &= i \int d^4 p_E \frac{f(-p_E^2)}{(-p_E^2 - \Delta)^n}, \\ \int d^4 p \frac{f(\vec{p}^2)}{(\vec{p}^2 - \Delta + i\epsilon)^n} &= 2\pi^2 i \int_0^\infty dp_E \frac{p_E^3 f(-p_E^2)}{(-p_E^2 - \Delta)^n}. \end{aligned} \quad (695)$$

19. PAULI-VILLARS REGULARIZATION

Pauli-Villars regularization (PVR) adds fictions ghost particles with a mass Λ that is assumed to be large. So PVR changes the UV theory. These ghost particles have the wrong statistics (fermions that act like Bosons) or have the wrong sign kinetic terms. They are designed to exactly cancel divergences. The best way to see how this works is with an example, which I pull from Schwartz Sec. B.4.2. Consider the divergent integral

$$I = \int \frac{d^4 p}{(\vec{p}^2 - m^2 + i\epsilon)^2}. \quad (696)$$

Now rewrite this integral as

$$I = \lim_{\Lambda \rightarrow \infty} \int d^4 p \left[\frac{1}{(\vec{p}^2 - m^2 + i\epsilon)^2} - \frac{1}{(\vec{p}^2 - \Lambda^2 + i\epsilon)^2} \right]. \quad (697)$$

As $\Lambda \rightarrow \infty$ the new term approaches zero, which is why the equality above is true. Also this means the new term is small, because Λ and therefore its denominator is big. At the end of a calculation we take $\Lambda \rightarrow \infty$ to recover the true result. But when we calculate observables Λ cancels out and taking this limit is trivial: we do nothing. This is very elegant. I don't want to keep writing the $\lim$ sign, so from now on let it be implicitly understood that $\Lambda \rightarrow \infty$.

$$I = \int d^4p \left[\frac{1}{(\vec{p}^2 - m^2 + i\epsilon)^2} - \frac{1}{(\vec{p}^2 - \Lambda^2 + i\epsilon)^2} \right]. \quad (698)$$

I Wick rotate (Eq. 694).

$$I = 2\pi^2 i \int_0^\infty dp_E p_E^3 \left[\frac{1}{(p_E^2 + m^2)^2} - \frac{1}{(p_E^2 + \Lambda^2)^2} \right] \quad (699)$$

$$I = 2\pi^2 i \int_0^\infty dr r^3 \left[\frac{1}{(r^2 + m^2)^2} - \frac{1}{(r^2 + \Lambda^2)^2} \right] \quad (700)$$

Both terms have the same form

$$\star = \int_0^a dx \frac{x^3}{(x^2 + b^2)^2}. \quad (701)$$

$$u = u(x) = x^2, \quad u(0) = 0, \quad u(a) = a^2, \quad \frac{du}{2} = x dx \quad (702)$$

$$\star = \frac{1}{2} \int_0^{a^2} du \frac{u}{(u + b^2)^2} \quad (703)$$

$$y = u + b^2, \quad y(0) = b^2, \quad y(a^2) = a^2 + b^2, \quad dy = du \quad (704)$$

$$\star = \frac{1}{2} \int_{b^2}^{a^2+b^2} dy \frac{y - b^2}{y^2} \quad (705)$$

$$2\star = \int_{b^2}^{a^2+b^2} \frac{dy}{y} - b^2 \int_{b^2}^{a^2+b^2} \frac{dy}{y^2} \quad (706)$$

$$2\star = \ln \left(\frac{a^2 + b^2}{b^2} \right) - \frac{a^2}{a^2 + b^2} \quad (707)$$

$$\boxed{2 \int_0^a dx \frac{x^3}{(x^2 + b^2)^2} = \ln \left(\frac{a^2 + b^2}{b^2} \right) - \frac{a^2}{a^2 + b^2}} \quad (708)$$

Applying this identity to I yields

$$I = \pi^2 i \lim_{\Lambda \rightarrow \infty} \lim_{a \rightarrow \infty} \left[\ln \left(\frac{a^2 + m^2}{m^2} \right) - \ln \left(\frac{a^2 + \Lambda^2}{\Lambda^2} \right) + \frac{a^2}{a^2 + \Lambda^2} - \frac{a^2}{a^2 + m^2} \right]. \quad (709)$$

I have once again made the limit $\Lambda \rightarrow \infty$ explicit, because there is a subtle issue with the order of limits. To get the right answer I must take $a \rightarrow \infty$ first, while holding Λ finite.

$$I = \pi^2 i \lim_{\Lambda \rightarrow \infty} \lim_{a \rightarrow \infty} \left[\ln \left(\frac{a^2}{m^2} \right) - \ln \left(\frac{a^2}{\Lambda^2} \right) + \frac{a^2}{a^2} - \frac{a^2}{a^2} \right] \quad (710)$$

$$I = \pi^2 i \lim_{\Lambda \rightarrow \infty} \lim_{a \rightarrow \infty} \left[\ln \left(\frac{\cancel{a^2} \Lambda^2}{m^2 \cancel{a^2}} \right) + \cancel{1} - \cancel{1} \right] \quad (711)$$

$$I = \pi^2 i \lim_{\Lambda \rightarrow \infty} \lim_{a \rightarrow \infty} \ln \left(\frac{\Lambda^2}{m^2} \right) = \pi^2 i \lim_{\Lambda \rightarrow \infty} \ln \left(\frac{\Lambda^2}{m^2} \right) \quad (712)$$

I make $\Lambda \rightarrow \infty$ implicit again. The final result is

$$\boxed{\int \frac{d^4p}{(\vec{p}^2 - m^2 + i\epsilon)^2} = i\pi^2 \ln \left(\frac{\Lambda^2}{m^2} \right)}. \quad (713)$$

20. VACUUM POLARIZATION

Let's do another example of loop integration: the scalar analog of QED vacuum polarization (also called the photon self energy). I follow along with Schwartz Sec. 16.1. The Lagrangian is

$$\mathcal{L} = -\frac{\Phi(\Box + m^2)\Phi}{2} + \frac{g}{3!}\Phi^3. \quad (714)$$

I label the diagram $\star$ and draw it in Fig. 20. I write the integral

$$\star = \frac{(ig)^2 S}{(2\pi)^{4n}} \int d^4p d^4q \frac{i}{\vec{p}^2 - m^2 + i\epsilon} \frac{i}{\vec{q}^2 - m^2 + i\epsilon} \delta^{(4)}(\vec{P} + \vec{q} - \vec{p}) \delta^{(4)}(\vec{p} - \vec{K} - \vec{q}), \quad (715)$$

where S is the symmetry factor and n is the number of independent loop momenta. I showed how to find S in Sec. 17

$$S = \frac{6 \times 3 \times 2}{(3!)^2 2!} = \frac{1}{2} \quad (716)$$

I finish writing the integral.

$$\star = \frac{g^2}{2(2\pi)^{4n}} \delta^{(4)}(\vec{P} - \vec{K}) \int \frac{d^4p}{(\vec{p}^2 - m^2 + i\epsilon)[(\vec{p} - \vec{P})^2 - m^2 + i\epsilon]} \quad (717)$$

So $n = 1$.

$$\star = \frac{g^2}{2(2\pi)^4} \delta^{(4)}(\vec{P} - \vec{K}) \int \frac{d^4p}{(\vec{p}^2 - m^2 + i\epsilon)[(\vec{p} - \vec{P})^2 - m^2 + i\epsilon]} \quad (718)$$

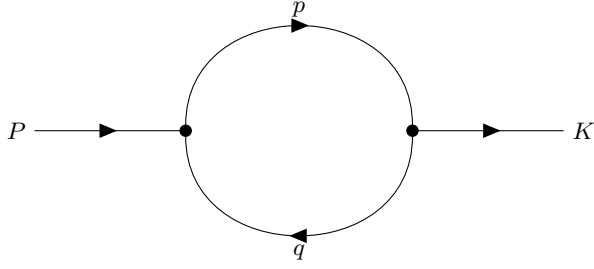


FIG. 20. ✧

Eq. 718 is the integral for a 1 to 1 “scattering process.” That is a particle moving “freely.” Of course there is no such thing as a free particle. A particle is a field excitation invariant to Poincaré transforms, and the other fields are always present everywhere. As long as it is coupled to those fields (and all fields are coupled to gravity) the particle will be influenced by those other fields. In this simple model the scalar field interacts with itself. But the vacuum polarization diagram is not just used to calculate 1 to 1 scattering. It is also used to find corrections to the propagator. Consider the free Feynman propagator from Sec. 5.1

$$\mathcal{G}_F^{(0)}(\vec{x} - \vec{y}) = \langle 0 | T \phi(\vec{x}) \phi(\vec{y}) | 0 \rangle = \frac{i}{(2\pi)^4} \int d^4 p \frac{e^{i\vec{p} \cdot (\vec{y} - \vec{x})}}{\vec{p}^2 - m^2 + i\epsilon} = \int \frac{d^4 p}{(2\pi)^4} e^{i\vec{p} \cdot (\vec{y} - \vec{x})} \mathcal{G}_F^{(0)}(\vec{p}), \quad (719)$$

where

$$\mathcal{G}_F^{(0)}(\vec{p}) = \frac{i}{\vec{p}^2 - m^2 + i\epsilon}, \quad (720)$$

is the Fourier transform of the propagator. $\mathcal{G}_F^{(0)}(\vec{p})$ is what actually shows up in Feynman diagrams like ✧. The subscript 0 reminds us that the free propagator is the lowest order term in the perturbative expansion of the exact propagator

$$\mathcal{G}_F(\vec{x} - \vec{y}) = \langle \Omega | T \Phi(\vec{x}) \Phi(\vec{y}) | \Omega \rangle = \int \frac{d^4 p}{(2\pi)^4} e^{i\vec{p} \cdot (\vec{y} - \vec{x})} \mathcal{G}_F(\vec{p}). \quad (721)$$

We can use the ✧ diagram to find corrections to Eq. 720. To facilitate this I redefine the diagram. Following Schwartz I take

$$\star \rightarrow \frac{\star}{\delta^{(4)}(\vec{P} - \vec{K})} \quad (722)$$

so

$$\star = \frac{g^2}{2(2\pi)^4} \int \frac{d^4 p}{(\vec{p}^2 - m^2 + i\epsilon)[(\vec{p} - \vec{P})^2 - m^2 + i\epsilon]}. \quad (723)$$

In $\mathcal{G}_F(\vec{p})$ the momentum $\vec{p}$ is off-shell (we integrate over all $\vec{p}$ in Eq. 721). Thus $\vec{P}$ in ✧ is off-shell: $\vec{P}^2 \neq m^2$.

Regularizing ✧ is great fun and will take more complicated mathematical technology than the simple derivative regularization I used in Sec. 17. I start by using Feynman parameters to transform the integrand: I use the identity

$$\frac{1}{AB} = \int_0^1 \frac{dx}{[A + (B - A)x]^2}, \quad (724)$$

with $A = (\vec{p} - \vec{P})^2 - m^2 + i\epsilon$ and $B = \vec{p}^2 - m^2 + i\epsilon$.

$$\star = \frac{g^2}{2(2\pi)^4} \int d^4 p \int_0^1 dx \frac{1}{[(\vec{p} - \vec{P})^2 - m^2 + i\epsilon + [\vec{p}^2 - (\vec{p} - \vec{P})^2]x]^{-2}} \quad (725)$$

$$\star = \frac{g^2}{2(2\pi)^4} \int d^4 p \int_0^1 \frac{dx}{f(\vec{p}, x)^2} \quad (726)$$

$$f = f(\vec{p}, x) = (\vec{p} - \vec{P})^2 - m^2 + i\epsilon + [\vec{p}^2 - (\vec{p} - \vec{P})^2]x \quad (727)$$

$$f = \vec{p}^2 + 2(x - 1)\vec{P} \cdot \vec{p} + (1 - x)\vec{P}^2 - m^2 + i\epsilon \quad (728)$$

Completing the square means

$$ax^2 + bx + c = a(x + d)^2 + e, \quad d = \frac{b}{2a}, \quad e = c - \frac{b^2}{4a}. \quad (729)$$

I apply this to f by taking

$$x \rightarrow \vec{p}, \quad a \rightarrow 1, \quad b \rightarrow 2(x - 1)\vec{P}, \quad c \rightarrow (1 - x)\vec{P}^2 - m^2 + i\epsilon. \quad (730)$$

$$f = [\vec{p} + (x - 1)\vec{P}]^2 + x(1 - x)\vec{P}^2 - m^2 + i\epsilon \quad (731)$$

Because the definition in Eq. 729 is in terms of scalars and f is composed of 4-vectors I verified this identity in Mathematica. Following Schwartz I introduce the shorthand

$$\Delta = m^2 - x(1 - x)\vec{P}^2. \quad (732)$$

$$f = [\vec{p} + (x - 1)\vec{P}]^2 - \Delta + i\epsilon \quad (733)$$

$$\star = \frac{g^2}{2(2\pi)^4} \int d^4 p \int_0^1 \frac{dx}{[\vec{p} + (x - 1)\vec{P}]^2 - \Delta + i\epsilon]^2} \quad (734)$$

$$\star = \frac{g^2}{2(2\pi)^4} \int_0^1 dx \int \frac{d^4 p}{[\vec{p} + (x - 1)\vec{P}]^2 - \Delta + i\epsilon]^2} \quad (735)$$

I substitute

$$\vec{q} = \vec{p} + (x - 1)\vec{P}, \quad d^4 q = d^4 p. \quad (736)$$

This is a constant shift as far as the momentum integral is concerned. Because the integral is over all of momentum space the bounds of integration do not change.

$$\star = \frac{g^2}{2(2\pi)^4} \int_0^1 dx \int \frac{d^4 q}{(\vec{q}^2 - \Delta + i\epsilon)^2}. \quad (737)$$

Using Feynman parameters and completing the square is often the first step of regularization. It doesn't cure the divergence, but it does tidy up the integral, making the actual regularization easier.

I have already regularized this integral using PVR (Eq. 713)

$$\star = \frac{i\pi^2 g^2}{2(2\pi)^4} \int_0^1 dx \ln\left(\frac{\Lambda^2}{\Delta}\right) = -\frac{ig^2}{32\pi^2} \int_0^1 dx \ln\left(\frac{\Delta}{\Lambda^2}\right), \quad (738)$$

where Λ is the UV cutoff. In PVR it is the mass of a fictitious high energy ghost particle.

$$\star = -\frac{ig^2}{32\pi^2} \int_0^1 dx \ln\left(\frac{m^2 - x(1-x)\vec{P}^2}{\Lambda^2}\right) \quad (739)$$

I follow Schwartz by switching to the special case $m = 0$ at this point.

$$\star = -\frac{ig^2}{32\pi^2} \int_0^1 dx \ln\left(-x(1-x)\frac{\vec{P}^2}{\Lambda^2}\right) \quad (740)$$

I do the integral in Mathematica

$$\star = \frac{ig^2}{32\pi^2} \left[2 - \ln\left(-\frac{\vec{P}^2}{\Lambda^2}\right) \right]. \quad (741)$$

The finite part is not physically important and can be absorbed into Λ .

$$\star = -\frac{ig^2}{32\pi^2} \ln\left(-\frac{\vec{P}^2}{\Lambda^2 e^2}\right) \quad (742)$$

Λ is the UV cutoff so I can redefine it as long as it stays big. I take $\Lambda \rightarrow \Lambda e^{-1}$.

$$\star = -\frac{ig^2}{32\pi^2} \ln\left(-\frac{\vec{P}^2}{\Lambda^2}\right). \quad (743)$$

The coefficient $-\frac{ig^2}{32\pi^2}$ is physically meaningful and regulator independent. It is useful to view this diagram as a function of the external momentum

$$\star(\vec{p}) = -\frac{ig^2}{32\pi^2} \ln\left(-\frac{\vec{p}^2}{\Lambda^2}\right). \quad (744)$$

20.1. Renormalization: Loop Correction to the Propagator & Coupling Constant

In QED the vacuum polarization is used to calculate corrections to the photon propagator and the electric charge, i.e. the coupling between Dirac spinors and electromagnetic fields. To understand how this works I use my result in Eq. 744 to work out the scalar analogues in this section, following Schwartz Sec. 16.1.1. I start with the propagator.

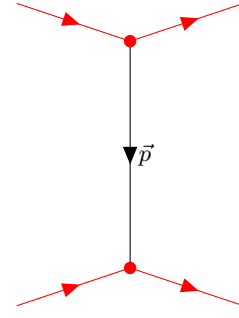


FIG. 21. Lowest order unbounded propagator $\mathcal{G}^{(0)}(\vec{p})$.

Consider the Fourier transform of the Feynman propagator, $\mathcal{G}_F(\vec{p})$, in Eq. 721. I gave its lowest order (free field theory) approximation in Eqs. 719 & 720. Now I am not concerned with the Feynman propagator but a related object: the unbounded (it has no boundary condition) propagator. The real and momentum space (Fourier transformed) versions are still related by

$$\mathcal{G}(\vec{x} - \vec{y}) = \int \frac{d^4 p}{(2\pi)^4} e^{i\vec{p} \cdot (\vec{y} - \vec{x})} \mathcal{G}(\vec{p}). \quad (745)$$

The lowest order free field approximation is represented by the diagram in Fig. 21. This is a new type of Feynman diagram I have not used before. We turn it into an expression by following a modified version of the Feynman rules in Sec. 11.1. We ignore the red vertices: we do not add factors of ig , we do not add momentum conserving delta functions, and we do not integrate over internal momenta connected to them. Internal momenta still get propagators:

$$\frac{i}{\vec{p}^2 - m^2} \quad (746)$$

but since these are unbounded there is no $i\epsilon$. Also since we followed Schwartz and assumed $m = 0$, in this case we get $i/\vec{p}^2$. The result is

$$\boxed{\mathcal{G}^{(0)}(\vec{p}) = \frac{i}{\vec{p}^2}}. \quad (747)$$

The first correction is second order (in g)

$$\mathcal{G}^{(2)}(\vec{p}) = \mathcal{G}^{(0)}(\vec{p}) + \Delta\mathcal{G}^{(2)}(\vec{p}). \quad (748)$$

The leading correction comes from the diagram in Fig. 22. Notice that the internal (black) momentum connected to both red vertices is the same: $\vec{p}$. We treat the black vertices using the original momentum space Feynman rules

$$\Delta\mathcal{G}^{(2)}(\vec{p}) = \frac{(ig)^2 S}{(2\pi)^4} \int d^4 q \frac{i}{\vec{p}^2} \frac{i}{\vec{q}^2} \frac{i}{(\vec{q} - \vec{p})^2} \frac{i}{\vec{p}^2}. \quad (749)$$

I already took care of momentum conservation at each black vertex so there is no need for delta functions. S is the symmetry factor. There are two $\vec{p}$ propagators because there

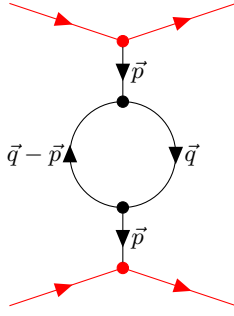


FIG. 22. The leading correction to the interacting propagator: $\Delta\mathcal{G}^{(2)}(\vec{p})$.

are two $\vec{p}$ lines. This is the same S I calculated in Eq. 716: $S = 1/2$.

$$\Delta\mathcal{G}^{(2)}(\vec{p}) = -\frac{g^2}{2(2\pi)^4\vec{p}^4} \int \frac{d^4q}{\vec{q}^2(\vec{q}-\vec{p})^2}. \quad (750)$$

I compare this with Eq. 723 with $m = \epsilon = 0$

$$\star(\vec{p}) = \frac{g^2}{2(2\pi)^4} \int \frac{d^4q}{\vec{q}^2(\vec{q}-\vec{p})^2}, \quad (751)$$

$$\Delta\mathcal{G}^{(2)}(\vec{p}) = -\frac{\star(\vec{p})}{\vec{p}^4}. \quad (752)$$

I plug in the regularized vacuum polarization from Eq. 744

$$\Delta\mathcal{G}^{(2)}(\vec{p}) = \frac{ig^2}{32\pi^2\vec{p}^4} \ln\left(-\frac{\vec{p}^2}{\Lambda^2}\right). \quad (753)$$

To second order

$$\mathcal{G}^{(2)}(\vec{p}) = \frac{i}{\vec{p}^2} \left[1 + \frac{g^2}{32\pi^2\vec{p}^2} \ln\left(-\frac{\vec{p}^2}{\Lambda^2}\right) \right]. \quad (754)$$

That is how we find corrections to the propagator. Now we will use the second order propagator to renormalize the coupling g .

To renormalize g we work in a region where $\vec{p}$ is spacelike: $\vec{p}^2 < 0$. We define $Q^2 = -\vec{p}^2$ so $Q^2 > 0$. We define

$$\mathcal{M}(Q) = \frac{(ig)^2}{i} \mathcal{G}(\vec{p}). \quad (755)$$

To second order

$$\mathcal{M}^{(2)}(Q) = \frac{(ig)^2}{i} \mathcal{G}^{(2)}(\vec{p}), \quad (756)$$

$$\mathcal{M}^{(2)}(Q) = \frac{g^2}{Q^2} \left[1 - \frac{g^2}{32\pi^2 Q^2} \ln\left(\frac{Q^2}{\Lambda^2}\right) \right]. \quad (757)$$

The coupling constant has mass dimension $[g] = 1$. Also $[\vec{p}] = [Q] = 1$. It is better to renormalize with a dimensionless coupling so let's define: $G = g/Q$, where $[G] = 0$.

$$\mathcal{M}^{(2)}(Q) = G^2 - \frac{G^4}{32\pi^2} \ln\left(\frac{Q^2}{\Lambda^2}\right). \quad (758)$$

We define a renormalized coupling G_R with the renormalization condition

$$\boxed{G_R^2 = \mathcal{M}(Q_0)}, \quad (759)$$

where Q_0 is a particular value we have to pick. By enforcing this renormalization condition order by order, we generate a power series for the bare coupling G in terms of G_R

$$G^{(2)} = G^{(0)} + \Delta G^{(2)}. \quad (760)$$

To lowest order

$$G_R^2 = \mathcal{M}^{(0)}(Q_0) = G^2, \implies \boxed{G^{(0)} = G_R}. \quad (761)$$

When we pick the renormalization condition we are constrained to pick one where the renormalized and bare couplings are the same, to lowest order. I calculate the leading correction.

$$\mathcal{M}^{(2)}(Q_0) = G_R^2 \quad (762)$$

$$G^2 - \frac{\lambda G^4}{32\pi^2} \ln\left(\frac{Q_0^2}{\Lambda^2}\right) = G_R^2 \quad (763)$$

I use λ to track order.

$$(G^{(0)} + \lambda \Delta G^{(2)})^2 - \frac{\lambda}{32\pi^2} \ln\left(\frac{Q_0^2}{\Lambda^2}\right) (G^{(0)} + \lambda \Delta G^{(2)})^4 = G_R^2 \quad (764)$$

$$(G^{(0)} + \lambda \Delta G^{(2)})^2 - \frac{\lambda G^{(0)4}}{32\pi^2} \ln\left(\frac{Q_0^2}{\Lambda^2}\right) = G_R^2 \quad (765)$$

$$(G_R + \lambda \Delta G^{(2)})^2 - \frac{\lambda G_R^4}{32\pi^2} \ln\left(\frac{Q_0^2}{\Lambda^2}\right) = G_R^2 \quad (766)$$

$$G_R^2 \left(1 + \lambda \frac{\Delta G^{(2)}}{G_R} \right)^2 - \frac{\lambda G_R^4}{32\pi^2} \ln\left(\frac{Q_0^2}{\Lambda^2}\right) = G_R^2 \quad (767)$$

$$\left(1 + \lambda \frac{\Delta G^{(2)}}{G_R} \right)^2 - \frac{\lambda G_R^2}{32\pi^2} \ln\left(\frac{Q_0^2}{\Lambda^2}\right) = 1 \quad (768)$$

$$\chi + 2\lambda \frac{\Delta G^{(2)}}{G_R} - \frac{\lambda G_R^2}{32\pi^2} \ln\left(\frac{Q_0^2}{\Lambda^2}\right) = \chi \quad (769)$$

$$2 \frac{\Delta G^{(2)}}{G_R} = \frac{G_R^2}{32\pi^2} \ln\left(\frac{Q_0^2}{\Lambda^2}\right) \quad (770)$$

$$\boxed{\Delta G^{(2)} = \frac{G_R^3}{64\pi^2} \ln\left(\frac{Q_0^2}{\Lambda^2}\right)} \quad (771)$$

The leading correction is infinite. Which is disturbing because I assumed it was much smaller than the leading order term in this calculation. But that's okay. Its not elegant, but it is the correct way to do renormalization. QFT lives on the frontier of physics, and still has rough spots. The second order coupling is

$$G^{(2)} = G_R + \frac{G_R^3}{64\pi^2} \ln \left(\frac{Q_0^2}{\Lambda^2} \right). \quad (772)$$

I approximate G^2 to second order.

$$G^2 = \left[G_R + \frac{\lambda G_R^3}{64\pi^2} \ln \left(\frac{Q_0^2}{\Lambda^2} \right) \right]^2 \quad (773)$$

$$G^2 = G_R^2 \left[1 + \frac{\lambda G_R^2}{64\pi^2} \ln \left(\frac{Q_0^2}{\Lambda^2} \right) \right]^2 \quad (774)$$

$$G^2 = G_R^2 + \frac{\lambda G_R^4}{32\pi^2} \ln \left(\frac{Q_0^2}{\Lambda^2} \right) \quad (775)$$

I plug this into $\mathcal{M}^{(2)}(Q)$ (Eq. 758) and expand to second order.

$$\mathcal{M}^{(2)}(Q) = G^2 - \frac{\lambda G^4}{32\pi^2} \ln \left(\frac{Q^2}{\Lambda^2} \right) \quad (776)$$

$$\begin{aligned} \mathcal{M}^{(2)}(Q) &= G_R^2 + \frac{\lambda G_R^4}{32\pi^2} \ln \left(\frac{Q_0^2}{\Lambda^2} \right) \\ &\quad - \frac{\lambda}{32\pi^2} \ln \left(\frac{Q^2}{\Lambda^2} \right) \left[G_R^2 + \frac{\lambda G_R^4}{32\pi^2} \ln \left(\frac{Q_0^2}{\Lambda^2} \right) \right]^2 \end{aligned} \quad (777)$$

$$\mathcal{M}^{(2)}(Q) = G_R^2 + \frac{\lambda G_R^4}{32\pi^2} \ln \left(\frac{Q_0^2}{\Lambda^2} \right) - \frac{\lambda G_R^4}{32\pi^2} \ln \left(\frac{Q^2}{\Lambda^2} \right) \quad (778)$$

I no longer need λ , so I set it to 1.

$$\mathcal{M}^{(2)}(Q) = G_R^2 + \frac{G_R^4}{32\pi^2} \ln \left(\frac{Q_0^2}{Q^2} \right) \quad (779)$$

$$\mathcal{M}^{(2)}(Q) = G_R^2 + \frac{G_R^4}{32\pi^2} \ln \left(\frac{Q_0^2}{Q^2} \right) \quad (780)$$

The bare coupling is not observable, but $\mathcal{M}(Q)$ is, which is why we found a finite expression for it. We just have to set a benchmark Q_0 and find $\mathcal{M}(Q_0)$ before we can predict measurement outcomes.

21. DIMENSIONAL REGULARIZATION

I prefer hard-cutoff regularization because it is physically the most intuitive: we don't know what's going on above

the UV cutoff. But hard-cutoffs destroy symmetries, like gauge invariance. Dimensional regularization preserves all the symmetries we care about and moreover it is the most elegant and important regularization technique in QFT. In scalar field theory we can get by without it (we don't have many symmetries), but we should use it to understand how to use it in QED, and in non-Abelian theories. It is so important that I cannot leave it out of my notes. I follow Schwartz Sec. B.3.

Consider the integral

$$I(d) = \int \frac{d^d p}{(\vec{p}^2 - \Delta + i\epsilon)^2}, \quad (781)$$

where d is the number of dimensions. Roughly speaking (think polar coordinates in 2/3D)

$$I(d) \sim \int_1^\infty dp \frac{p^{d-1}}{p^4} = \int_1^\infty \frac{dp}{p^{5-d}}. \quad (782)$$

I am only concerned with large p . The integrand in Eq. 781 does not diverge at $\vec{p} = 0$. To sidestep the divergence at the origin I made the lower bound 1 (instead of 0). Also $\vec{p}$ is a 4-vector so assuming large p is speaking roughly indeed. The statement can be made rigorous by Wick rotating to the Euclidean momentum (Sec. 18). With this simple power counting we see that $I(d)$ converges for $d < 4$ and diverges for $d \geq 4$. So if we are working in 4D, maybe we can switch to 2D, integrate, and analytically continue back to 4D. That's the essence of dimensional regularization.

The Minkowski metric in 5 dimensions is

$$\eta_{\alpha\beta} = \text{diag}(1, -1, -1, -1, -1) \quad (783)$$

The generalization to d dimensions is obvious: each spatial dimension gets a -1. There is still only one time dimension. The metric obeys

$$\eta^{\alpha\beta} \eta_{\alpha\beta} = d. \quad (784)$$

The differential solid angle in d dimensions is

$$d\Omega_d = d\phi \prod_{n=1}^{d-2} d\theta_n \sin \theta_n, \quad (785)$$

where $0 \leq \phi < 2\pi$ and $0 \leq \theta_n < \pi$. It's easy to see that this formula works in 2 & 3 dimensions:

$$d\Omega_2 = d\phi \prod_{n=1}^0 d\theta_n \sin \theta_n = d\phi, \quad d\Omega_3 = \sin \theta_1 d\theta_1 d\phi. \quad (786)$$

The volume element decomposes into

$$d^d r = d\Omega_d r^{d-1} dr \quad (787)$$

just like in 2, $d^2 r = d\Omega_2 r dr = d\phi dr r$ and 3, $d^3 r = d\Omega_3 r^2 dr = d\phi d\theta_1 dr \sin \theta_1 r^2$, dimensions. The volume of a d -sphere with radius R is

$$V_d = \int_{r \leq R} d^d r = \int d\Omega_d \int_0^R dr r^{d-1} = \Omega_d \int_0^R dr r^{d-1}, \quad (788)$$

$$V_d = \frac{\Omega_d R^d}{d}, \quad (789)$$

where Ω_d is the analogue of the unit sphere surface area in d dimensions. The 2D surface area is the circumference, $\Omega_2 = 2\pi$, and the 2D volume is the area, $V_2 = \pi R^2$. In 3D Archimedes proved $\Omega_3 = 4\pi$ so

$$V_3 = \frac{4}{3}\pi R^3. \quad (790)$$

But what if we don't know Ω_d in d dimensions? How do we define Ω_d for non-integer d ? We get it using a beautiful trick. It is so pretty that I cannot stop myself from including it in these notes. We start with the Gaussian integral

$$\sqrt{\pi} = \int_{-\infty}^{\infty} dx e^{-x^2}. \quad (791)$$

I square both sides

$$\begin{aligned} \pi &= \left(\int_{-\infty}^{\infty} dx e^{-x^2} \right)^2 = \int_{-\infty}^{\infty} dx e^{-x^2} \int_{-\infty}^{\infty} dy e^{-y^2} = \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} d^2x e^{-x^2-y^2} = \int_0^{2\pi} d\phi \int_0^{\infty} dr r e^{-r^2} = \\ &= 2\pi \int_0^{\infty} dr r e^{-r^2} = \Omega_2 \int_0^{\infty} dr r^{2-1} e^{-r^2}. \end{aligned} \quad (792)$$

I generalize to d dimensions.

$$\pi^{d/2} = \Omega_d \int_0^{\infty} dr r^{d-1} e^{-r^2}, \quad (793)$$

and substitute

$$x = x(r) = r^2, \quad x(0) = 0, \quad x(\infty) = \infty, \quad \frac{dx}{2\sqrt{x}} = dr, \quad (794)$$

$$2\pi^{d/2} = \Omega_d \int_0^{\infty} dx x^{d/2-1} e^{-x}. \quad (795)$$

I recognize an old friend: the Gamma function,

$$\Gamma(z) = \int_0^{\infty} dx x^{z-1} e^{-x}. \quad (796)$$

Some algebra yields

$$2\pi^{d/2} = \Omega_d \Gamma\left(\frac{d}{2}\right), \quad (797)$$

$$\Omega_d = \frac{2\pi^{d/2}}{\Gamma(d/2)}. \quad (798)$$

The gamma function has simple poles at all integers $n \leq 0$. A useful expansion around 0 is

$$\lim_{\epsilon \rightarrow 0} \Gamma(\epsilon) = \frac{1}{\epsilon} - \gamma_E + \mathcal{O}(\epsilon), \quad (799)$$

where $\gamma_E = 0.577216$ is the Euler–Mascheroni constant.

Now I have all the tools I need to solve the integral in Eq. 781.

$$I(d) = \int \frac{d^d p}{(\vec{p}^2 - \Delta + i\epsilon)^2}, \quad (800)$$

I Wick rotate using Eq. 691

$$I(d) = i \int \frac{d^d p_E}{(p_E^2 + \Delta)^2}. \quad (801)$$

I use Eq. 787

$$\begin{aligned} I(d) &= i \int d\Omega_d dp_E \frac{p_E^{d-1}}{(p_E^2 + \Delta)^2} = \\ &= i \int d\Omega_d \int_0^{\infty} dp_E \frac{p_E^{d-1}}{(p_E^2 + \Delta)^2} = i \Omega_d \int_0^{\infty} dp_E \frac{p_E^{d-1}}{(p_E^2 + \Delta)^2}. \end{aligned} \quad (802)$$

I plug in Eq. 798

$$I(d) = \frac{2i\pi^{d/2}}{\Gamma(d/2)} \int_0^{\infty} dp_E \frac{p_E^{d-1}}{(p_E^2 + \Delta)^2}. \quad (803)$$

I rescale assuming $\Delta > 0$.

$$I(d) = \frac{2i\pi^{d/2}}{\Gamma(d/2)} \int_0^{\infty} dr \frac{r^{d-1}}{(r^2 + \Delta)^2} \quad (804)$$

$$r = \sqrt{\Delta} x, \quad x(0) = 0, \quad x(\infty) = \infty, \quad dr = \sqrt{\Delta} dx \quad (805)$$

$$I(d) = 2i\pi^{d/2} \frac{\Delta^{d/2-2}}{\Gamma(d/2)} \int_0^{\infty} dx \frac{x^{d-1}}{(x^2 + 1)^2} \quad (806)$$

$$\int_0^{\infty} dx \frac{x^{d-1}}{(x^2 + 1)^2} = \frac{2-d}{4\sin(\pi d/2)} \pi \quad (807)$$

$$I(d) = i\pi^{1+d/2} \frac{2-d}{2\Gamma(d/2)\sin(\pi d/2)} \Delta^{d/2-2} \quad (808)$$

I use the identity (Schwartz Eq. B.32)

$$\sin(\pi x) = \frac{1-x}{\Gamma(x)\Gamma(2-x)} \pi, \quad (809)$$

$$I(d) = \frac{i\pi^{d/2}}{\Delta^{2-d/2}} \Gamma\left(\frac{4-d}{2}\right). \quad (810)$$

Finally

$$\int \frac{d^d p}{(\vec{p}^2 - \Delta + i\epsilon)^2} = \frac{i\pi^{d/2}}{\Delta^{2-d/2}} \Gamma\left(\frac{4-d}{2}\right). \quad (811)$$

I also work out

$$\mathbb{I} = \int d^d p \frac{\vec{p}^2}{(\vec{p}^2 - \Delta + i\epsilon)^2}. \quad (812)$$

I Wick rotate using Eq. 695

$$\mathbb{I} = -i \int d^d p_E \frac{p_E^2}{(p_E^2 + \Delta)^2} = -i \int d^d r \frac{r^2}{(r^2 + \Delta)^2}. \quad (813)$$

$$\mathbb{I} = -i \Omega_d \int_0^\infty dr \frac{r^{d+1}}{(r^2 + \Delta)^2} = \frac{2\pi^{d/2}}{i\Gamma(d/2)} \int_0^\infty dr \frac{r^{d+1}}{(r^2 + \Delta)^2} \quad (814)$$

$$r = \sqrt{\Delta} x, \quad x(0) = 0, \quad x(\infty) = \infty, \quad dr = \sqrt{\Delta} dx \quad (815)$$

$$\mathbb{I} = \frac{2\pi^{d/2}}{i\Gamma(d/2)} \Delta^{d/2-1} \int_0^\infty dx \frac{x^{d+1}}{(x^2 + 1)^2} \quad (816)$$

$$\int_0^\infty dx \frac{x^{d+1}}{(x^2 + 1)^2} = \frac{d\pi}{4\sin(\pi d/2)} \quad (817)$$

$$\mathbb{I} = \frac{\pi^{1+d/2} d \Delta^{d/2-1}}{2i\Gamma(d/2) \sin(\pi d/2)} = \frac{\pi^{d/2} d \Delta^{d/2-1}}{i(2-d)} \Gamma(1 + 1 - d/2) \quad (818)$$

$$\mathbb{I} = \frac{d\pi^{d/2}}{2i\Delta^{1-d/2}} \Gamma\left(\frac{2-d}{2}\right) \quad (819)$$

$$\boxed{\int d^d p \frac{\vec{p}^2}{(\vec{p}^2 - \Delta + i\epsilon)^2} = \frac{d\pi^{d/2}}{2i\Delta^{1-d/2}} \Gamma\left(\frac{2-d}{2}\right)} \quad (820)$$

In general (Schwartz Eq. B.35)

$$\int \frac{d^d p}{(2\pi)^d} \frac{\vec{p}^{2a}}{(\vec{p}^2 - \Delta + i\epsilon)^b} = \frac{i(-1)^{a-b}}{(4\pi)^{d/2} \Delta^{b-a-d/2}} \frac{\Gamma(a+d/2)\Gamma(b-a-d/2)}{\Gamma(b)\Gamma(d/2)}. \quad (821)$$

Another useful integral identity is

$$\boxed{\int \frac{d^d p}{(2\pi)^d} \frac{p^\mu p^\nu}{(\vec{p}^2 - b)^n} = \frac{\eta^{\mu\nu}}{d} \int \frac{d^d p}{(2\pi)^d} \frac{\vec{p}^2}{(\vec{p}^2 - b)^n}} \quad (822)$$

Schwartz gives an interesting argument from Lorentz invariance for this identity. But I did not find his argument to be intuitive and obviously true, and when I have more time I should try to prove it for myself.

21.1. Example: Dimensional Regularization in QED

Now we have the mathematical technology to regularize integrals. Even though these are Real Scalar Field Theory Notes I will use an example from QED. The QED interaction Lagrangian is

$$\mathcal{L}_1 = -e\bar{\psi}\gamma^\mu\psi A_\mu, \quad (823)$$

where ψ is the electron field, γ is a Dirac Matrix, A is the electromagnetic field, and the coupling constant e is the electric charge. The dimensions are

$$[A] = 1, \quad [\gamma] = [e] = 0, \quad [\psi] = \frac{3}{2}, \quad (824)$$

which yields

$$[\mathcal{L}] = [\mathcal{L}_1] = [e\gamma\psi^2 A] = [\cancel{e\gamma}] + [\psi^2 A] = [\psi^2 A] = 2[\psi] + [A] = 4. \quad (825)$$

In d dimensions

$$\mathcal{L}_1 = -\mu^{(4-d)/2} e\bar{\psi}\gamma^\mu\psi A_\mu, \quad (826)$$

where the coupling is $\mu^{(4-d)/2}e$. Adding μ is conventional: we use it to keep e dimensionless. It will disappear when we take $d \rightarrow 4$. The units are

$$[A] = \frac{d-2}{2}, \quad [\psi] = \frac{d-1}{2}, \quad [\mu] = 1, \quad [\gamma] = [e] = 0, \quad (827)$$

so

$$[\mathcal{L}] = d. \quad (828)$$

Now let's regularize the divergent integral

$$I = \frac{e^2}{(2\pi)^4} \int \frac{d^4 p}{(\vec{p}^2 - \Delta + i\epsilon)^2}. \quad (829)$$

This diverges in 4D. I rewrite this integral in d dimensions

$$I(d) = \mu^{4-d} \frac{e^2}{(2\pi)^d} \int \frac{d^d p}{(\vec{p}^2 - \Delta + i\epsilon)^2}, \quad (830)$$

so $I(4) = I$. Notice that $e \rightarrow \mu^{(4-d)/2}e$. For values of d where this integral does not diverge I can integrate it using Eq. 811.

$$I(d) = \mu^{4-d} \frac{e^2}{(2\pi)^d} \frac{i\pi^{d/2}}{\Delta^{2-d/2}} \Gamma\left(\frac{4-d}{2}\right) \quad (831)$$

$$I(d) = \frac{ie^2\mu^{4-d}}{(4\pi)^{d/2}\Delta^{2-d/2}} \Gamma\left(\frac{4-d}{2}\right) \quad (832)$$

I return to 4D.

$$I = I(4) = \lim_{\epsilon \rightarrow 0} I(4 - \epsilon) \quad (833)$$

Let the limit $\epsilon \rightarrow 0$ be implicit.

$$I = I(4 - \epsilon) \quad (834)$$

$$I = \frac{ie^2 \mu^\epsilon}{(4\pi)^{2-\epsilon/2} \Delta^{\epsilon/2}} \Gamma\left(\frac{\epsilon}{2}\right) \quad (835)$$

$$I = \frac{ie^2}{16\pi^2} \left(\frac{4\pi\mu^2}{\Delta}\right)^{\epsilon/2} \Gamma\left(\frac{\epsilon}{2}\right) \quad (836)$$

I want to absorb the 2 into ϵ , but I can't. We see from Eq. 799 that the scale of ϵ affects the value of I :

$$\Gamma\left(\frac{\epsilon}{2}\right) = \frac{a}{\epsilon} - \gamma_E. \quad (837)$$

I expand around $\epsilon = 0$.

$$I = \frac{ie^2}{16\pi^2} \left[\frac{2}{\epsilon} + \ln\left(\frac{\tilde{\mu}^2}{\Delta}\right) \right], \quad (838)$$

where

$$\tilde{\mu}^2 = 4\pi e^{-\gamma_E} \mu^2, \quad (839)$$

where e is Euler's number, not the electric charge. Finally

$$\frac{e^2}{(2\pi)^4} \int \frac{d^4 p}{(\vec{p}^2 - \Delta + i\epsilon)^2} = \frac{ie^2}{16\pi^2} \left[\frac{2}{\epsilon} + \ln\left(\frac{\tilde{\mu}^2}{\Delta}\right) \right], \quad (840)$$

or

$$\boxed{\frac{1}{i\pi^2} \int \frac{d^4 p}{(\vec{p}^2 - \Delta + i\epsilon)^2} = \frac{2}{\epsilon} + \ln\left(\frac{\tilde{\mu}^2}{\Delta}\right)}. \quad (841)$$

That's dimensional regularization.